

5 Interactions and topology for itinerant electrons

5.1 Introduction

5.1.1 Outline

As outlined in the previous chapter of this book, the single-particle band structure description for electrons in solids can break down completely due to the electron-electron interactions. For example, fully occupied Hubbard bands develop from a half-filled single-particle band, such that the system turns from a metal into an insulator (Section 4.5). Intriguingly, one cannot capture this behavior via a parameter-free description based on first principles. Instead, one has to apply instructive toy models as the Hubbard Hamiltonian (eq. (4.26)), that can explain the qualitative behavior of the electrons in the solid straightforwardly. The Hubbard model can then act as a starting point to implement further relevant interactions such as the spin-orbit interaction (eq. (4.44)) or electron-lattice couplings (Section 4.6). This eventually leads to the explanation of other observed effects such as polaron hopping or orbital ordering.

These type of model approaches proved to be extremely successful to get a systematic understanding of a multitude of effects observed in the correlated oxides and chalcogenides. Nevertheless, the approach is incomplete by construction, e.g., ignoring the long-range part of the Coulomb interaction. It is believed that this incompleteness hides additional interesting properties such as the still not understood origin of high temperature superconductivity (Section 4.1). In other words, it is believed that the incompleteness of the developed toy models explains most of the missing understanding of solid state properties. Hence, the partial failure in quantitative understanding does not indicate any failure of the first principle assumptions as given by the Schrödinger or Dirac equation, but points to inappropriate approximations during the solution of these equations.

Therefore, a central task is to develop adequate toy models for the approximate description of the multiple degrees of freedom in a solid. These toy models must be combined with the knowledge, that can be gained by first-principle methods mostly based on the density functional theory (DFT) approach. DFT can be employed to narrow down the adequate parameter regimes. This way, the toy models get a predictive character to be confirmed by experiments.

In the following final chapter of this book, additional toy models are introduced, that apply to the more itinerant s-type or p-type electrons, as present, e.g., in doped semiconductors (Figs. 2.8 and 5.1(a)). Hence, the starting configuration of the tight binding model, that is used for the oxides and basically localizes electrons at single atomic sites, has to be replaced by a starting configuration operating with Bloch states

or, even more simplified, with plane wave states $\psi(\vec{x}) \propto e^{i\vec{k}\vec{x}}$ as for free electrons. Consequently, the phase of the electron wave function can be decisive for the behavior of the system as discussed for some simple systems already in Section 1.4 and 1.5.

In first principle methods, electrons in crystalline solids are described as Bloch waves, i.e., each quasiparticle is mapped to a single particle wave function $\psi_j(\vec{x}) = u_{n\vec{k}}(\vec{x}) \cdot e^{i\vec{k}\vec{x}}$ (eq. (2.90)). Thereby, n is the band index, $\vec{k}$ the wave vector, $u_{n\vec{k}}(\vec{x})$ a function that exhibits the periodicity of the lattice vectors, and $j \in \mathbb{N}$ a counting index. A typical band structure (Fig. 5.1(a)) displays the energy of these single particle states as a function of the wave vector $\vec{k}$, that is the primary good quantum number in a periodic crystal. A rather successful model to treat the resulting quasiparticle states is to fill pairs of electrons with opposite spin into the states of lowest energy up to the Fermi level E_F at $T = 0$ K. At larger T , one can use the Fermi–Dirac distribution function (eq. (1.39)) for filling. An excitation of the solid, e.g., by absorbing a photon, is then described as a transition of one electron from an occupied to an unoccupied single-particle state (Section 2.3.1). The calculated short-range atomically periodic parts $u_{n\vec{k}}(\vec{x})$ as well as the long-range wave parts $e^{i\vec{k}\vec{x}}$ of the Bloch state are, hereby, very realistic descriptions. They can be imaged, e.g., by scanning tunneling spectroscopy (STS) (Fig. 5.1(b)–(f)), where the visibility of the long-range part typically requires scattering at defects in order to fix the phase of the wave in real space. The band structure model also describes a large part of other solid state properties adequately such as, e.g., conductivity σ . However, several predictions of the band structure model are qualitatively wrong. This was outlined in the previous chapter of the book (Chapter 4) and will be the focus of this chapter of the book, too.

We will again go beyond the band structure description, but in contrast to the previous chapter, we will start from rather simple band structures as derived from s-type and p-type electrons. We will systematically discuss the influence of different types of interactions beyond the band structure description. These interactions partly change the behavior of the electron system completely. They are electron-phonon, electron-disorder and electron-electron interaction. Some examples have already been discussed in previous sections. For instance, weak localization (Section 1.4.3.7) is a consequence of the interplay of crystalline disorder with the phase of a single particle wave function at the Fermi level, eventually leading to localization of electrons via constructive interference. This effect, not present in the single-particle band structure description, drives the system insulating, which would be a metal according to its single-particle band structure. Another example is the Mott–Hubbard transition (Section 4.5.3), where the electron-electron interaction (described by the Hubbard U) drives the single-particle metal into an insulator.

In this chapter, we will extend the learned approaches used to describe such phase transitions by analyzing several other phase transitions of the electron system, that are driven by the strength of a particular interaction. Thereby, we will also employ a novel classification scheme of electron phases, which is topology. It boils down the properties of an electron phase, e.g., conductivity, to a single integer number deduced

from the Schrödinger or Dirac equation. Such a number can also be found from a single-particle band structure, such that the single particle states determine the collective properties as, e.g., surface conductivity directly. Importantly, the number called the topological index makes the found behavior robust against small continuous changes of the Hamiltonian as, e.g, the interaction of the electrons with disorder. Thus, it allows distilling of robust properties already from the incomplete descriptions as via first-principle methods or adequate toy models without including small disturbing interaction terms explicitly. More intriguingly, from a practical point of view, the novel classification scheme discovered useful properties of solids, that were simply overlooked for decades, albeit the corresponding solids were available and experimental hints to these particular properties were present in measured data.

5.1.2 Approximations of the band structure model

Before starting the description of the interaction effects, we recall, once more, the approximations leading to the description of electrons by Bloch waves (Section 4.2). A solid consists of N_A nuclei and $b \cdot N_A$ electrons with b being the average number of electrons per atom. Quantum-mechanically, this ensemble is described by the many-particle wave function $\Psi_{\text{total}}(\vec{x}_1, \vec{x}_2, \dots, \vec{x}_{N_A}, \vec{x}_{N_A+1}, \dots, \vec{x}_{N_A+bN_A}, t)$.¹ The function Ψ_{total} must be a solution of the Schrödinger equation or the Dirac equation including all energy terms of the system. Additionally, it must obey the exchange statistics of fermions and bosons.

The first approximation separates the electron system from the nuclei (Born–Oppenheimer-approximation) (Section 4.2). The electronic wave function, that reads $\Psi_{\text{el}}(\vec{x}_{N_A+1} \dots \vec{x}_{N_A+bN_A})$, is calculated for fixed positions of the nuclei. The interaction potential between the nuclei is then reconstructed from the electron energy at different fixed positions of the nuclei, such that the solutions of the resulting Schrödinger equation for the nuclei reveal the excitation energies of the nuclear system (phonons). The central argument in favor of the separation is, that the electrons are much lighter than the nuclei, such that they can follow the motion of the nuclei rather instantaneously. One can estimate the error in energy made by the separation $\Psi_{\text{total}} = \Psi_{\text{nuclei}} \cdot \Psi_{\text{el}}$ as $\Delta E = 1/b \cdot (m_e/M_n)^{0.75} \cdot E_{\text{electrons}} = 1/b \cdot (m_e/M_n)^{0.25} \cdot E_{\text{nuclei}}$ with m_e and M_n being the mass of electron and nucleus, respectively, and $E_{\text{electrons}}$ and E_{nuclei} the energies of the electron and the nuclear system, respectively. Thus, ΔE is small with respect to E_{nuclei} and $E_{\text{electrons}}$. Nevertheless, the properties of the system can be dramatically changed by the small contribution ΔE of the electron-phonon-interaction. Prominent examples are superconductivity as discussed in more basic textbooks of condensed matter physics and charge density waves as described in Section 5.2.1.

¹ Here, we ignore the spin degree of freedom of each electron and nucleus, for the sake of simplicity.

The second approximation is the separation of the electron wave function in single particle states:

$$\Psi_{\text{el}}(\vec{x}_1, \dots, \vec{x}_{bN_A}) = P_{\leftrightarrow} \left(\prod_{i=1}^{bN_A} \psi_i(\vec{x}_i) \right), \quad (5.1)$$

$$E_{\text{electrons}} = \sum_{i=1}^{bN_A} E_i. \quad (5.2)$$

Hereby, $P_{\leftrightarrow}$ makes the required permutations in order to get an antisymmetric wave function Ψ_{el} as necessary for fermions. The twofold occupation of each single particle state by electrons with opposite spin (Pauli principle) is a direct consequence of this antisymmetry. The separation into single particle states works, if the individual electrons are independent, i.e., if the Coulomb interaction between the electrons gets negligible. This is not the case, since typical Coulomb interaction energies between electrons in a solid are as large as the kinetic energies of the electrons (1–10 eV) (Section 4.3). However, using an iterative process within the so-called local density approximation, one can approximate these interactions by an additional, effective electrostatic potential for a particular electron. In the most simple, quasi classical approximation, the potential $\Phi_{\text{electrons}}^j(\vec{x})$ for the electron in state ψ_j reads:

$$\Phi_{\text{electrons}}^j(\vec{x}) = \frac{1}{4\pi\epsilon_0} \cdot \int \int \int \frac{n_{e,j}(\vec{y})}{|\vec{y} - \vec{x}|} d^3\vec{y}. \quad (5.3)$$

Here, $n_{e,j}(\vec{y}) = \sum_{i \neq j} |\psi_i(\vec{y})|^2$ is the electron density given by all the other electrons. The potential energy $\Delta E_{\text{pot},j}$ of the electron j in the electric field of all other electrons is then: $\Delta E_{\text{pot},j} = \langle \psi_j(\vec{x}) | \Phi_{\text{electrons}}^j(\vec{x}) | \psi_j(\vec{x}) \rangle$. The iterative process to determine $\Delta E_{\text{pot},j}$ starts with a set of single particle wave functions, calculates the electron density $n_{e,j}(\vec{y})$ and the corresponding $\Phi_{\text{electrons}}^j(\vec{x})$ from these wave functions, solves the Schrödinger equation with $\Phi_{\text{electrons}}^j(\vec{x})$ via matrix diagonalisation, and uses the novel wave functions from the diagonalisation to repeat the whole loop. The loop repetition stops, if the changes in energy per loop are small. Notice, that the diagonalisation is an approximation, too, since it requires an infinitely large $n \times n$ matrix to be exact, but uses only a finite number of matrix elements in practice.

Moreover, the loop neglects the required antisymmetrisation of Ψ_{el} , that would lead, in second quantisation, to terms of the form:

$$\hat{H}_{\text{ee}} = \sum_{i,j,k,l} u_{ijkl} \hat{c}_i^\dagger \hat{c}_j^\dagger \hat{c}_k \hat{c}_l \quad (5.4)$$

with

$$u_{ijkl} = \int \int \int \left(\int \int \int \psi_i^*(\vec{x}) \cdot \psi_j^*(\vec{y}) \cdot \frac{e^2}{4\pi\epsilon_0 \cdot |\vec{x} - \vec{y}|} \cdot \psi_k(\vec{x}) \cdot \psi_l(\vec{y}) d^3\vec{y} \right) d^3\vec{x}$$

with $\hat{c}_i$ ($\hat{c}_i^\dagger$) being annihilation (creation) operators of electrons. Notice that, in eq. (5.3), only the terms u_{ijj} contribute. The terms u_{ijji} are the exchange terms (Sec-

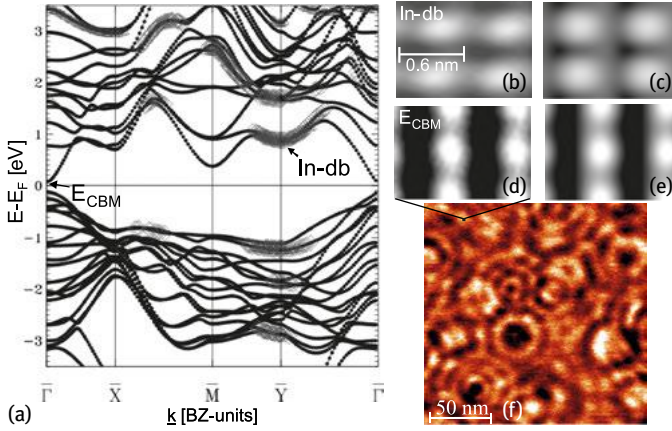


Fig. 5.1: (a) Band structure of a slab of an InAs crystal. The bands are projected to the (110) surface, i.e., only their k_x and k_y value are displayed. Each point is a single Bloch state as calculated by density functional theory (DFT) within the local density approximation (LDA). The Bloch states are equidistant in $\vec{k}$ space. The conduction band minimum E_{CBM} and a surface state (In-db) are marked. (b), (d) Measured $|u_{n\vec{k}}(\vec{x})|^2$ using scanning tunneling spectroscopy (STS) for the states marked in (a) as In-db (db = dangling bond) (b) and E_{CBM} (d). More precisely, the image displays the sum of all $|u_{n\vec{k}}(\vec{x})|^2$ for all states at the chosen energy. (c), (e) Calculated sum of all $|u_{n\vec{k}}(\vec{x})|^2$ at the energies marked in (a). (f) Large-scale STS image at energy $E - E_F = 0.05$ eV. The concentric rings are a 2D-cut through the 3D standing electron waves, that form spherical standing waves around defects $|\psi_i(|\vec{x}|)|^2 \propto 1/|\vec{x}|^2 \cdot \cos^2(|\vec{k}| \cdot |\vec{x}| + \varphi)$ ($|\vec{x}|$: distance to defect, φ : phase shift). The wave length $\lambda = 2\pi/|\vec{k}|$ is ~ 40 nm; (d) is a zoom of (f) as marked. (a)–(e) [101], (f) [102].

tion 4.4.1). All other terms are called correlation terms. Within the local density approximation, these exchange and correlation terms are approximated as an additional electrostatic potential $\Phi_{xc}(\vec{x})$, called the exchange-correlation potential. $\Phi_{xc}(\vec{x})$ can be approximated analytically for a free electron gas (no nuclei) as a functional of the electron density $n_e(\vec{x})$. This offers a reasonable starting point for the development of adequate potentials in the presence of nuclei. A multitude of approximate functionals $\Phi_{xc}(n_e(\vec{x}))$ have been derived for different solids with different potentials of nuclei in order to approximate the electron system as good as possible.²

Using these functionals of $\Phi_{xc}(n_e(\vec{x}))$, the loop described above can be used to determine the single particle wave functions iteratively. Formally, these single particle wave functions are variational parameters only, but they approximate the real single particle wave functions adequately for most cases.

² This approach can be extended by combining it with the Hubbard model (Section 4.5) via including an on-site term U into the DFT calculation in a method called LDA+U (LDA: local density approximation) or by the inverse mapping of the DFT band structure to an adequate local model, that is called dynamical mean-field theory (DMFT).

The band structure in Fig. 5.1(a) and the atomically periodic parts of the squared single particle wave functions in Fig. 5.1(c) and (e), as calculated by such a method, e.g., agree excellently with the experimental data (Fig. 5.1(b) and (d)). The method gets worse, if the electron-electron interaction gets stronger with respect to the kinetic energy of the electrons. This is the case for the strongly localized d- and f-electrons in oxides, where the Hubbard model reveals better results than a DFT approach (Section 4.5). For nearly free electron systems (itinerant electron states), the electron-electron interaction dominates with respect to the kinetic energy at low electron density, in particular, if the dimension of electron motion is reduced to 2 or 1 or in strong magnetic fields. This either leads to localized electron phases (Wigner crystal, Section 5.5.2.1) or it implies, that the excitations of the system are genuinely of a many-particle type as described in Section 5.5.2.3.

A third approximation within the band structure description is the perfectly periodic lattice potential. This implies the lattice periodicity of $|\psi_i(\vec{x})|^2$ for each electronic state i , i.e., the description by extended Bloch waves. In real crystals, the periodicity is not perfect. Firstly, the crystal has boundaries. This leads to additional states, the so-called surface states (not discussed here), that are the solutions with an imaginary $\vec{k}$ component perpendicular to the surface (eq. (1.28) in Section 1.4.3). Moreover, the interior of the crystal contains defects such as vacancies, dislocations, or foreign atoms. For many properties of the solid as, e.g., the heat capacity, these defects are only a minor perturbation. For other properties as, e.g., the low temperature conductivity, they are constitutional (e.g., Section 1.4.3.7). There are regimes, where the disorder changes the properties of the system completely, as within the quantum Hall regime: the disorder in the atomic lattice leads to a Hall resistance R_{Hall} of nearly always the $1/n$ -fold ($n \in \mathbb{N}$) of the natural constant h/e^2 (h : Planck's constant) with a measured precision of $\Delta R_{\text{Hall}}/R_{\text{Hall}} \approx 10^{-10}$ (Section 5.3.2). This spectacular precision also led to the more abstract description of the whole system by the already mentioned single integer index on the base of a topological analysis. This type of analysis has afterwards been extended to other solids, where it led to the prediction of dissipationless electron transport at surfaces and edges. The corresponding edge transport was indeed subsequently found in experiments (Section 5.4).

In summary, there are three major approximations for the description of solid state electrons as Bloch waves, i.e., one neglects electron-phonon, electron-disorder, and the dynamics of electron-electron interactions. All three of them imply novel phases and phenomena of the electron system not covered by the band structure description as exemplary described in the following.³ The most complex description results from the electron-electron interaction as partly already covered in Section 4.5–4.7. An

3 Another approximation is the separation of spin and spatial degrees of freedom of the electrons, that neglects the spin-orbit interaction (Section 4.5.4), itself originating from the relativistic description of spins within the Dirac equation. This leads to additional phenomena, e.g., the Rashba effect (e.g. [103]), that are beyond the scope of this book.

experimentally simple approach to study the consequences of such interactions are so-called artificial atoms or quantum dots (Section 5.5.1). Their investigation basically matches the few particle description of exchange effects in Section 4.4.

We will start with a consequence of the electron-phonon interaction (Section 5.2), will then describe some consequences of the electron-disorder interaction (Section 5.3) including the topological description of solids (Section 5.4). Finally, we will discuss the electron-electron interaction starting with the quantum dots (Section 5.5.1), before we proceed with higher dimensions employing itinerant s- or p-type electrons (Section 5.5.2). For the latter, we will learn how quasiparticles can emerge in solids prone to interaction effects, which are qualitatively different from the constituting particles. As an example, we discuss composite fermions, which can exhibit a charge, which is only a fraction of the electron charge e and a mass that depends on the electron density n_e .

5.2 Electron-Phonon Interaction

Since the electron-phonon interaction is small with respect to the electronic and phononic energy, it can be described via perturbation theory. In second quantisation, one finds for the electron system:

$$\hat{H}_{e,ph} = \sum_{n,n',\vec{k},\vec{q},\vec{G},j} M_{\vec{q}+\vec{G},\vec{k}}^{j,n,n'} \cdot (\hat{b}_{\vec{q},j}^\dagger + \hat{b}_{-\vec{q},j}) \cdot \hat{c}_{n',\vec{k}-\vec{q}-\vec{G}}^\dagger \cdot \hat{c}_{n,\vec{k}}. \quad (5.5)$$

This corresponds to a process, where an electron is initially in state n , $\vec{k}$ ($\hat{c}_{n,\vec{k}}$) and then scattered into state n' , $\vec{k} - \vec{q} - \vec{G}$ ($\hat{c}_{n',\vec{k}-\vec{q}-\vec{G}}^\dagger$), while it creates a phonon ($\hat{b}_{\vec{q},j}^\dagger$) or destroys it ($\hat{b}_{-\vec{q},j}$). Here, $\vec{k}$ and $\vec{q}$ are the wave vectors of the initial electrons and phonons, which must be summed up across the whole first Brillouin zone for both partners. The term $\vec{G}$ is a reciprocal lattice vector, describing so-called Umklapp scattering processes. Thereby, the translational symmetry of the crystal lattice implies a modified momentum conservation, that applies only modulus reciprocal lattice vector $\vec{G}$, since the translational symmetry is not continuous, but only periodic with the lattice vectors. The indices n , n' are electron band indices and j ($= 1, 2, 3$) sums the three possible, linearly independent vibrational directions of the phonons. The creation and annihilation operators $\hat{b}_{\vec{q},j}^\dagger$ and $\hat{b}_{\vec{q},j}$ for the phonons as well as $\hat{c}_{n,\vec{k}}^\dagger$ and $\hat{c}_{n,\vec{k}}$ for the electrons are linked to each other via matrix elements $M_{\vec{q}+\vec{G},\vec{k}}^{j,n,n'}$. The sum, thus, considers all processes as shown in Fig. 5.2(a) and (b), i.e., scattering of an electron via phonon creation or via phonon annihilation.

The calculation of the matrix elements is laborious and is not shown here. For a lattice with a single atom of mass M_n within the unit cell, it reads:

$$M_{\vec{q}+\vec{G},\vec{k}}^{j,n,n'} = \sqrt{\frac{\hbar \cdot N_A}{V_{uc}^2}} \cdot \frac{1}{\sqrt{M_n \cdot \omega_{ph,j}(\vec{q})}} \cdot i \cdot (\vec{q} + \vec{G}) \cdot \vec{e}_j(\vec{q}) \cdot \vec{\Phi}_{\vec{q}+\vec{G}}^{el} \cdot \langle n', \vec{k} - \vec{q} - \vec{G} | n, \vec{k} \rangle \quad (5.6)$$

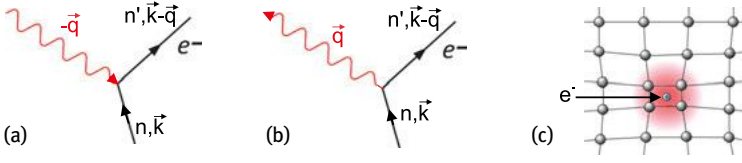


Fig. 5.2: (a) Process of phonon annihilation. (b) Process of phonon creation: straight black lines mark electronic states, curved red lines are phonon states. (c) Illustration of electron-phonon interaction.

with V_{uc} being the volume of the unit cell, N_A the number of atoms, $\omega_{ph,j}(\vec{q})$ the phonon frequency, $\vec{e}_j$ the unit vector in the direction of vibration of the phonon, $\langle n', \vec{k} - \vec{q} - \vec{G} | n, \vec{k} \rangle$ the overlap integral of the two atomically periodic parts of the contributing electron states, and $\tilde{\Phi}_{\vec{q}+\vec{G}}^{el}$ the Fourier component of the ionic potential acting on the electron and corresponding to the wave vector $\vec{q} + \vec{G}$.

Generally, the dynamical modification of the ionic potential due to the vibration leads to a mixing of similar single-particle states of the electrons. The process is particularly efficient, if the mass of the ion is small ($\omega_{ph,j} \propto 1/\sqrt{M_n} \Rightarrow 1/\sqrt{M_n} \cdot \omega_{ph,j} \propto M_n^{-0.25}$) and, if the Fourier component of the ionic potential, that corresponds to the scattering vector, is large. Moreover, scattering is favored for longitudinal phonons ($\vec{e}_j \parallel \vec{q}$) except for Umklapp scattering ($\vec{G} \neq 0$). Finally, the scattering is favored for electron states of the same band, because states in the same band exhibit similar orbital wave functions increasing the overlap integral. Consequently, the overlap integral is often approximated by $\delta_{nn'}$.

Using Coulomb potentials for the ions $\Phi^{el}(\vec{x}) \propto 1/|\vec{x}|$, – a good approximation in insulators-, one finds $\tilde{\Phi}_{\vec{q}}^{el} \propto 1/\vec{q}^2$ and, therefore, at small phonon energies ($\omega_{ph} \propto |\vec{q}|$) $M_{\vec{q},\vec{k}} \propto |\vec{q}|^{-1.5}$ and at large phonon energies ($\omega_{ph} \approx \text{const.}$) $M_{\vec{q},\vec{k}} \propto |\vec{q}|^{-1}$, i.e., the interaction with longitudinal phonons close to the center of the Brillouin zone, called Γ , is strongest. In metals, the ionic potentials are strongly screened by the conduction electrons leading approximately to $\Phi^{el}(\vec{x}) \propto \delta(\vec{x}) \Rightarrow \tilde{\Phi}_{\vec{q}}^{el} = \text{const.}$, i.e., to $M_{\vec{q},\vec{k}} \propto |\vec{q}|^{0.5}$ ($M_{\vec{q},\vec{k}} \propto |\vec{q}|$) for small (large) phonon energies, such that phonons at the Brillouin zone boundary contribute most strongly.

Often, the electron-phonon interaction is illustrated as in Fig. 5.2(c). The charge of the passing electron locally strains the atomic lattice. When the electron is gone, the strain relaxes, thereby inducing a lattice vibration. Of course, the electron can also decelerate an already existing vibration, if passing at the right vibrational phase. This corresponds to a phonon annihilation.

The most simple consequence of the electron-phonon interaction is a change of the single particle energy of the electrons. For s-type states, that can be described well by a parabolic dispersion ($E = \hbar^2 \vec{k}^2 / (2m^*)$) with effective mass m^* , the electron-phonon interaction changes m^* . Table 5.1 shows that the measured mass is up to 50% larger than calculated, while neglecting the electron-phonon interaction, i.e.,

Tab. 5.1: Effective mass m_{LDA}^* as calculated without electron-phonon interaction using the local density approximation and experimentally determined effective mass m_{Exp}^* for s-type states at the Fermi level within simple metals (m^* is given in units of the bare electron mass m_e).

metal	Li	Na	K	Rb	Cs	Al
$m_{\text{LDA}}^* (m_e)$	1.54	1.01	1.07	1.19	1.53	1.05
$m_{\text{Exp}}^* (m_e)$	2.22	1.24	1.21	1.37	1.80	1.49

the electrons are more difficult to move due to their interaction with phonons. This corresponds to the idea that the electron is accompanied by a lattice strain during its movement through the crystal, that has to be accelerated together with the electron. This is similar to the polaronic effect discussed in Section 4.7.

A second, intuitively simple consequence of the electron-phonon interaction is the electrical resistivity, that, at room temperature, is typically dominated by the scattering of electrons at phonons. Within the Boltzmann approximation, one firstly calculates, with the help of Fermi's Golden Rule (eq. (2.99)), the transition probabilities between the single particle states $n\vec{k}$ and $n'\vec{k}'$ of the electrons: $W_{n\vec{k} \rightarrow n'\vec{k}'} \propto |\langle n'\vec{k}' | \hat{H}_{e,\text{ph}} | n\vec{k} \rangle|^2$. After convoluting with the distribution functions (Fermi–Dirac for electrons, Bose–Einstein for phonons), one finds after a lengthy calculation that the temperature dependence of the resistivity $\rho(T)$ for scattering at acoustic phonons reads approximately [104]:

$$\begin{aligned} \rho(T) &\propto T^5 & \text{for } T \ll \Theta_D \\ \rho(T) &\propto T & \text{for } T \gg \Theta_D \end{aligned} \quad (5.7)$$

with Θ_D being the Debye temperature of the material. Such a temperature dependence has indeed been found for simple metals such as Na and Cu. For longitudinal optical (LO) phonons, we have $\omega_{\text{ph}}(\vec{q}) \approx \omega_{\text{ph,LO}} \approx \text{const}$, i.e., $M_{\vec{q},\vec{k}} \propto |\vec{q}|^{-1} \cdot \omega_{\text{ph,LO}}^{-0.5}$ leading to $\rho(T) \propto T$. Acoustic phonons in piezoelectric materials are a peculiarity, that exhibit a T^4 dependence of the resistivity at low temperature.

Besides these purely quantitative consequences of the electron-phonon interaction (heavier mass and limitation of the mean free path or, more formally, modification of the real and imaginary part of the self energy), one also finds qualitative changes of the electron system driven by the electron-phonon interaction, i.e., phase transitions. The most prominent example is the superconducting phase appearing at low temperature. The explanation within BCS theory considers electron-phonon interactions in second order perturbation theory, i.e., $\hat{H}_{e,\text{ph}}$ is described by processes as shown in Fig. 5.3.

It turns out that these processes, effectively coupling two electrons, lead to a reduction of the energy of the electrons, if the energy of the contributing phonon is larger than the energy difference between the initial and final state of each electron. Due to this effective binding energy, two electrons form a stable pair with bosonic character

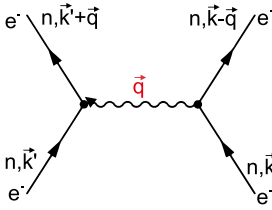


Fig. 5.3: Scattering process as derived within second order perturbation theory of the electron-phonon interaction, eventually leading to Cooper pairs.

(spin = 1 or 0) such that these bosons (Cooper pairs) can condense, within a Bose–Einstein like transition, into a phase where all bosons occupy the energetically lowest state in a phase coherent manner. This phase is the celebrated superconducting condensate with the favorable property of zero electrical resistance. Note that although the electron-phonon interaction is weak (Section 5.1.2), second order perturbation theory in this weak interaction can be decisive for the behavior of the system.

5.2.1 Peierls transition

The time period between the discovery of superconductivity (1908) and its explanation (1957) by the BCS theory amounts to five decades, which is largely due to the intuitively unexpected dominance of processes within second order perturbation theory.⁴ For a long period, a phase transition that appears in first order perturbation theory was thought to be a likely explanation. This phase transition, called the Peierls transition, will be discussed in the following.

Experimentally, one observes for some materials that the electron charge distribution gets inhomogeneous below a certain temperature, i.e., it forms a static wave pattern, whose wave length is half the Fermi wave length λ_F . Figure 5.4(b) shows a scanning tunneling microscopy (STM) image of such a charge density wave. Besides the atomic periodicity, that corresponds to the known $|u_{n\vec{k}}(\vec{x})|^2$ of the Bloch waves, one recognizes a periodicity with the threefold atomic distance, which is rather precisely $\lambda_F/2$. At higher temperature, this additional periodicity disappears (Fig. 5.4(c)).⁵ Simultaneously with the appearance of the charge density wave, one observes a lateral shift of the nuclei by about 1% of the lattice constant exhibiting a static wave pattern of the nuclear shifts (atomic superstructure) with wave length $\lambda_F/2$ as well. This is called a frozen phonon, since the wave pattern corresponds to a standing phonon wave with frequency $\omega_{ph} = 0/s$, i.e., with energy $E_{ph} = 0$ meV. This phonon is, thus, already excited at temperature $T = 0$ K, i.e., it is frozen.

⁴ The electron-phonon interaction is a weak perturbation, that might lead one to believe that it is negligible in second order perturbation theory.

⁵ STM approximately maps the contours of constant charge density of all states between the Fermi level of the sample and the Fermi level of the tip, that are different due to an applied tunneling voltage.

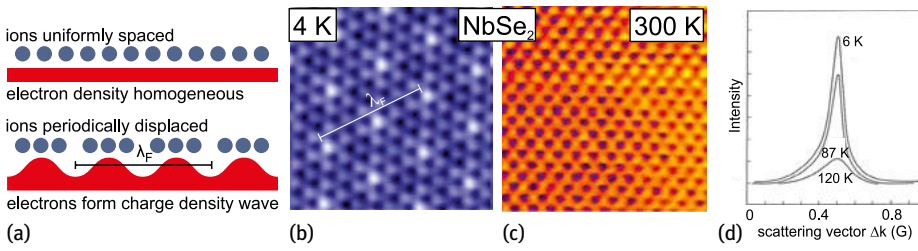


Fig. 5.4: (a) Model of atomic arrangement above (top) and below (bottom) the transition temperature of the Peierls transition. The shifts of the blue atoms (ions) are largely exaggerated. The Fermi wave length λ_F is marked. The pink area indicates the resulting distribution of electron charge. (b) Scanning tunneling microscopy (STM) image of NbSe₂ below the Peierls transition temperature. Besides the atomic periodicity, one observes a long-range periodicity with period $\lambda_F/2$. (c) STM image of NbSe₂ above the Peierls transition temperature. Only the atomic periodicity is visible. (d) Intensity of x-ray diffraction of K₂Pt(CN₄)Br_{0.3} as a function of the polar scattering angle (scattering vector Δk in units of the reciprocal lattice vector G) at different temperatures as marked. The strength of the peak, that belongs to the charge density wave, disappears at the Peierls transition temperature, (d) [105].

Figure 5.4(a) schematically shows the shift of the nuclei (ions) and Fig. 5.4(d) displays x-ray diffraction data for the diffraction spot belonging to a charge density wave as a function of temperature. The transition temperature is $T \approx 100$ K, around which the peak gets weaker and broader with increasing temperature, i.e., the charge density wave continuously disappears, while the domains of phase coherent charge density wave areas get smaller. Besides these observations, that principally could also indicate an entropy driven structural phase transition, one finds that the phonon at wave length $\lambda_F/2$ (wave vector: $|\vec{q}| = 2 \cdot |\vec{k}_F|$) exhibits a phonon energy reduction as shown in Fig. 5.5(a) and that the electron system exhibits a band gap E_{Gap} at the Fermi level of order 10–100 meV (Fig. 5.5(b)). The latter can be confirmed, e.g., by measuring the temperature dependence of the electrical conductivity $\sigma(T)$ (Section 1.6.3). Figure 5.5(c) shows $\sigma(T)$ of a certain material in the inset. The Peierls transition is indicated by the maximum of the conductivity. Above 250 K, σ decreases with increasing temperature as usual for metals, since the increasing number of phonons increases the scattering probability for the electrons. Below 250 K, σ increases with increasing T as usual for insulators, since the number of excited charge carriers into the conduction band increases with T . The Arrhenius plot in the main image shows a nearly linear behavior below 250 K, such that an energy barrier for excitations of $E_{\text{Gap}} = 80$ meV can be deduced according to $\sigma(T) \propto \exp(-E_{\text{Gap}}/k_B T)$ (k_B : Boltzmann constant). Hence, a band gap developed at E_F because of the Peierls transition. With the help of tunneling spectroscopy, the band gap can also be probed directly.

Most materials showing a Peierls transition exhibit either a strongly one dimensional (1D) structure, i.e., electronic dispersion and electrical conductivity are much stronger in one direction than in the two perpendicular ones, or large parts of the Fermi

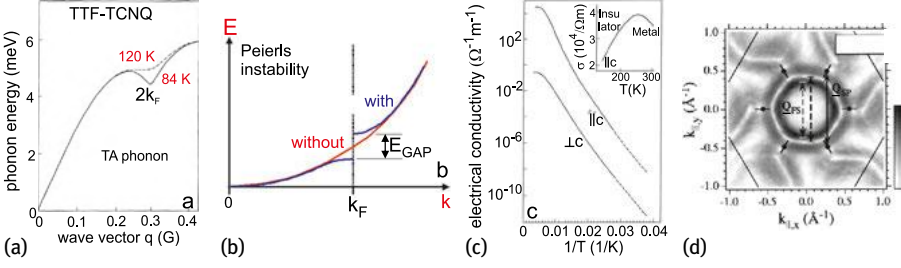


Fig. 5.5: (a) Phonon dispersion of the transversal acoustic (TA) branch of TTF-TCNQ at two temperatures slightly below and slightly above the Peierls transition temperature as determined by inelastic neutron diffraction. (b) Schematic of the electron dispersion $E(\vec{k})$ close to E_F above (red) and below (blue) the Peierls transition temperature. (c) Electrical conductivity of $\text{K}_2\text{Pt}(\text{CN}_4)\text{Br}_{0.3}$ below the Peierls transition temperature displayed as an Arrhenius plot. The linearity shows activated behavior of the type $\sigma(T) \propto \exp(-E_{\text{Gap}}/k_B T)$ with a deduced band gap of $E_{\text{Gap}} \approx 80$ meV. The two curves probe two different crystallographic directions. The inset shows the conductivity in linear $\sigma(T)$ representation. (d) A 2D section through the 3D Fermi surface for a material that shows a 2D Peierls transition (determined by photoelectron spectroscopy). The data are differentiated, i.e., the gradient of the photoelectron intensity $I_{\text{el}}, \nabla_{\vec{k}} I_{\text{el}}(E = E_F, k_x, k_y)$, is displayed in order to increase the visibility. Black lines mark the Brillouin zone boundaries. The vectors $\vec{Q}_{FS}$ and $\vec{Q}_{SP}$ are the preferential scattering vectors of the electron-phonon interaction leading to the Peierls transition. (a), (c) [105], (d) [106].

line of a largely two-dimensional (2D) material consist of parallel lines. The latter is shown in Fig. 5.5(d), where the inner Fermi line is mostly a hexagon, where the opposite edges of the hexagon run parallel.

The driving force of the Peierls transition is displayed in Fig. 5.4(a). The creation of the frozen phonon leads to periodic areas of increased positive charge by the ionic cores. Arranging the negative electronic charge density (pink area) in a way, such that the maxima are at these points gains Coulomb energy $E_{\text{e-ion}}$. However, the frozen phonon is not the most favorable arrangement of the ions such that one has to pay elastic energy E_{elastic} in order to create the frozen phonon. The decisive question, thus, is, if the energy difference $\Delta E = E_{\text{e-ion}} - E_{\text{elastic}}$ is negative, i.e., favorable to realize a charge density wave. This, interestingly, is always the case for 1D electron systems, as will be shown below.

The second question is about the favorable wave length. For the answer, one has to consider that the electron-phonon interaction, in first order perturbation theory, corresponds to scattering processes as described by eq. (5.5). This requires that the initial state of the electron is occupied while the final state is empty. Without additional energy and at $T = 0$ K, this is only possible at E_F . The corresponding process is displayed for a 1D system in Fig. 5.6(a). Obviously, the favorable wave vector of the phonon is $\vec{q} = 2 \cdot \vec{k}_F$. This fits with the experimental wave length of the frozen phonons. In addition, the resulting superposition of the two electronic states with opposite wave vector $(\vec{k}_F, -\vec{k}_F)$ creates a standing wave with a charge density $(|\psi_i(\vec{x})|^2)$ of the exper-

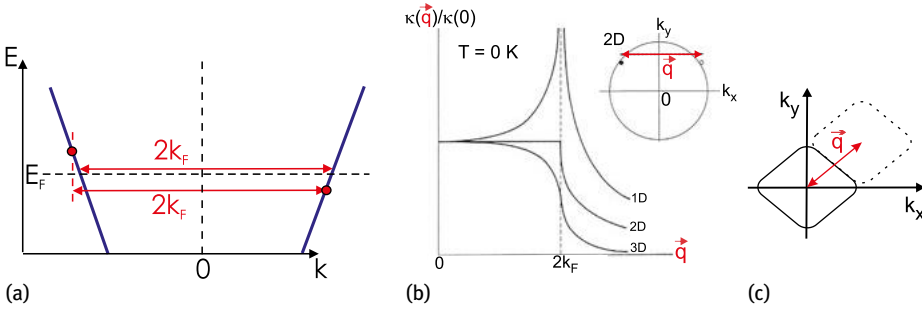


Fig. 5.6: (a) Scattering processes (red double arrows) leading to a charge density wave within a 1D system. The system is displayed by its $E(k)$ dispersion around E_F (blue lines). (b) Polarizability $\kappa(q)$ for a 1D, a 2D and a 3D electron system with parabolic, isotropic dispersion. Inset: Fermi circle for 2D electrons with one indicated phononic scattering vector $\vec{q}$. (c) Illustration of Fermi nesting: Using the indicated phononic wave vector $\vec{q}$, one can combine all states on the Fermi line, that overlap after the Fermi line is additionally shifted by $\vec{q}$ (dashed line) with respect to the original Fermi line (solid line). (b) [105].

imentally observed periodicity $\lambda_F/2$. The charge density wave at this wave length can be supported by processes as marked by the two red dots in Fig. 5.6(a). These processes also lead to a standing wave with periodicity $\lambda_F/2$, but require the energy difference between the two states to be provided, e.g., by annihilating a phonon.

Formally, we will restrict our description of the Peierls transition to 1D. We start with the electrostatic attraction $E_{e-ion} = \delta\rho_{c,q} \cdot \tilde{\Phi}_q$, where $\delta\rho_{c,q}$ and $\tilde{\Phi}_q$ are the amplitude of the charge density wave of the electrons and the amplitude of the electrostatic potential of the frozen phonon, respectively, both at wave vector q . The charge density wave is described as the polarization of the sea of electrons by the potential of the frozen phonons: $\delta\rho_{c,q} = \kappa(q) \cdot \tilde{\Phi}_q$. The missing polarizability function $\kappa(q)$ is now calculated in first order perturbation theory. The corresponding novel many-particle wave function $\tilde{\Psi}_{q,el}$ of the charge density wave with wave vector q reads:

$$\tilde{\Psi}_{q,el} = \Psi_{0,el} + \sum_{\pm k} \frac{\langle \Psi_{k,el}^{\pm} | \hat{H}_{e,ph} | \Psi_{0,el} \rangle}{\epsilon_k - \epsilon_{k \pm q}} \Psi_{k,el}^{\pm}. \quad (5.8)$$

Thereby, $\Psi_{0,el}$ is the ground state many-particle wave function without electron-phonon interaction, $|\Psi_{k,el}^{\pm}\rangle = \hat{c}_{k \pm q}^{\dagger} \hat{c}_k |\Psi_{0,el}\rangle$, and ϵ_k , $\epsilon_{k \pm q}$ the single particle energies of the unperturbed electronic states.⁶ The created charge density wave $\delta\rho_c(x)$ is the difference between the charge densities of the two many-particle wave functions $\tilde{\Psi}_{q,el}$ and $\Psi_{0,el}$. Using $\hat{H}_{e,ph} \propto \sum_q \sqrt{q} \cdot \tilde{\Phi}_q (\hat{b}_{\pm q} + \hat{b}_{\mp q}^{\dagger}) \hat{c}_{k \pm q}^{\dagger} \hat{c}_k$ as for a material interacting via longitudinal acoustic phonons (page 257ff.), we find after replacing the electron

⁶ A state at k is transferred to a state at $k \pm q$.

operators $\hat{c}_k^\dagger$ and $\hat{c}_k$ by Fermi–Dirac distribution functions according to $(1 - f_k), f_k$:⁷

$$\kappa(q) \propto \sqrt{q} \cdot \left[\sum_k \frac{f_k \cdot (1 - f_{k+q})}{\epsilon_k - \epsilon_{k+q}} + \sum_k \frac{f_k(1 - f_{k-q})}{\epsilon_k - \epsilon_{k-q}} \right]. \quad (5.9)$$

Here, f_k is the value of the Fermi–Dirac distribution function for single particle states with wave vector k . In eq. (5.9), we restricted ourselves to phonon creation processes, that do not depend on the Bose–Einstein distribution function of the phonons, i.e., they are possible even at $T = 0$ K. The first (second) term describes processes, that transfer electrons from k to $k + q$ ($k - q$).

Using a parabolic dispersion of the electrons $\epsilon_k = \frac{\hbar^2 k^2}{2m^*}$, we find $\epsilon_k - \epsilon_{k\pm q} = -\frac{\hbar^2}{2m^*}(q^2 \pm 2kq)$. For $T = 0$ K (Fermi function = step function), we can additionally select the k -space intervals where the electron is scattered from an occupied state ($f_k = 1$) to an unoccupied state $f_{k\pm q} = 0$. The initially occupied electron states can be, at most, $\pm q$ away from the closest Fermi wave vector. This leads to:

$$\begin{aligned} \kappa(q) &\propto -\frac{2m^*}{\hbar^2} \sqrt{q} \cdot \left[\sum_{k=k_F-q}^{k_F} \frac{1}{q^2 + 2kq} + \sum_{k=-k_F}^{-k_F+q} \frac{1}{q^2 - 2kq} \right] \\ &\propto -\frac{2m^*}{\hbar^2} \sqrt{q} \cdot \left[\int_{k_F-q}^{k_F} \frac{1}{q^2 + 2kq} dk + \int_{-k_F}^{-k_F+q} \frac{1}{q^2 - 2kq} dk \right] \end{aligned} \quad (5.10)$$

In the second line, we replaced the sum by an integral, exploiting the high density of states in k space for an extended crystal. By substituting $\tilde{k} = -k$ for the second term, one can verify straightforwardly that both terms are identical leading to:

$$\kappa(q) \propto -\frac{4m^*}{\hbar^2} \sqrt{q} \cdot \int_{k_F-q}^{k_F} \frac{1}{q^2 + 2kq} dk. \quad (5.11)$$

In order to simplify the denominator, we substitute $\tilde{k} = q^2 + 2kq$ with $\frac{d\tilde{k}}{dk} = 2q$ and corresponding replacement of the integration limits resulting in:

$$\kappa(q) \propto -\frac{4m^*}{\hbar^2} \frac{1}{2\sqrt{q}} \cdot \int_{q^2+2(k_F-q)q}^{q^2+2k_Fq} \frac{1}{\tilde{k}} d\tilde{k} \quad (5.12)$$

$$= -\frac{2m^*}{\hbar^2} \frac{1}{\sqrt{q}} \cdot \ln \left(\frac{q^2 + 2k_Fq}{2k_Fq - q^2} \right) = -\frac{2m^*}{\hbar^2} \frac{1}{\sqrt{q}} \cdot \ln \left(\frac{q + 2k_F}{2k_F - q} \right). \quad (5.13)$$

⁷ Applying the annihilation operator $\hat{c}_k$ to a many particle state returns a factor, that is the occupation probability of the corresponding single-particle state, i.e., f_k , while applying the creation operator $\hat{c}_k^\dagger$ returns a factor describing the emptiness of the state, i.e., $(1 - f_k)$. This averages over many realizations via the Fermi–Dirac distribution function, hence, it leads a mean-field description of the interaction term $\hat{H}_{e,ph}$.

Obviously, the polarizability diverges towards $-\infty$ for $q = 2 \cdot k_F$, i.e., the energy gain $E_{\text{e-ion}} = \kappa(q)\tilde{\Phi}_q^2$ for a charge density wave coupled to a frozen phonon of wave vector $2k_F$ diverges, too. Consequently, the charge density wave is the ground state at $T = 0$ K, independent of the required finite elastic energy E_{elastic} . Generally, the divergence is caused by the divergence of the perturbational terms for the two contributing electron states at k_F and $-k_F$ having the same energy.⁸ More descriptively, the energy gain arises, because the two states k_F and $-k_F$ can form two linearly independent standing waves with maxima either on the high density area of the frozen phonon or on the low density area of the frozen phonon (Fig. 5.4(a)) without energy penalty. Formation of one of the standing waves gains energy, while the other is losing energy. Since only one of the two waves must be occupied at k_F (at E_F , the Fermi function is $f_k(E_F) = 1/2$), the system gains energy by occupying the energetically favorable standing wave. This process is directly represented by an experimentally observed band gap at E_F .

For $T > 0$ K, the Fermi function $f_k(E)$ is not a step function anymore such that additional processes contribute. States above E_F are partly occupied and states below E_F are partly empty, such that (quasi-)elastic scattering processes are enabled at these energies, too. Importantly, the sign of the perturbational term changes at $\epsilon_k = \epsilon_{k \pm q}$ (eq. (5.8)), such that repulsive and attractive terms contribute at finite T . Hence, these extra processes do not stabilize the charge density wave additionally. Formally, one approximates the electron dispersion by straight lines around E_F , i.e., $\epsilon_k \propto k$, and neglects the influence of the thermally excited phonon bath (phonon annihilation), such that one gets after keeping the Fermi functions of eq. (5.9) and considering the phonon energy $\hbar\omega_{\text{ph}}(q)$ explicitly as the energy difference between initial and final state:⁹

$$\kappa(2k_F) \propto \sqrt{2k_F} \mathcal{D}(E_F) \int_0^{\tilde{x}} \frac{\tanh x}{x} dx \quad (5.14)$$

$$= \sqrt{2k_F} \mathcal{D}(E_F) \ln(2.28 \cdot \tilde{x}) \quad (5.15)$$

with

$$\tilde{x} = \frac{\hbar\omega_{\text{ph}}(2k_F)}{k_B T} \quad (5.16)$$

with $\mathcal{D}(E_F)$ being the electron density of states at E_F , that is independent of energy for the approximated linear, one dimensional $E(k)$ dispersion close to E_F and determines how many states are available for scattering. The energy gain does not diverge anymore. It, moreover, decreases with increasing T . This is due to the increasing contribu-

⁸ Notice that perturbation theory requires that the matrix element of two degenerate states with the perturbing Hamiltonian $\tilde{H}_{\text{e,ph}}$ must be zero, that is always possible, such that a more detailed analysis including the overlap integral in the matrix element is required. However, this does not change the general observation and will not be performed here.

⁹ $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$.

tion of scattering processes, that are not exchanging a momentum of $2\hbar k_F$, such that there is no clear selection anymore for a favorable wave vector of the frozen phonon. Additionally, the changing sign of the interaction term contributes. Hence, $E_{e-ion} = \kappa^2(2k_F)\tilde{\Phi}_{2k_F}$ must be compared with the loss of elastic energy $E_{elastic}$.

Therefore, one approximates the temperature independent, elastic energy by a parabolic function around the equilibrium positions of the nuclei (linear chain with springs between the atoms) (eq. (4.45)). Hence, one takes $E_{elastic} \approx 1/2 \cdot \tilde{D} \cdot u_q^2$ with u_q being the amplitude of the displacement of the nuclei from the equilibrium position and $\tilde{D}$ being an effective spring constant representing the binding forces between the atoms. Moreover, the amplitude of the electrostatic potential $\tilde{\Phi}_q$ is assumed to be linear in the displacement amplitude $\tilde{\Phi}_q = A_{ion} \cdot u_q$ (eq. (4.46)). This is correct for small displacements (Section 4.6.2). The term A_{ion} is proportional to the ionic charge. Setting the temperature dependent Coulomb energy E_{e-ion} equal to the elastic energy $E_{elastic}$, one eventually gets an estimate for the Peierls transition temperature T_C , that reads

$$T_C \approx 2.28 \cdot \hbar \omega_{ph}(2k_F)/k_B \cdot \exp\left(-\frac{\chi \tilde{D}}{A_{ion}^2 \cdot \mathcal{D}(E_F)}\right) \quad (5.17)$$

with χ being a material dependent constant. As expected, a large density of states at E_F ($\mathcal{D}(E_F)$) and a large ionic charge (large A_{ion}) are favorable for the Peierls transition, while a rigid lattice (large $\tilde{D}$) is not favorable. Moreover, a large phonon energy helps, since this allows that the electron states can couple across a larger energy region by $2k_F$ scattering such that more electrons can gain from the charge density wave.

The charge density wave, in turn, is attractive for the created frozen phonon, which explains the energy gain of the phonon, as visible in Fig. 5.5(a). Moreover, the frozen phonon leads to a novel unit cell of the atomic lattice with reciprocal unit vector $2k_F$. This induces a band gap at the novel Brillouin zone boundary k_F , as usual for Brillouin zone boundaries. The electronic states close to k_F that are in phase (or 180° shifted) with respect to the frozen phonon periodicity will gain (lose) energy. This explains the band gap of Fig. 5.5(b) from a slightly different perspective and accordingly the corresponding $\sigma(T)$ in Fig. 5.5(c).

For electron systems in higher dimensions, the polarizability $\kappa(q)$ at $T = 0$ K does not diverge at $2k_F$. The result for electron systems with isotropic and parabolic dispersion, which leads to a Fermi circle (Fermi sphere) in 2D (3D), is shown in Fig. 5.6(b). The constant behavior of $\kappa(q)$ in 2D is clarified by the inset. Each phonon vector $q < 2k_F$ can couple exactly two possible pairs of electronic states such that the corresponding density of states $\mathcal{D}(E_F, \vec{k})$ is infinitesimally small, being zero in the thermodynamic limit. Consequently, the transition temperature is $T = 0$ K. However, if one increases the corresponding (joint) density of states belonging to a particular $\vec{q}$, a Peierls transition can also appear in higher dimensions. Figure 5.6(c) illustrates a 2D-Fermi line, where complete sections of the Fermi line can be coupled by a single phonon vector. The coupled density of states gets larger than zero and a Peierls transition takes place at finite T . For the system shown in Fig. 5.5(d), the Fermi line exhibits a hexagon

in the Brillouin zone center, such that coupling vectors such as the one displayed as $\vec{Q}_{\text{FS}}$ can lead to three charge density waves in three different directions. This qualitatively explains the charge density wave pattern of Fig. 5.4(b), albeit the details are more complex and are still not settled completely (e.g. [107]). In particular, these details decisively include the wave vector dependence of $M_{\vec{q}+\vec{G},\vec{k}}^{j,n,n'}$ in real systems.

In closing this section, we briefly discuss two final aspects:

1. Since the electron-electron repulsion for extended states is usually weaker for parallel than for antiparallel spins due to the exchange term (Section 4.4.1), the charge density wave is often accompanied by a spin density wave, i.e., the maxima of the charge density wave are alternately filled preferentially with electrons of spin $\uparrow$ and electrons of spin $\downarrow$. This results in a spin density wave with twice the wave length of the charge density wave, respectively, with wave vector k_F . Mostly, the spin density wave appears at lower temperature than the charge density wave, such that one observes two phase transitions.
2. Using the arguments above, the existence of the charge density wave can be explained, but not its relative phase with respect to the atomic lattice. If the wave length is $\lambda_F/2 = n/m \cdot a$ with n, m being integer numbers ($n > m$) and a the lattice vector, this is called a commensurable charge density wave. Then, it is obvious, that the maximum of the charge density wave is at an ionic center where it can profit most strongly from the highest positive charge density of the ions. If, in contrast, the ratio of $\lambda_F/2$ and a is not a rational number, then there is no optimal position. The energy of the system is independent of the relative phase between the atomic lattice and the charge density wave. Such a charge density wave necessarily includes regions, where the charge density maximum sits above an ion core and regions, where the charge density maximum is located exactly between two ion cores. This situation does not change at all, if one changes the position of the charge density wave with respect to the atomic lattice. Consequently, the charge density wave can be moved through the crystal without any energy penalty. This idea has originally been pursued as an explanation for the disappearing electrical resistance of superconductors, but turned out to be wrong. In reality, each deviation from the perfect lattice (atomic defects) destroys the equivalence of all relative phases, since the defect energetically prefers either a minimum or a maximum of the charge density wave. This is called pinning of the charge density wave at the defect. It leads to the insulating behavior as shown in Fig. 5.5(c).

5.3 Consequences of potential disorder

In this section, we will discuss consequences of disorder within the atomic lattice. The most intriguing result in this respect is the quantum Hall effect. Due to the presence of the disorder and independent of the details of the disorder, one gets an extremely precise measurement of natural constants. The relation h/e^2 (e : electron charge,

h : Planck's constant) can be determined with an error of less than 10^{-10} by probing the Hall resistance of an imperfect 2D crystal. In fact, this effect defines the standard unit Ω for the resistance and defines together with the Josephson effect the SI-unit Ampère. A novel definition of the mass (kg) based on natural constants has been launched in 2018. It replaces the primary kilogram in Paris. Probably, the quantum Hall effect will play an important role in the resulting measurement protocols.

Prior to discussing the quantum Hall effect in Section 5.3.2, we will shortly introduce disorder induced localization as already discussed in Section 1.6 with respect to the resulting transport properties. Here, we will focus more on the resulting wave function properties.

5.3.1 Metal-insulator transition

One can not avoid deviations from the perfect arrangement of atoms in an ideal crystal, even if the production processes are continuously improved in order to decrease these deviations. The central reason is the relation between atomic mobility and entropy. Mostly, crystals are grown by slow cooling from the melt, i.e., the crystal gets enough time to realize the configuration of minimum free energy. But since diffusion processes depend exponentially on temperature, the waiting time gets too large at a certain minimum temperature, such that the atomic arrangement is basically frozen at that temperature during cooling. A finite temperature, however, increases the entropic contribution to the free energy, such that deviations from the ideal arrangement appear in equilibrium. At lower temperature, the minimum of free energy exhibits less deviations, but the mobility of the atoms is already too low to transfer the crystal from the current local minimum of internal energy into the global one. An ideal crystal, thus, requires either no energy barriers between local and global minima or a way to construct the crystal artificially atom by atom. The latter is possible, e.g., with a scanning tunneling microscope (STM), but only for a few hundred atoms in reasonable time. Partly, the disorder is even constitutional for the functionality of the material, e.g., if one dopes a semiconductor by foreign atoms, i.e., by donors or acceptors.

In 3D, there are basically two types of phase transitions driven by disorder. Metals become insulators (Section 1.6 and Fig. 4.27) and insulators become metallic. Recall that an insulator is a material with conductivity $\sigma = 0$ S/m at $T = 0$ K and a metal is a material with $\sigma > 0$ S/m at $T = 0$ K. However, this is practically useless, since one can not measure at $T = 0$ K. In practice, one measures the temperature dependence of the conductivity or resistivity ρ . If ρ decreases with decreasing temperature, the material is called metallic, following the idea that the density of charge carriers is constant, but the scattering probability decreases with decreasing temperature, e.g., due to the decreasing number of phonons or due to the decreasing phase space for electron-electron scattering. If ρ is increasing with decreasing temperature, the material is called an insulator (or semiconductor), for which the charge carrier density gets significantly

reduced with decreasing temperature (page 242, Section 1.6). This practical definition is not unambiguous, since some materials exhibit non-monotonic $\rho(T)$ dependence (Fig. 4.27), that we ignore in the following for the sake of simplicity.

Figure 5.7(a) shows a successful experiment for an alloy. The resistance $R(T)$ is recorded repeatedly, while the crystal is bombarded by α particles. This increases the disorder within the crystal continuously and induces a change in the slope of $R(T)$, that initially decreases and finally increases with decreasing temperature. The metal-insulator transition (MIT) is defined at the point of disorder, where the sign of the slope of $R(T)$ changes.

As already discussed in Section 1.6, the reason for the insulating properties at higher disorder is not the formation of a band gap at E_F , as e.g., during a Mott–Hubbard transition (Fig. 4.14), but the creation of localized states. The probability density of these states is concentrated in a certain area. Consequently, they barely contribute to the electrical transport between two macroscopically separated electrodes and its contribution to the transport decreases with decreasing T (Section 1.6.3). An example is shown in Fig. 5.7(b)–(e). For a regular electrostatic potential, as shown in

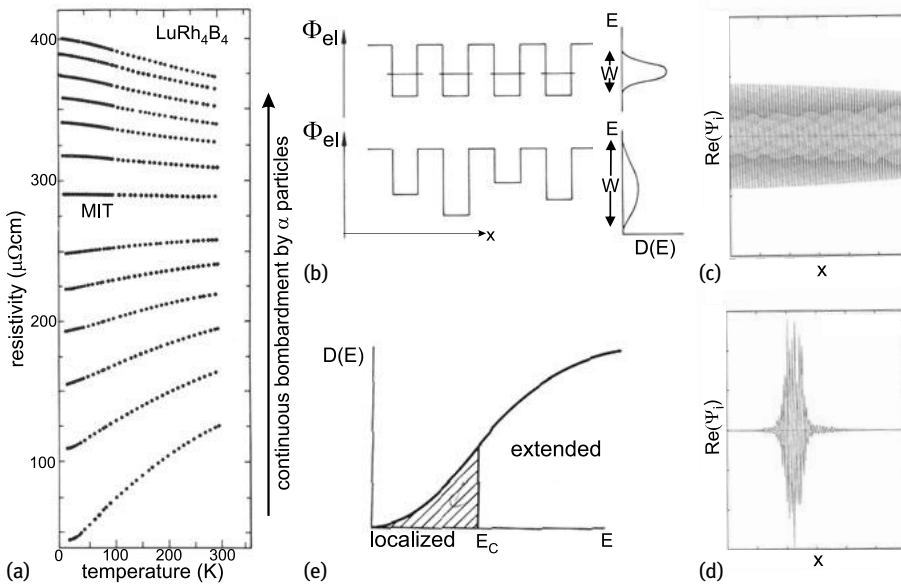


Fig. 5.7: (a) Resistivity as a function of temperature for LuRh_4B_4 . The data are recorded continuously from the bottom to the top while the crystal is bombarded by α particles. (b) Schematic of the electrostatic potentials $\Phi_{\text{el}}(x)$ of a 1D crystal without (top) and with (bottom) potential disorder. The corresponding density of states $\mathcal{D}(E)$ is plotted on the right. (c) Real part of $\psi_i(x)$ for an extended state. (d) Real part of $\psi_i(x)$ for a localized state. (e) Density of states $\mathcal{D}(E)$ of a disordered crystal with localized states (hatched area). E_C is called the critical energy or the mobility edge of the metal-insulator transition (similar to Fig. 1.28). (a) R. C. Dynes et al. in [108].

the top part of Fig. 5.7(b), one gets Bloch states and a band width W , that depends on the strength of the hopping parameter t (eq. (4.33)). Varying the atomic potentials, as in the bottom part of Fig. 5.7(b), leads to areas that are on average more attractive to the electrons than others. Localizing the probability amplitude $\psi_i(\vec{x})$ of a single particle state in these areas costs kinetic energy (stronger curvature of $\psi_i(\vec{x})$), but this can be compensated by the gain in electrostatic energy. Thus, one gets some states, that are localized in these areas (example in Fig. 5.7(d)). Their probability amplitude decreases roughly exponentially with distance from the center of mass (eq. (1.98)). However, other states are still extended similar to Bloch states. They only exhibit a slightly varying amplitude of $\psi_i(\vec{x})$ with position (Fig. 5.7(c)). Recall that different (stationary) states have to be orthogonal. This avoids states to be located in the same area.¹⁰ It turns out that the localized states have energies at the rims of the bands as one would expect, since these states mostly explore the extrema of the potential landscape. Hence, there is a transition between localized states at lowest energies in the band and extended states towards the energetic center of the band as displayed in Fig. 5.7(e). These two types of states are separated by the critical energy E_C , also called the mobility edge in Section 1.6. Moreover, the bands get broader due to these localized states as shown on the right of Fig. 5.7(b). Localized states have indeed been observed in experiments. A very direct experimental observation, where a light wave instead of an electron wave is localized within a disorder potential, is shown in Fig. 5.8(b).

In order to calculate states within a disorder potential, one typically uses tight-binding models (Section 4.5) reading:

$$\hat{H} = \sum_{n,m} t_{n,m} \hat{c}_n^\dagger \hat{c}_m + \sum_n \Phi_{n,\text{el}} \hat{c}_n^\dagger \hat{c}_n \quad (5.18)$$

with n, m counting the lattice positions and hopping amplitudes $t_{n,m}$, that are mostly taken to be non-zero only for neighboring atoms. $\Phi_{n,\text{el}}$ are the spatially varying electrostatic potentials of the individual atoms, that in the most simple calculations are distributed randomly (Fig. 5.7(b)). Notice that the potential disorder term is distinct from the Hubbard U term in eq. (4.26). The former operates with a single occupation operator $\hat{n}_n = \hat{c}_n^\dagger \hat{c}_n$, while the later uses two occupation operators $\hat{n}_n \hat{n}_n$, hence counting double occupation of a site n .

The probability amplitudes of the different single particle states within the disorder potential are then determined by matrix diagonalization, where Wannier states localized at the atoms (Section 4.5) are used as a starting base for the diagonalization. The energy interval, where one finds localized states,¹¹ increases with increasing disorder. This is displayed in Fig. 5.8(a). The physical reason for the increasing

10 Additionally, the electron-electron repulsion naturally avoids that all states are localized in the areas of strongest attractive ionic potential.

11 A state is called localized, if $|\psi_i(\vec{x})|^2$ decreases with the distance from the center of mass $\vec{x}_0$ of the probability density of the state. Mostly, one can fit an exponential function $\overline{\psi_i}(|\vec{x} - \vec{x}_0|) \propto$

energy interval of localization is, that one can curve the probability density increasingly strongly, since the increasing gain in potential energy at larger disorder increasingly compensates for the loss in kinetic energy by the wave function curvature. The stronger curvature implies a better localization of states on small areas such that more localized states, orthogonal to each other, become possible. The existence of localized and extended states does not depend on the details of the chosen disorder. Within the calculation of Fig. 5.8(a), a simple cubic atomic lattice is used with a hopping rate between neighbors of $t = 1/6$. The disorder is the randomly chosen potential $\Phi_{n,\text{el}} \in [-\bar{W}/2, +\bar{W}/2]$. For disorder strength $\bar{W} < 2.6$, both, localized and extended states exist. Hence, it depends on E_F , if the system is an insulator (localized states at E_F) or a metal (extended states at E_F). This implies a metal-insulator transition with E_F as the control parameter. Alternatively, the disorder strength $\bar{W}$ can be used as the control parameter as in the experiment of Fig. 5.7(a). Above $\bar{W} = 2.6$, the electrons at E_F are always localized independent of the position of E_F .

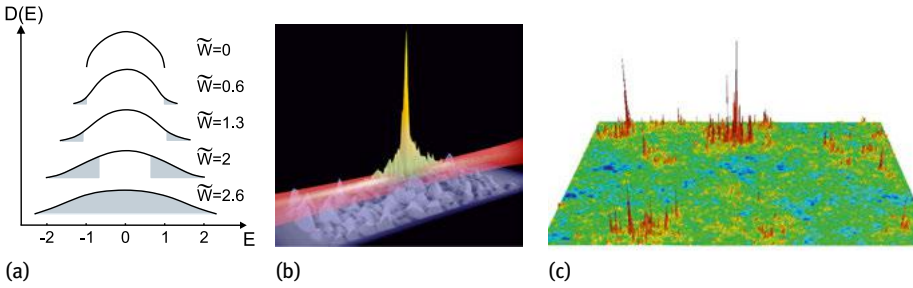


Fig. 5.8: (a) Calculated density of states $\mathcal{D}(E)$ of a cubic lattice for increasing disorder $\bar{W}$ of the electrostatic potential of the atoms as marked ($\Phi_{n,\text{el}} \in [-\bar{W}/2, +\bar{W}/2]$). Energy intervals of localized states are filled in gray. (b) Experimentally determined $|\psi_i(\vec{x})|^2$ of a localized state (yellow-green). The disorder potential $\Phi_{\text{el}}(\vec{x})$ is shown below in violet. The experiment is performed with light waves instead of electron waves, which facilitates the imaging. (c) Calculated $|\psi_i(\vec{x})|^2$ of a critical state at the transition energy from localized to extended states. (b) [109], (c) courtesy of A. Mildenberger, see also [110].

Note that the order parameter of these phase transitions is not temperature, but electron density or disorder. As other phase transitions, it exhibits universal properties, that do not depend on the details of the system. An important example is the exponent of the localization length as a function of energetic distance from the mobility edge E_C . Close to E_C , one finds: $\xi(E) \sim |E - E_C|^{\nu_C}$ with ν_C being the critical exponent, that only depends on the general symmetries of the Hamiltonian, such as time-reversal symmetry, charge conjugation symmetry, chiral symmetry, and the spatial dimension d

$\exp(|\vec{x} - \vec{x}_0|/\xi)$ (eq. (1.98)) to the state at large $|\vec{x}|$. ξ is called the localization length, that diverges at the transition between localized and extended states.

of the electronic system. The study of such universalities of so-called quantum phase transitions is a scientific subject in itself, which is beyond the scope of this book. An interesting aspect are the fractal properties of the critical state at E_C . This state is neither a 0D state, i.e., a localized state existing only in a finite area, nor a 3D state, i.e., a state that is extended across the whole crystal, but a filamentary state with a non-integer dimension in between. This dimension can be calculated by the so called box counting method (see, e.g., Wikipedia) applied to the probability density of an electron $|\psi_i(\vec{x})|^2$. A calculated $|\psi_i(\vec{x})|^2$ of a critical state at E_C is shown in Fig. 5.8(c). It exhibits so-called multi-fractal properties, i.e., the fractal dimension (calculated by the box counting method) differs for different momenta of the probability distribution, i.e., the dimension of $|\psi_i(\vec{x})|^{2n}$ depends on the chosen $n \in \mathbb{N}$. The distribution of the fractal dimensions for different n , however, is universal, i.e., it depends only on the general symmetries of the Hamiltonian and the details are not important.

The opposite behavior to metals, where disorder drives the system insulating, is observed for semiconductors and insulators, where foreign atoms can drive the system metallic. As discussed in more basic textbooks of condensed matter physics, a doping atom can be approximated as a Coulomb potential $\Phi_{el}(\vec{x}) \propto 1/|\vec{x}|$ for one electron (or one hole), analogously to the hydrogen atom, but with reduced binding energy $E_{\text{Bind}} = m^*/m_e \varepsilon^2 \cdot 13.6 \text{ eV}$ (m^* : effective mass, ε : dielectric constant) (see also Section 2.3.3).¹² At $T = 0 \text{ K}$ and low donor density, the electrons are bound to the donor and only get mobile after thermal excitation into the conduction band. Thus, according to our definition, the material is an insulator. However, if the density of donors (or acceptors) is increased, neighboring Coulomb potentials overlap as shown in Fig. 5.9(a). The resulting bound states appear at different energies, i.e., we get a so-called impurity band of localized states, where the probability amplitudes of the states start to get distributed across several doping atoms. However, the electrons are still localized with their center of mass at the corresponding donor. This tendency is supported by the electron-electron repulsion, that avoids having several electrons at the same donor. The situation is similar to the Mott–Hubbard transition of a half filled band (Section 4.3 and 4.5). Within the localized regime, the electrical transport is governed by thermally induced hopping of the electrons between localized states as indicated by the red arrow in Fig. 5.9(a) (Section 1.6.3). The transport described by matrix elements as $\langle \psi_i | \hat{H}_{e,ph} | \psi_j \rangle$, thus, depends strongly on temperature. One finds, e.g., $\sigma(T) \propto \exp(-\beta/\sqrt[4]{T})$ within the model of variable range hopping as outlined in Section 1.6.3.2 (eq. (1.110)).

Increasing the donor density further, naturally leads to a decreasing potential maximum around the donors as shown in Fig. 5.9(b), such that the majority of the states belonging to the donors are not bound anymore. Alternatively, one can employ the knowledge from Section 4.5. One can argue in analogy that the gain via hopping t

¹² This applies to a parabolic band, where m^* is well defined.

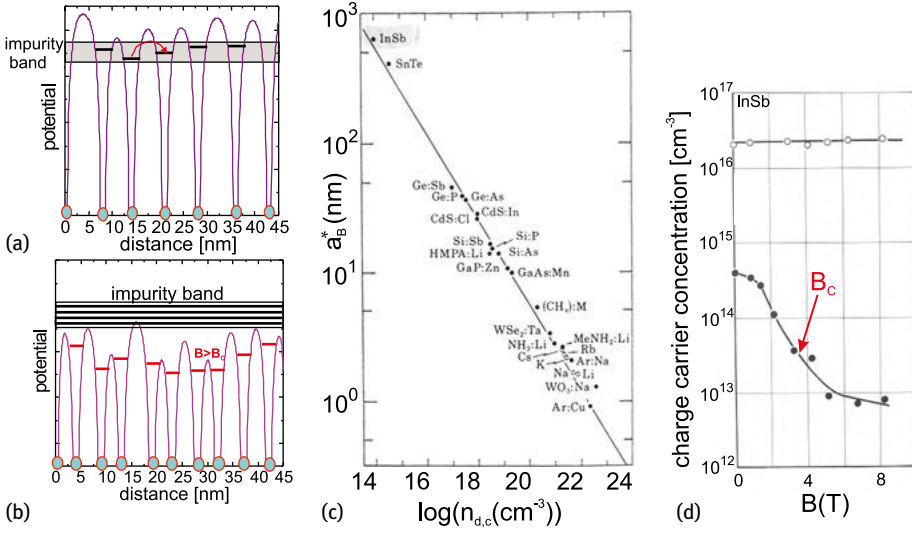


Fig. 5.9: (a) Disorder potential resulting from a row of donors (Coulomb potentials) with varying distances. Donors are marked by turquoise circles at the bottom. The thicker black, horizontal lines mark the energies of bound electron states. The greyish rectangle around these states marks the impurity band. A hopping process is indicated by a red arrow. (b) Same as (a) at larger donor density. Extended states are marked by continuous black lines. Above a critical magnetic field B_c , the states get localized as marked by red, horizontal lines. (c) Double logarithmic plot of the critical donor density $n_{d,c}$ as a function of the Bohr radius of the donor atoms a_B^* . The materials are labeled as crystal:donor or only as crystal, if little dependence on the dopant material is found. The line displays the relation $a_B^* = 0.27 \cdot n_{d,c}^{-1/3}$. (d) $\vec{B}$ field dependence of effective charge carrier density of InSb as determined by the Hall effect at two different donor concentrations (full circles: $n_d = 4 \cdot 10^{14}/\text{cm}^{-3}$, empty circles: $n_d = 2 \cdot 10^{16}/\text{cm}^{-3}$). The critical $\vec{B}$ field B_c is marked. (c) [111].

(eq. (4.28)) gets larger than the energy penalty via on-site interaction U (eq. (4.27)), while replacing the atomic potentials from Section 4.5 by the randomly distributed donor potentials. Since the bandwidth W increases with t , the impurity band additionally starts to overlap with the original conduction band. Consequently, we get occupied, extended states with E_F within the overlapping bands as required for a metal. The transition, where E_F passes from localized to extended states, such that the system turns from an insulator into a metal, is also called Mott transition in analogy to Section 4.3. The order parameter is the dopant density and the transition appears at the critical dopant density $n_{d,c}$. The value of $n_{d,c}$ turns out to be related to the Bohr radius of the donor $a_B^* = \epsilon \cdot m_e/m^* \cdot a_B$ ($a_B = 0.053 \text{ nm}$: Bohr radius of H atom):

$$a_B^* \cdot n_{d,c}^{1/3} = 0.27. \quad (5.19)$$

This so-called Mott criterion is excellently fulfilled for many dopants in different materials as shown in Fig. 5.9(c). Although a bit oversimplified, one can argue that the

states get extended if the Bohr radius of neighboring, bound electrons overlap.¹³ Thus, also doped semiconductors exhibit a disorder driven phase transition, now from the insulator to the metal, with the dopant density n_d as the control parameter.

At donor (or acceptor) concentrations slightly above $n_{d,c}$, one can drive the metal back into an insulator by applying a magnetic field $\vec{B}$.¹⁴ The central reason is, that the probability amplitude of the bound electrons gets laterally squeezed by the magnetic field in all directions perpendicular to the field, such that the effective hopping amplitude t gets smaller (eq. (4.28)). The squeezing can be understood semiclassically. While the Bohr radius corresponds to the classical radius of the electron paths encircling the positive ionic core, electrons in $\vec{B}$ field get additionally moved in circles with a radius called the cyclotron radius $r_c = mv/eB$ (v : electron velocity, $B = |\vec{B}|$). Quantization of this radius results in the so called Landau radius (also called magnetic length) $l_B = \sqrt{\hbar/(eB)} = 29 \text{ nm} \cdot B[T]^{-1/2}$ of the circular movement in the ground state, that decreases with increasing $\vec{B}$ independent of v . As soon as $l_B < a_B^*$, the wave function decreases in lateral extension such that the Coulomb potential of the donor can bind the electron despite the reduced potential maximum around the donor. These magnetically confined states are marked as red lines in Fig. 5.9(b). Since the resulting reduction of free charge carriers additionally increases the screening length λ_{screen} and, thereby, the Coulomb potential strength $\Phi_{\text{Coul}}(\vec{x}) \propto 1/|\vec{x}| \cdot \exp(-|\vec{x}|/\lambda_{\text{screen}})$, respectively, the maxima around each donor, one observes a relatively sharp transition at the critical field B_C . Such a transition is called magnetic freeze-out¹⁵ and can be probed experimentally, if a_B^* is large as, e.g., in InSb (Fig. 5.9(d)). The remaining carrier density above B_C originates from thermally excited electrons, which thus disappear at $T \rightarrow 0 \text{ K}$. Consequently, the $\vec{B}$ field can also be a control parameter for a metal insulator transition (see also Section 1.5.1.2). Notice that $\vec{B}$ is an elegant parameter to tune t , which can drive the Mott–Hubbard transition (Section 4.5). However, the disorder in the system and the fact that the orbital states in a donor potential are only separated by a few meV, such that multiple bands contribute, makes the interpretation of the data more complex.

5.3.2 Quantum Hall effect

In 2D, there are many more metal-insulator transitions driven by a $\vec{B}$ field, where each insulating phase is accompanied by a quantized value of the Hall resistance R_{Hall} .

¹³ A more detailed calculation has to consider the effect of electron-electron repulsion as well, similar to Section 4.5.

¹⁴ Formally, $\vec{B}$ is the magnetic induction.

¹⁵ The magnetic-freeze-out effect is also relevant for stars and the creation of the universe, since $\vec{B}$ obviously influences the transition from a plasma made of electrons and nuclei to atoms, but at much larger $\vec{B}$, since for atoms we have $a_B^* \approx 0.05 \text{ nm}$.

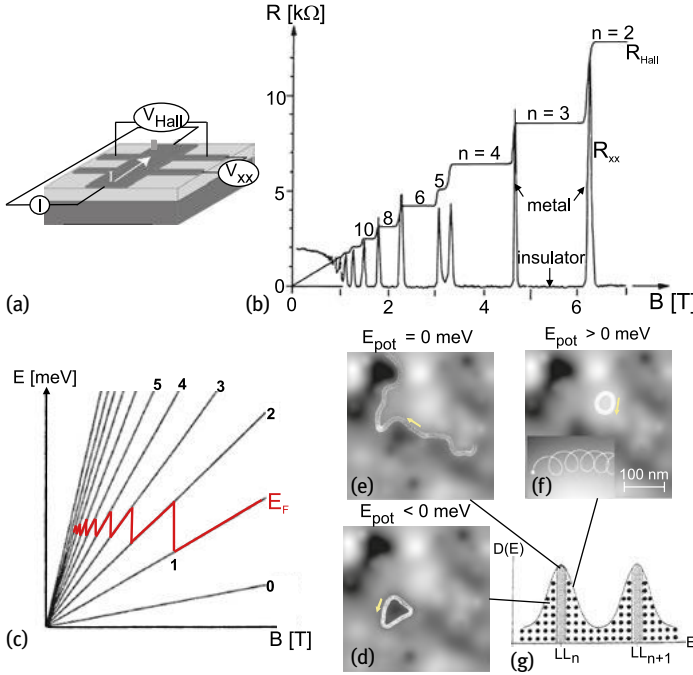


Fig. 5.10: (a) Sketch of a setup to measure the quantum Hall effect. (b) Longitudinal resistance $R_{xx} = V_{xx}/I$ and Hall resistance $R_{Hall} = V_{Hall}/I$ of a 2D electron system as a function of $\vec{B}$ field. (c) Landau levels LL_n as a function of $\vec{B}$ field with Fermi energy E_F marked in red. (d)–(f) Potential disorder (grey scale image) and classically resulting electron paths (white lines) within a magnetic field of $B = 6$ T at different potential energies E_{pot} as marked. The Inset in (e) shows a magnification of the cycloid path of an electron. (g) Density of states $\mathcal{D}(E)$ of a 2D electron system in $\vec{B}$ field with marked Landau levels LL_n , LL_{n+1} and assigned states from (d)–(f). (b) after https://www.nobelprize.org/nobel_prizes/physics/laureates/1998/press.html.

R_{Hall} is the ratio of the voltage V_{Hall} measured perpendicular to the direction of current I and the current I . The quantized Hall resistance $R_{Hall} = V_{Hall}/I$ appears to be $R_{Hall} = h/e^2 \cdot 1/n = 25.8 \text{ k}\Omega/n$, with I being the current through the 2D system and n being a positive integer. Figure 5.10(b) shows the measurement for a 2D system of GaAs using the setup in Fig. 5.10(a). The areas with $R_{xx} \approx 0 \Omega$ are the insulating regions (explanation below), that exhibit plateaus of $R_{Hall}(B)$ with the values $h/e^2 n$, where n gets smaller with increasing B .

In order to understand the quantum Hall effect (QHE), we firstly consider the behavior of 2D electronic states in the $\vec{B}$ field. The electrons move in circles, if a $\vec{B}$ field perpendicular to the 2D sample is applied. This leads to the well known quantization of the kinetic energy (rotational energy) described by the Landau energies E_n^{LL} . Formally, it is a quantization of the orbital momentum, which for a parabolic dispersion $E = \hbar^2 k^2 / 2m^*$ leads to $E_n^{LL} = (n + 1/2) \cdot \hbar e B / m^*$. In 2D, we get a complete quanti-

zation, if the $\vec{B}$ field is not parallel to the 2D plane. The corresponding energy levels, equidistant by $\hbar\omega_c = \hbar e B_\perp / m^*$ ($B_\perp$: $\vec{B}$ field perpendicular to 2D electron system), are called Landau levels and are highly degenerate. Each Landau level can be occupied by one electron per area A of a flux quantum $\Phi_0 = B_\perp \cdot A = h/2e$.¹⁶ This leads to a degeneracy density (number of states per m^2) of $n_{\text{LL}} = 1/A = 2 \cdot e B_\perp / h$. The degeneracy of a Landau level, thus, increases with $B_\perp$ such that, at constant electron density n_e , the number of occupied Landau levels decreases with increasing $B_\perp$. The Fermi level E_F resides within the m th Landau level with $m = \lfloor n_e / n_{\text{LL}} \rfloor$ ¹⁷ and jumps to a lower Landau level, if the summed degeneracy density of all Landau levels at lower energy is larger than n_e . The movement of E_F within the Landau levels at constant n_e is depicted as a red line in Fig. 5.10(c). We will see that the jumps between the Landau levels are important for the QHE, but that we will need the additional ingredient of disorder to smooth these jumps such that the QHE becomes a robust feature. The electrostatic potential of the atoms is mostly irrelevant as long as the magnetic length l_B is significantly larger than the lattice constant, i.e., up to $B_\perp \approx 1000$ T. In the following, we will restrict ourselves to magnetic fields perpendicular to the 2D electron system, such that $B = B_\perp$.

The required potential disorder appears naturally because of the charged dopants within doped semiconductors, where the quantum Hall effect is usually observed. This potential disorder is typically long-range (correlation length: ~ 0.1 – 1 μm) as shown as a greyscale pattern in Fig. 5.10(d)–(f), in particular, if the dopants are placed remotely (typically by about 100 nm) from the 2D electron system (2DES). Hence the correlation length is large with respect to l_B , respectively r_c . The originally degenerate states of one Landau level are then simply distributed at different energies. This can be qualitatively understood by a classical consideration. If the cyclotron radius is much smaller than the correlation length of the potential disorder, the electron will be alternately accelerated and decelerated during its circular path by the in-plane electric field corresponding to the potential disorder. Consequently, the radius of curvature $r_c \propto v$ changes continuously. This results in a cycloid path as shown in the inset of Fig. 5.10(f). Importantly, the cycloid path moves the electron perpendicular to the direction of the electric field, i.e., along equipotential lines of the disorder potential. In fact, also quantum mechanically, one finds single particle states with $|\psi_i(\vec{x})|^2$ located along an equipotential line of the potential disorder. Perpendicular to the equipotential line, $|\psi_i(\vec{x})|^2$ exhibits a full width at half maximum of about $\sqrt{2n+1} \cdot l_B$ and n nodes for Landau level n . In particular, the potential energy of the equipotential line,

16 Recall that the flux quantum is defined for Cooper pairs leading to the factor of two in the denominator of the flux quantum. The corresponding phase preserving area encircled by a single electron contains a flux of $\Phi = h/e = 2\Phi_0$. This area can be occupied by two electrons, one electron for each spin direction. Notice that $\Phi = h/e$ is the smallest flux unit that can be introduced into a system of coherently moving electrons.

17 $\lfloor x \rfloor$ is the largest integer that is smaller than $x \in \mathbb{R}$.

belonging to a particular $|\psi_i|^2$, adds to the kinetic energy of the electron given by the Landau level energy E_n^{LL} . Thus, states with small (large) potential energy occupy the area along equipotential lines around valleys (hills) of the potential disorder. Such states are sketched in Fig. 5.10(d) and (f). They are localized since the equipotential lines are closed contours. However, in the energetic center of the potential landscape, there is one state (equipotential line) that traverses the whole crystal, i.e., it is extended (Fig. 5.10(e)). In fact, one can show rigorously, that an infinitely large 2D system with disorder has exactly one extended state per Landau level. Figure 5.11(a)–(b) maps the $|\psi_i(\vec{x})|^2$ of localized (a) and extended (b) states by scanning tunneling spectroscopy.

Within Fig. 5.10(g), the density of states $\mathcal{D}(E)$ is plotted where states of one Landau energy correspond to one peak, that, however is broadened by the different potential energies of the states, such that the peak widths reflects the strength of the potential disorder.

With increasing $\vec{B}$, E_F shifts within a particular Landau level from high to low energies, if $n_e = \text{const.}$, since n_{LL} increases for each Landau level. Consequently, E_F always coincides with localized states, i.e., the 2D system is an insulator ($\sigma(T = 0 \text{ K}) = 0 \text{ S}$), except if E_F passes the extended state of the particular Landau level and, thus, the 2D system gets metallic.

Notice that conductivity σ and resistivity ρ are (2×2) tensors in a $\vec{B}$ field, since the current in one direction implies a voltage in two perpendicular directions. With $\vec{j} = \sigma \cdot \vec{E}$ and $\rho = \sigma^{-1}$, one straightforwardly finds $\sigma_{xx} = 0 \text{ S} \iff \rho_{xx} = \frac{\sigma_{xx}}{\sigma_{xx}^2 + \sigma_{xy}^2} = 0 \Omega$, as long as $\sigma_{xy} \neq 0 \text{ S}$, i.e., counter-intuitively insulators exhibit zero resistance. Hence, the B -field regions with $R_{xx} \approx 0 \Omega$ in Fig. 5.10(b), indeed, can be explained by E_F being in the area of localized states, where conductivity is suppressed. In contrast, if E_F is at the energy of an extended state, a peak in $R_{xx}(B)$ is observed.

The remaining conductivity at $T > 0 \text{ K}$, if E_F is at localized states, depends on the energy distance of E_F to the extended state. Since this distance gets decreasingly smaller with decreasing potential disorder, the insulating behavior gets more pronounced with increasing disorder, as long as the scattering length is significantly larger than l_B such that Landau quantization appears.

Quantitatively, the distance between Landau levels is relatively small, i.e., $\hbar\omega_c = 0.18 \text{ meV/T} \cdot B m_e / m^*$, such that the disorder has to be firstly reduced in order to separate the Landau levels and, moreover, systems with small m^* , as e.g., GaAs ($m^* = 0.067 \cdot m_e$) are preferred. In metals, so far, the quantum Hall effect has not been observed, since the life time of electrons in metals is typically shorter than the time required for a full cyclotron orbit. Hence, there is no quantization of the circular orbit and, thus, no Landau levels. Here, the dynamic disorder prohibits the QHE.

An additional complication is the spin degree of freedom of the electrons, whose degeneracy is lifted in $\vec{B}$ field, such that one gets two different spin energies for the same Landau level (Zeeman effect). The energy distance is the Zeeman energy $E_Z = g\mu_B B$ with g being the effective gyromagnetic factor of the electrons in the material

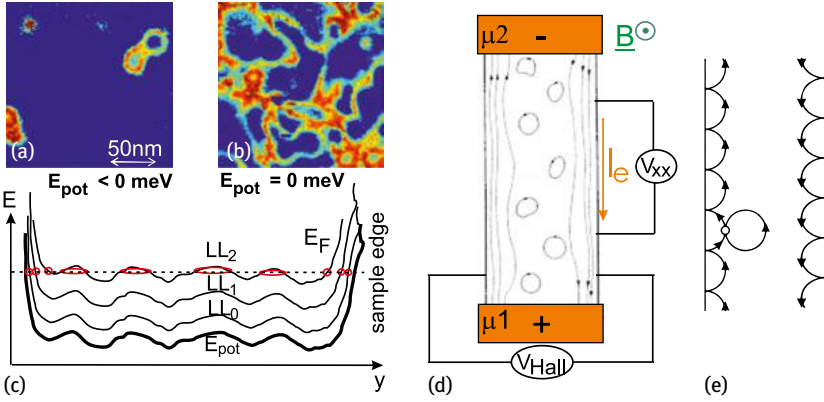


Fig. 5.11: (a) Localized states and (b) extended states of the lowest Landau level LL_0 as probed by scanning tunneling spectroscopy at $B = 12$ T and $T = 0.3$ K. (c) Thick, curved line: electrostatic potential E_{pot} for a sample with edges and disorder. Thinner curved lines: potential energy is shifted by the kinetic energy of the Landau level LL_n . The resulting states at E_F belonging to the third Landau level LL_2 are shown as red ellipses. The red circles at the edges are edge states. (d) Top view of the drifting electron paths for states at E_F (lines with arrows). Electrodes are drawn as orange rectangles. The Hall voltage V_{Hall} , the longitudinal voltage V_{xx} , the potential of the two electrodes μ_1 , μ_2 ($\mu_1 > \mu_2$), and the electron current $I_e = -I$ are indicated. (e) Schematic showing that scattering can not change the direction of propagation of the electron (thick lines with arrows) at the edge as long as the distance between the scattering centers (small circle) is larger than the cyclotron radius. (a), (b) [112].

and the Bohr magneton $\mu_B = 9.27 \cdot 10^{-24}$ J/T. Each spin level exhibits a degeneracy density $n_{\text{LL}}/2 = eB/h$ including one extended state. The pairs of metallic states are indeed visible in Fig. 5.10(b) left and right of the Hall Plateaus, which are marked by $n = 5$ and $n = 3$.

The quantized Hall resistance (plateaus) within the insulating regimes is explained most straightforwardly by the existence of edge states. Figure 5.11(c) displays the electrostatic potential $E_{\text{pot}}(y)$ perpendicular to the direction of current flow with disorder and the edges of the sample (thick line). The disorder potential is additionally displayed after adding the kinetic energy of the Landau levels, hence, $E_{\text{pot}}(y) + (n+1/2)\hbar\omega_c$ as marked by LL_n (thin lines). The cycloid paths at E_F are running along equipotential lines of Landau level LL_2 , as shown by red lines in Fig. 5.11(c). They are localized at hills of the potential disorder and, thus, the 2D system is insulating. Towards the edge of the sample, $E_{\text{pot}}(y)$ must increase such that it matches the vacuum energy (or another type of confinement energy) somewhere at the edge. The region between the edge and interior of the sample is thereby as large as about the screening length of the material being 100–1000 nm for semiconducting 2D systems in $\vec{B}$ field. Within this region, all energy lines of Landau levels, that are in the interior of the sample below E_F , must cross E_F . The corresponding states at these crossing points (marked by red circles in Fig. 5.11(c)) perform its cycloid path along

the equipotential line at the edge. Thus, the corresponding state called an edge state surrounds the whole 2D sample in one preferential direction. These paths together with the closed paths of the localized states at E_F are shown as viewed from the top in Fig. 5.11(d). Obviously, the edge states connect different electrodes and, thus, can transport electrons albeit only in one direction. For the displayed polarity of the electrodes, only the three edge states at the right edge can transport electrons as indicated by the arrow marked I_e . Within these edge channels, the electrons, moreover, can not be backscattered as shown in Fig. 5.11(e). Even if they are scattered, the Lorentz force will turn them back into the same direction, i.e., they will never move backwards on scales larger than the cyclotron radius, at least, as long as the scattering length is larger than about twice the cyclotron radius. Since one can not get a potential difference along the edge channels, if the electrons are not backscattered, one measures $V_{xx} = 0$ V between two electrodes along the edge channel, i.e., one finds indeed the insulating behavior, $R_{xx} = V_{xx}/I = 0 \Omega$, as discussed above. The potential difference applied between the two orange electrodes, injecting and removing electrons, drops exclusively at the beginning of the second electrode (μ_1), where the electrons get adapted to μ_1 by inelastic scattering processes within the metal. This is indeed measurable.¹⁸

Since the electrons are injected only in one of the edge channels, one gets a potential difference between the edges, i.e., a Hall voltage. Formally, this Hall voltage is $\mu_2 - \mu_1$, since the left channel equilibrates with μ_1 and the right channel equilibrates with μ_2 (Fig. 5.11(d)). The relation of this Hall voltage to the current can be calculated realizing that the edge channels are one-dimensional ballistic conductors (Section 1.4.1), where all electrons between μ_1 and μ_2 contribute to the current. Each channel, thus, contributes $I_n = \int_{\mu_1}^{\mu_2} \mathcal{D}_R(E) \cdot e \cdot v(E) dE = e/h \cdot (\mu_2 - \mu_1) = e^2/h \cdot V_{\text{ext}}$ with velocity $v(E) = dE/dk \cdot \hbar^{-1}$, 1D density of states of spin polarized electrons moving in the same direction $\mathcal{D}_R(E) = (dE/dk \cdot 2\pi)^{-1}$ (eq. (1.13)), and external voltage $V_{\text{ext}} = (\mu_2 - \mu_1)/e$. Hence, each 1D channel (edge state) contributes with e^2/h to the conductivity at the edge.

Since $V_{\text{ext}} = V_{\text{Hall}}$, the Hall conductance is simply $G_{\text{Hall}} = I/V_{\text{Hall}} = ne^2/h$ with n being the number of edge channels and, thus, the number of occupied extended states in the interior of the sample. The Hall resistance is the inverse of G_{Hall} , if $\sigma_{xx} = 0$ S, i.e., $R_{\text{Hall}} = h/e^2n$ in agreement with the experimental data of Fig. 5.10(b).

Only if an extended state from the interior is at E_F , the current partly runs along this state such that G_{Hall} is not quantized anymore. Backscattering can take place within the extended state at the many crossing points of the extended state as visible in Fig. 5.11(b). Consequently, the longitudinal resistance R_{xx} is larger than 0Ω and

¹⁸ To show the position of voltage drop, one covers the 2D system with a thin film of suprafluid He. The energy that gets dissipated by the electrons at the position of voltage drop heats the sample locally, such that an increased pressure within the He film appears inducing a little fountain at this point [113].

R_{Hall} is not quantized anymore. If B is further increased, E_F gets below the energy of the extended state, such that the sample becomes insulating again ($\sigma_{xx} = 0$ S) and the number of edge states is reduced by one. Consequently, G_{Hall} is dropped by e^2/h . This explains the steps in the Hall resistance straightforwardly by largely employing classical considerations of the electron paths. Quantum mechanics was only necessary to explain the development of Landau levels via the quantization of the kinetic energy and for the derivation of the universal conductance e^2/h of a ballistic channel (Section 1.4.1).

This explanation, however, raises a number of questions. Most importantly, there is no fundamental reason that the edge states do not get gapped at E_F , such that the system becomes an insulator with $\sigma = 0$ S. A mechanism to gap a 1D electron system (1DES) has been shown, e.g., in Section 5.2.1. In principle, details of the edge properties can distinguish, if the 1DES gets gapped or not. Experiments, however, show that the quantized plateaus of R_{Hall} are very robust with respect to the details of the edge and are found in many different materials. Hence, the question arises, if there is a more fundamental reason to explain the quantum Hall effect. This is the topic of the next section.

5.4 Topological analysis of electron systems in solids

There is a more general way to describe the quantum Hall effect. It is based on a so-called topological number, the Chern number n_{Ch} . The Chern number is an integer. It can be described as a geometrical feature of the Hilbert space of the wave functions in two dimensions exposed to a magnetic field. Thus, it is called topological.¹⁹

The approach is more difficult to illustrate. However, it secures the quantized Hall conductance, independent of the details of the edges. Most favorably, the topological approach is transferable to other systems, which has been done very successfully within the last decade. For example, it led to a totally new class of materials, that can be insulating in the bulk, but are necessarily conducting at the surface. These new type of materials are called strong 3D topological insulators (3DTIs) and several tens of 3DTIs have been verified experimentally. In principle, this is the third possible type of conductance in solids besides metals and insulators. Topologically, this class is distinct from all other materials by a single number ν_0 . This number is $\nu_0 = 1$ for 3DTIs, i.e., for the insulators, that necessarily have surface states on all surfaces, and it is $\nu_0 = 0$ for all other insulators. For metals, ν_0 is not a useful index, but other topological indices apply, that are not discussed in this book.²⁰

¹⁹ Topology is a mathematical classification scheme of geometrical objects. It allows, e.g., to distinguish three-dimensional objects by their number of holes. Therefore, one calculates a specific surface integral, that returns the integer n , the number of holes (Section 5.4.1).

²⁰ A key word is, e.g., Weyl semimetals.

Many other new classes of materials (described by other Hamiltonians) have been predicted based on the approach. They provide a number of exotic properties as, e.g., they contain quasiparticles called Majorana fermions, that are only half a fermion, i.e., the probability of finding an electron in a certain area is exactly $1/2$. These particles, of course, can only exist in multiples of two. Interestingly, they are maximally entangled in terms of the constituent electrons, which renders them a favorite candidate for fault tolerant quantum computation (Chapter 3).

Another intriguing aspect of the topological approach is its transferability to other periodic systems. For example, one can build a lattice of optical waveguides, such that the geometry guarantees that the light propagates exclusively along the edges of the geometry. This might be exploited to concentrate the light as required for lasers (Section 2.3.2.4). The approach of topology also works for phonons or for magnons. Since the topological approach is rather new, it is far from being studied exhaustively. The game is partly even completely open, e.g., with respect to possible applications of the novel properties resulting from this approach.

In order to get an insight into the approach, we will follow the historical path of its discovery. Thus, we firstly sketch the reasoning for the quantum Hall effect, which was one of the first solid state systems described in terms of topology.²¹ Afterwards, we will describe the distillation of topological numbers for 2D systems and 3D systems without a magnetic field and without electron-electron interaction. A more rigorous treatment of the topological approach would require a field theoretical description, that, however, is too mathematical for an experimental course. Moreover, it would require a long-term experience with this type of description to deliver any intuitive insight. Thus, the arguments will be partly heuristic, only requiring a solid quantum mechanical understanding. However, following the quotations, the interested reader is guided to the papers settling the field theoretical approach.

5.4.1 What is topology?

In topology, as far as it concerns solid state physics, one deals with geometrical properties of objects in mathematically defined spaces. Such a space is, e.g., the $\vec{k}$ space for

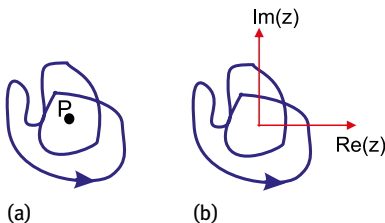


Fig. 5.12: (a) Path (blue line with arrow), which winds twice around the point P counterclockwise. (b) Same curve as in (a) with a coordinate system of the complex plane centered in point P.

²¹ Nobel prize in Physics 2016, www.nobelprize.org.

electrons, that describes a complete set of quantum numbers, i.e., the relevant Hilbert space. The aim is to classify objects in such spaces according to its geometrical properties, such that different objects of the same class can be transformed into each other by stretching and bending. In contrast, tearing and gluing is needed to transfer an object to another class. Obviously, this classification implies a robustness to stay in the same class with respect to weak perturbations such as stretching or bending.

The art is to firstly show that such a classification exists and, subsequently, to define a calculation, whose output unambiguously puts the object to a certain class.

An example is illustrated in Fig. 5.12. Consider a closed line in space, that avoids the point of observation P. Then, a clearly posed question is how often does the line wind around the point P, before it closes? The answer for Fig. 5.12(a) is obviously twice.

Next, we have to put this answer on solid ground. Experimentally, one would position an observer at point P. The person would follow an object along the closed path visually once and one would count, how often he has turned around its axis.

Mathematically, one defines the path as the function $z(t)$ in the complex plane as depicted in Fig. 5.12(b), where $t \in [0, 1)$, $t \in \mathbb{R}$ runs around the whole curve once and $z \in \mathbb{C}$ is the corresponding point in the complex plane. As usual, one can write:

$$z(t) = |z(t)| \cdot e^{i\varphi(t)}. \quad (5.20)$$

Now one can define the integral:

$$Q_I(z) = \frac{1}{2\pi i} \int_0^1 \frac{dz(t)/dt}{z(t)} dt. \quad (5.21)$$

Q_I is exactly the answer to our question for each path $z(t)$ as one sees as follows:

$$Q_I(z) = \frac{1}{2\pi i} \int_0^1 \frac{d}{dt} \ln(z(t)) dt \quad (5.22)$$

$$= \frac{1}{2\pi i} [\ln(z(t))]_0^1 = \frac{1}{2\pi i} \ln \left(\frac{|z(1)|e^{i\varphi(1)}}{|z(0)|e^{i\varphi(0)}} \right) \quad (5.23)$$

$$= \frac{1}{2\pi i} \cdot (\ln(e^{i\varphi(1)}) - \ln(e^{i\varphi(0)})) = \frac{\varphi(1) - \varphi(0)}{2\pi}. \quad (5.24)$$

If we define the function $\varphi(t)$ as continuous, such that it does not jump from 2π to 0 after one turn, $|Q_I(z)|$ is the number of turns, we were asking for. Moreover, if the turns are counterclockwise, $Q_I(z)$ is positive and, if the turns are clockwise, $Q_I(z)$ is negative. Mathematicians can show now that a path turning n times clockwise can be transformed into a path turning $n + m$ times clockwise and m times counterclockwise without tearing and gluing. But we do not go into this discussion.

Importantly, we have classified all closed paths in 2D space avoiding the point P (the origin) into classes with identical numbers of counterclockwise turns around the origin. Obviously $Q_I(z) \in \mathbb{Z}$ and, thus, one calls $Q_I(z)$ a $\mathbb{Z}$ -type topological invariant: the winding number.

5.4.2 Quantum Hall effect in terms of topology

Now, we apply this concept to the quantum Hall effect, where the geometrical space that we consider, is the Hilbert space of wave functions in magnetic field.

To do so, we start with deriving the Hall conductivity quantum-mechanically. Therefore, we firstly calculate the Hall current for a single electron being within a Bloch state, i.e., within a crystalline solid (eq. (2.90))

$$\psi_{n\vec{k}}(\vec{x}) = u_{n\vec{k}}(\vec{x}) \cdot e^{i\vec{k}\vec{x}} \quad (5.25)$$

with n being the band index and $\vec{k}$ being a wave vector. We will argue later that this description is also relevant in magnetic field. We use first order perturbation theory in the weak electric field $\vec{E} = E_x \cdot \vec{e}_x$ applied in the x direction to the sample. This corresponds to an electric potential

$$\Phi_{el}(x) = eE_x \cdot x = -i \cdot eE_x \cdot \frac{d}{dk_x}, \quad (5.26)$$

where we used the operator replacement $x = -i \cdot d/dk_x$, that can be verified by representing the wave function as a superposition of plane waves.

The perturbed wave function $|n\rangle$ than reads:

$$|n\rangle = |n_0\rangle - \sum_{m_0 \neq n_0} \frac{|m_0\rangle \langle m_0 | ieE_x \cdot \frac{d}{dk_x} | n_0 \rangle}{E_{n_0} - E_{m_0}}, \quad (5.27)$$

where $|n_0\rangle, |m_0\rangle$ are solutions of the unperturbed Hamiltonian, i.e., Bloch states with energies E_{n_0} and E_{m_0} , respectively. The bracket $\langle || \rangle$ indicates spatial integration.

Next, we determine the velocity of the resulting state in y direction v_y , that is transversal to $\vec{E}$ as the Hall current:

$$\hat{v}_y = \langle n | v_y | n \rangle = \langle n_0 | v_y | n_0 \rangle - ieE_x \sum_{m_0 \neq n_0} \frac{\langle n_0 | v_y | m_0 \rangle \langle m_0 | \frac{d}{dk_x} | n_0 \rangle}{E_{n_0} - E_{m_0}} + \text{h.c.}, \quad (5.28)$$

where h.c. is the complex conjugate and the quadratic term in E_x is neglected. We use

$$\hat{v}_y = \frac{dy}{dt} = -\frac{i}{\hbar} [\hat{H}, \hat{y}], \quad (5.29)$$

where $[\cdot, \cdot]$ is the commutator and $\hat{H}$ is the Hamiltonian of the unperturbed system. The relation (5.29) can be verified straightforwardly by employing the Schrödinger equation $i\hbar d\psi/dt = \hat{H}\psi$. With eq. (5.29), one gets:

$$\langle n_0 | v_y | m_0 \rangle = -\frac{i}{\hbar} (\langle n_0 | \hat{H} \hat{y} | m_0 \rangle - \langle n_0 | \hat{y} \hat{H} | m_0 \rangle) = -\frac{i}{\hbar} \langle n_0 | y | m_0 \rangle \cdot (E_{n_0} - E_{m_0}). \quad (5.30)$$

Using again the operator replacement $y = -i \cdot d/dk_y$, we obtain:

$$\langle n_0 | v_y | m_0 \rangle = \frac{1}{\hbar} \left\langle n_0 \left| \frac{d}{dk_y} \right| m_0 \right\rangle \cdot (E_{n_0} - E_{m_0}) = \frac{-1}{\hbar} \left\langle \frac{dn_0}{dk_y} \middle| m_0 \right\rangle \cdot (E_{n_0} - E_{m_0}) \quad (5.31)$$

for all $m_0 \neq n_0$. After insertion into eq. (5.28), we find:

$$\hat{v}_y = \langle n_0 | v_y | n_0 \rangle + \frac{ieE_x}{\hbar} \sum_{m_0 \neq n_0} \left\langle \frac{dn_0}{dk_y} \middle| m_0 \right\rangle \left\langle m_0 \middle| \frac{dn_0}{dk_x} \right\rangle + h.c. \quad (5.32)$$

Using the same argument as in eq. (5.29)–(5.31), one can verify that the first term $\langle n_0 | v_y | n_0 \rangle$ vanishes. After exploiting the completeness relation $|m_0\rangle\langle m_0| = \mathbf{1}$, one gets

$$\hat{v}_y = \frac{ieE_x}{\hbar} \left(\left\langle \frac{dn_0}{dk_y} \middle| \frac{dn_0}{dk_x} \right\rangle - \left\langle \frac{dn_0}{dk_x} \middle| \frac{dn_0}{dk_y} \right\rangle \right). \quad (5.33)$$

One can finally verify straightforwardly (by applying the derivation with respect to k_x and k_y), that the plane wave part of the Bloch state $|n_0\rangle = u_{n\vec{k}}(\vec{x}) \cdot e^{i\vec{k}\vec{x}}$ does not contribute, such that we end up with

$$\hat{v}_y = \frac{ieE_x}{\hbar} \left(\left\langle \frac{du_{n\vec{k}}(\vec{x})}{dk_y} \middle| \frac{du_{n\vec{k}}(\vec{x})}{dk_x} \right\rangle - \left\langle \frac{du_{n\vec{k}}(\vec{x})}{dk_x} \middle| \frac{du_{n\vec{k}}(\vec{x})}{dk_y} \right\rangle \right). \quad (5.34)$$

Thus, in this so-called linear response theory, based on the Kubo formula, the transversal velocity in an electric field is a property of the periodic part of a Bloch state. To get the transversal current I_y in electric field E_x , one has to add up the contribution from all occupied states $u_{n\vec{k}}(\vec{x})$ with different $\vec{k}$ and n . The transversal current is, hence, only non-zero, if, for some $u_{n\vec{k}}(\vec{x})$, the derivative with respect to a $\vec{k}$ component (k_x or k_y) is different, if applied to the function or to the complex conjugate of the function. Moreover, this difference must be different for the k_x component and the k_y component. Finally, contributions from different $\vec{k}$ should not cancel. Naively, one might think that will never be the case, but we will see that this is not correct.

Before looking into this, we verify the correctness of the Bloch wave approach in a magnetic field. An elegant verification of the Bloch waves as stationary single particle states without a magnetic field is given by the translational operator $\hat{T}$:

$$\hat{T}(\vec{R}_n) = \exp(\vec{R}_n \cdot \nabla) \quad (5.35)$$

with the lattice vector $\vec{R}_n$. It shifts an arbitrary function by $\vec{R}_n$, i.e.,

$$\hat{T}(\vec{R}_n) \cdot f(\vec{x}) = f(\vec{x} + \vec{R}_n) \quad (5.36)$$

as one can verify, e.g., by Taylor expansion for the three coordinates.²² Thus, $\hat{T}(\vec{R}_n)$ commutes with the lattice periodic potential $\hat{\Phi}_{\text{el}}(\vec{x}) = \hat{\Phi}_{\text{el}}(\vec{x} + \vec{R}_n)$. It moreover commutes with ∇ , since ∇ commutes with itself and, thus, each arbitrary function of ∇ commutes with ∇ . Consequently, $\hat{T}(\vec{R}_n)$ commutes with the Schrödinger equation in periodic potential $\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + e\hat{\Phi}_{\text{el}}(\vec{x})$. This implies that eigenfunctions of the translational operator can be found such that they are eigenfunctions of the Schrödinger equation. These are exactly the Bloch states.²³

²² This is a general mathematical result by Lagrange known as the shift operator.

²³ We do not verify this. A verification can be found, e.g., in Ashcroft/Mermin Solid State Physics: First Proof of Bloch's theorem.

If a homogeneous magnetic field $\vec{B}$ is applied, the Schrödinger equation becomes $\hat{H}_{\vec{B}} = \frac{1}{2m}(i\hbar\nabla + e\vec{A}(\vec{x}))^2 + e\hat{\Phi}_{\text{el}}(\vec{x})$ with the magnetic vector potential $\vec{A}(\vec{x}) = -\frac{1}{2}(\hat{x} \times \hat{B})$ in symmetric gauge, where $\vec{B}$ is the magnetic induction measured in Tesla. Since $\hat{A}$ depends continuously on $\hat{x}$, $\hat{T}(\vec{R}_n)$ does not commute with $\hat{H}_{\vec{B}}$. However, a different operator

$$\hat{T}_B(\vec{R}_n) = \exp\left(\vec{R}_n \cdot \left(\nabla - \frac{e}{i\hbar}\vec{A}(\vec{x})\right)\right) \quad (5.37)$$

commutes with $i\hbar\nabla + e\vec{A}(\vec{x})$ as one can verify by showing that $\nabla - \frac{e}{i\hbar}\vec{A}(\vec{x})$ commutes with $i\hbar\nabla + e\vec{A}(\vec{x})$ and, thus, $i\hbar\nabla + e\vec{A}(\vec{x})$ commutes with any function of $\nabla - \frac{e}{i\hbar}\vec{A}(\vec{x})$. However, applying $\hat{T}_B$ to $\hat{\Phi}_{\text{el}}(\vec{x})$ shifts $\hat{\Phi}_{\text{el}}(\vec{x})$ by $\vec{R}_n$ and gives an additional phase factor, i.e.,

$$\hat{T}_B(\vec{R}_n)\hat{\Phi}_{\text{el}}(\vec{x}) = \exp\left(\vec{R}_n \cdot \frac{e}{i2\hbar}(\hat{x} \times \hat{B})\right) \cdot \hat{\Phi}_{\text{el}}(\vec{x} + \vec{R}_n). \quad (5.38)$$

In the next step, one can show that the phase that one adds up by a series of $\hat{T}_B(\vec{R}_n)$, where the corresponding $\vec{R}_n$ form a closed loop around an area $\vec{A}$ is simply $i \cdot \frac{e|\vec{B}|}{\hbar} \cdot \vec{A} = i \frac{2\pi e B \vec{A}}{\hbar}$. The derivation is similar as for the Aharonov–Bohm effect (Section 1.4.3.5). One employs

$$\oint \vec{A}(\vec{x})d\vec{x} = \iint \nabla \times \vec{A}(\vec{x})d\vec{A} = \iint \vec{B}d\vec{A} = |\vec{B}| \cdot A \cdot \text{sgn}(\vec{B}\vec{A}), \quad (5.39)$$

where the last step uses that $\vec{B}$ is constant and perpendicular to the plane of the quantum Hall sample ($\vec{A}$: area vector). A more rigorous treatment can be found elsewhere.²⁴ Since $e^{i \cdot 2\pi} = 1$, the phase vector vanishes, if the area $\vec{A}$ contains an even number of magnetic flux quanta, i.e., $\Phi = n \cdot h/e = 2n\Phi_0$.²⁵ This implies a regular lattice of points $\vec{R}_{n,\vec{B}}$, where $\hat{T}_B(\vec{R}_{n,\vec{B}})$ commutes with $\hat{H}_B$. The $\vec{R}_{n,\vec{B}}$ are related to each other by the line integral $\oint \frac{e}{\hbar}\vec{A}d\vec{x} = 1$ (eq. (5.38)).

One can hence define a new unit cell, that contains an even number of flux quanta and an integer number of the unit cells without magnetic field. This new unit cell is called the magnetic unit cell with lattice vectors $\vec{R}_{n,\vec{B}}$. The Schrödinger equation commutes with $\hat{T}_B(\vec{R}_{n,\vec{B}})$. Due to this quasi-periodicity, one gets Bloch waves consisting of a plane wave $e^{i\vec{k}\vec{x}}$ and a periodic function in the magnetic unit cell: $u_{n\vec{k}}(\vec{x})$. Thus, the derivation of v_y leading to eq. (5.34) is also valid for $\vec{B} \neq \vec{0}$ T.

However, as a result of the term $\frac{e}{i\hbar}\vec{A}(\vec{x})$ in $\hat{T}_B(\vec{R}_{n,\vec{B}})$, one obtains a more complex $u_{n\vec{k}}(\vec{x})$, that is the origin of $v_y \neq 0$ m/s. One can show that $u_{n\vec{k}}(\vec{x}) = |u_{n\vec{k}}(\vec{x})| \cdot \exp(i \cdot \Theta_{n\vec{k}}(\vec{x}))$ has a phase factor that requires:²⁶

$$\oint_{\text{m.u.c.}} \frac{d\Theta_{n\vec{k}}(\vec{x})}{d\vec{s}} d\vec{s} = -2\pi p, \quad (5.40)$$

²⁴ See, e.g., [114]: eqs. (7)–(10), [115]: eqs. (2.6)–(2.8).

²⁵ Notice that the flux is a quantity independent on the gauge of $\vec{A}(\vec{x})$.

²⁶ Similar argument as in eqs. (5.37)–(5.39), more rigorously in [115]: eqs. (2.9)–(2.13).

if one integrates counterclockwise around the magnetic unit cell (m.u.c.), where p is the number of pairs of flux quanta within the magnetic unit cell. More intuitively, p is simply the number of times the phase $\Theta_{n\vec{k}}(\vec{x})$ changes by 2π , if one moves around the magnetic unit cell. This is called the vorticity of the wave function. It is easy to imagine that the continuous change of the phase on a closed path by 2π surrounds a zero of the spatially continuous function $u_{n\vec{k}}(\vec{x})$, since it implies positive and negative values at $\Theta_{n\vec{k}}(\vec{x}) = 0$ and $\Theta_{n\vec{k}}(\vec{x}) = \pi$, respectively, on the surrounding path. Shrinking the closed path to the central point $\vec{z}_0$ would lead to a diverging kinetic energy (phase change per path length) of the wave function, except if $\psi_i(\vec{z}_0) = 0$. Hence, the only meaningful solution is $\psi_i(\vec{z}_0) = 0$. As a consequence, the applied magnetic field is imprinted in the geometrical properties of the wave function by zeroes, independent from the functional form of the electrostatic potential $\Phi_{\text{el}}(\vec{x})$. This vorticity of the wave function, that is related to the number of zeroes, is called a topological property, since it is given by the geometric construction of the magnetic unit cell.

By transforming to $\vec{k}$ space, we also get a magnetic Brillouin zone (MBZ) that is given by the reciprocal lattice of the magnetic unit cell (Fig. 5.13).

Finally, we write down the transversal current density j_y carried by a fully occupied, single band n within the MBZ, without spin degeneracy, by integrating eq. (5.34) across the MBZ:

$$\begin{aligned}\hat{j}_y &= -e \int \int_{\text{MBZ}} \frac{1}{(2\pi)^2} \hat{v}_y(\vec{k}) d\vec{k} \\ &= -e \int \int_{\text{MBZ}} \frac{1}{(2\pi)^2} \frac{ieE_x}{\hbar} \left(\left\langle \frac{du_{n\vec{k}}(\vec{x})}{dk_y} \middle| \frac{du_{n\vec{k}}(\vec{x})}{dk_x} \right\rangle - \left\langle \frac{du_{n\vec{k}}(\vec{x})}{dk_x} \middle| \frac{du_{n\vec{k}}(\vec{x})}{dk_y} \right\rangle \right) d^2\vec{k},\end{aligned}\quad (5.41)$$

where the factor $1/(2\pi)^2$ comes from the density of states in $\vec{k}$ -space.

By factorization aiming for $G_0/2 = e^2/h$ as a prefactor, i.e., the conductance carried by a ballistic spin polarized channel (eq. (1.23)), we get:

$$\hat{j}_y = \frac{e^2}{h} E_x \int \int_{\text{MBZ}} \frac{1}{2\pi i} \left(\left\langle \frac{du_{n\vec{k}}(\vec{x})}{dk_y} \middle| \frac{du_{n\vec{k}}(\vec{x})}{dk_x} \right\rangle - \left\langle \frac{du_{n\vec{k}}(\vec{x})}{dk_x} \middle| \frac{du_{n\vec{k}}(\vec{x})}{dk_y} \right\rangle \right) d^2\vec{k}. \quad (5.42)$$

Thus, according to the experimental result (Fig. 5.10(b)), the integral must be an integer at the plateaus of the transversal conductivity $\sigma_{xy} = j_y/E_x = G_{\text{Hall}}$ (Section 5.3.2). This integer is the Chern number n_{Ch} .

To show that it is indeed an integer, we firstly use Stokes theorem, i.e., we write:

$$\begin{aligned}\left\langle \frac{du_{n\vec{k}}(\vec{x})}{dk_y} \middle| \frac{du_{n\vec{k}}(\vec{x})}{dk_x} \right\rangle - \left\langle \frac{du_{n\vec{k}}(\vec{x})}{dk_x} \middle| \frac{du_{n\vec{k}}(\vec{x})}{dk_y} \right\rangle \\ = [\nabla_{\vec{k}} \times \langle u_{n\vec{k}}(\vec{x}) | \nabla_{\vec{k}} | u_{n\vec{k}}(\vec{x}) \rangle]_z := [\nabla_{\vec{k}} \times \vec{A}_{\text{Berry},n}(\vec{k})]_z,\end{aligned}\quad (5.43)$$

where $\nabla_{\vec{k}}$ is the gradient operator in $\vec{k}$ -space and the subscript z refers to the third component of the resulting vector. The vector

$$\vec{A}_{\text{Berry},n}(\vec{k}) = \langle u_{n\vec{k}}(\vec{x}) | \nabla_{\vec{k}} | u_{n\vec{k}}(\vec{x}) \rangle \quad (5.44)$$

is called the Berry connection of band n at wave vector $\vec{k}$.

By Stokes theorem, that can be applied, if the integrand is continuous, we would get:

$$\sigma_{xy} = \frac{j_y}{E_x} = \frac{e^2}{h} \cdot \frac{1}{2\pi i} \oint_{\text{dMBZ}} \vec{A}_{\text{Berry},n}(\vec{k}) \cdot d\vec{k} := \frac{e^2}{h} \cdot \frac{1}{2\pi i} \cdot \varphi_{\text{Berry},n} \quad (5.45)$$

with the boundary of the magnetic Brillouin zone dMBZ and the so-called Berry phase $\varphi_{\text{Berry},n}$. Since the MBZ is periodic, i.e., the left edge and the right edge as well as the upper and the lower edge are identical, the line integral along dMBZ is necessarily zero, i.e., $\varphi_{\text{Berry},n} = 0$. This is indeed an integer, but a rather boring one.

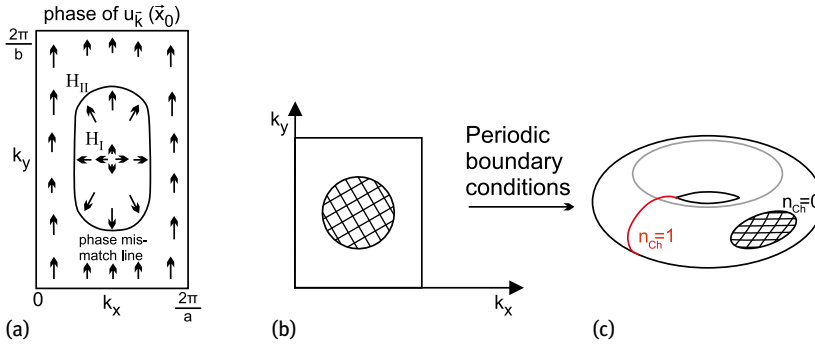


Fig. 5.13: (a) Magnetic Brillouin zone (MBZ) with phase $\theta_{n\vec{k}}(\vec{x} = \vec{x}_0)$ of the function $u_{n\vec{k}}(\vec{x} = \vec{x}_0)$ at a particular real space position $\vec{x}_0$. $\theta(\vec{k}) := \theta_{n\vec{k}}(\vec{x} = \vec{x}_0)$ is symbolized as a vector with a corresponding angle θ to the vertical axis. Periodic boundary conditions require that the phase is the same on the left and on the right as well as on the top and on the bottom of the MBZ. A particular $\vec{x}_0$ with a single zero in the MBZ is chosen and the required phase mismatch line separating area I and II is drawn. (b) Same as (a) with reduced complexity showing area I as a hatched circle. (c) Transformation of the MBZ to a torus which correctly displays the periodic boundary conditions (PBC). An additional phase mismatch line encircling the torus along one circumference is added in red. The Chern number n_{ch} is marked for both phase mismatch lines. (a) After [115].

However, $\vec{A}_{\text{Berry},n}(\vec{k})$ cannot be continuous in $\vec{k}$ space, since the zeros of $u_{n\vec{k}}(\vec{x})$ require a circulating phase around them also in $\vec{k}$ -space. This does not continuously fit to the periodic boundary conditions as sketched in Fig. 5.13(a). Figure 5.13(a) shows a $\vec{k}$ dependent representation of the phase $\theta_{n,\vec{k}}(\vec{x})$ (eq. (5.40)) for a selected $\vec{x} = \vec{x}_0$ as a vector with a corresponding angle to the vertical direction. One recognizes the requirement of a phase mismatch line by the clash of phases. One phase is encircling the zero and the other phase has to be the same on both sides due to the periodic boundaries. The

topology of closed lines of the winding vector and of closed lines of the periodic vector do not match. Consequently, the Bloch functions of the two areas I and II are different at the phase mismatch line according to:

$$u_{n\vec{k}}^I(\vec{x}) = u_{n\vec{k}}^{II}(\vec{x}) \cdot e^{i\Delta\theta_{n\vec{x}}(\vec{k})}. \quad (5.46)$$

Here, $\Delta\theta_{n\vec{x}}(\vec{k})$ describes the phase difference of wave functions at the same $\vec{k}$, but belonging to the two different areas, for a particular $\vec{x}$. To apply Stokes theorem, one has to integrate for area *I* and *II* separately. Taking into account that the line integral at the MBZ-boundary vanishes, using the definitions in eq. (5.43) and (5.46), and calculating the difference in Berry connection $\vec{A}_{\text{Berry},n}(\vec{k}) = \langle u_{n\vec{k}}(\vec{x}) | \nabla_{\vec{k}} | u_{n\vec{k}}(\vec{x}) \rangle$ between the $u_{n\vec{k}}^I$ and the $u_{n\vec{k}}^{II}$ explicitly, one gets:

$$\begin{aligned} \sigma_{xy} &= \frac{e^2}{h} \cdot \frac{1}{2\pi i} \oint_{\text{dPMM}} (\vec{A}_{\text{Berry},n}^I(\vec{k}) - \vec{A}_{\text{Berry},n}^{II}(\vec{k})) d\vec{k} \\ &= \frac{e^2}{h} \cdot \frac{1}{2\pi i} \oint_{\text{dPMM}} \iint i \nabla_{\vec{k}} \Delta\theta_{n\vec{x}}(\vec{k}) d^2\vec{x} d\vec{k} \end{aligned} \quad (5.47)$$

with dPMM being the phase mismatch line. Since $u_{n\vec{k}}$ has to be single-valued in area I and II for each $\vec{x}$, the path integral of the gradient has to be an integer multiple of 2π such that, we finally get

$$\sigma_{xy} = \frac{e^2}{h} \cdot n_{\text{Ch}} \quad (5.48)$$

with $n_{\text{Ch}} \in \mathbb{Z}$.

Thus, we have found a possibility to get a non-zero Chern number by the complex phase vector of the Bloch function according to eq. (5.40) and we have verified that it must be an integer. The requirement is that the integral in eq. (5.42) is continuous in both areas up to the phase mismatch line, such that Stokes theorem can be applied. This is the case, if the MBZ is fully occupied, which is equivalent to a fully occupied, spin-polarized Landau level. Hence, we describe the situation where E_F is in between two spin-polarized Landau levels. Determining the value of n_{Ch} gets too complicated for this introductory textbook, such that we restrict ourselves to describe the result. It turns out that the value of n_{Ch} depends on the topology of the mismatch line after transforming the MBZ to a torus and, thus, representing the periodic boundary conditions correctly (Fig. 5.13(c)). Only if the mismatch line is not contractible on the torus, is n_{Ch} different from zero (without argument). Thus, we are back at the winding number from the previous section, asking how many times does the mismatch line surrounds the torus, respectively how many mismatch lines surround it. Most of the mismatch lines surrounding zeros of $u_{n\vec{k}}$ are irrelevant for the Hall conductance, since they do not surround the $\vec{k}$ space torus. Only the ones that surround the torus contribute to the Hall conductivity. In other words, the integer value of σ_{xy} is related to the properties of particular zeros of the wave functions in $\vec{k}$ space, which require a non-contractible phase mismatch line.

Unfortunately, the above result is intuitive, but experimentally useless as described below. Nevertheless, it is a useful introduction into the general approach of topological considerations in solids. Using geometrical properties of the space, where the wave functions are defined (the MBZ), we have found a number n_{Ch} related to a measurable property ($\sigma_{xy} \cdot \frac{h}{e^2} = n_{\text{Ch}}$), that is necessarily an integer. The definition of such numbers is at the heart of the topological approach to solids.

Notice, that n_{Ch} is not related to the edge of the sample, but is a property of the wave functions of an infinite 2D system in a magnetic field. Hence, in a real sample with edges, the properties of the edge are not important for the measured value of V_{Hall} .

Finally, we have indeed found a robust prediction of the quantum Hall effect, that does not depend on the details of the edges. Counter-intuitively, it does not contradict our previous description that the current flows at the edge of the sample, but only implies that the origin of this edge current is given by the complex wave functions of the bulk of the sample. This surprising relation between edge properties and bulk properties is dubbed the bulk-boundary correspondence.

5.4.3 General approach to topology

Before proceeding with the quantum Hall effect, we dwell briefly on the generality of the topological approach. Figure 5.14 shows a so-called periodic table of topology. Hamiltonians which describe solids are classified according to fundamental symmetries as particle-hole symmetry (PHS), time-reversal symmetry (TRS) and sublattice symmetry (SLS), also called chiral symmetry. PHS is conserved, e.g., in superconductors, TRS is broken by application of an external or an internal magnetic field and SLS requires two identical atoms in the unit cell of the crystal as present, e.g., in graphene. The Hamiltonians are analyzed for different dimensions d , i.e., for situations where the electrons are only free to move in d linearly independent directions (Section 1.2.3). It is then analyzed by field theoretical means, if a non-trivial integer number for the conductivity is well defined. This eventually reveals results similar to eq. (5.48). The table shows that many of the Hamiltonians exhibit such a number. The quantum Hall case discussed above corresponds to the $d = 2$ unitary class and is $\mathbb{Z}$ topological, thus, exhibits a Chern number $n_{\text{Ch}} \in \mathbb{Z}$.

The analysis leading to topological tables is a task for theoretical physicists, but it shows that the approach is applicable to many other types of electron systems. The remaining experimental problem is then mainly to find real materials, that exhibit the robust transport properties predicted by the topological numbers, if these numbers are not zero.

		TRS	PHS	SLS	d=1	d=2	d=3
Standard (Wigner-Dyson)	A (unitary)	0	0	0	-	$\mathbb{Z}$	-
	AI (orthogonal)	+1	0	0	-	-	-
	AII (symplectic)	-1	0	0	-	$\mathbb{Z}_2$	$\mathbb{Z}_2$
Chiral (sublattice)	AIII (chiral unitary)	0	0	1	$\mathbb{Z}$	-	$\mathbb{Z}$
	BDI (chiral orthogonal)	+1	+1	1	$\mathbb{Z}$	-	-
	CII (chiral symplectic)	-1	-1	1	$\mathbb{Z}$	-	$\mathbb{Z}_2$
Bogliubov de Gennes	D	0	+1	0	$\mathbb{Z}_2$	$\mathbb{Z}$	-
	C	0	-1	0	-	$\mathbb{Z}$	-
	DIII	-1	+1	1	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}$
	CI	+1	-1	1	-	-	$\mathbb{Z}$

Fig. 5.14: Periodic table of topology: Ten symmetry classes of single-particle Hamiltonians are classified in terms of the presence or absence of time-reversal symmetry (TRS), particle-hole symmetry (PHS), and sublattice symmetry (SLS), which is relevant, if the unit cell of the crystal contains more than one atom. The absence of symmetries is denoted by 0, while the presence of these symmetries is denoted by +1 or -1, indicating the sign of the result of the squared symmetry operation. The first two columns contain terms used to characterize the Hamiltonian of a certain symmetry by different standards (2nd column) or by traditional terms including names of physicists (1st column). The last three columns show if a topological number is found within the three spatial dimensions symbolized by d . $\mathbb{Z}$ and $\mathbb{Z}_2$ indicate two different types of topological integer numbers, where $\mathbb{Z}$ is related to the Chern number. The dashes indicate the absence of a topological number. The classification ignores electron-electron interaction [116].

5.4.4 Quantum Hall effect and disorder

After this short excursion, we proceed by analyzing the quantum Hall effect.

The derivation of $\sigma_{xy} = \frac{e^2}{h} \cdot n_{\text{Ch}}$ required that the MBZ is densely filled. Otherwise, Stokes theorem is not applicable and we have no means to predict integer values of the transversal conductivity.

If the area of two magnetic flux quanta is commensurable with the lattice unit cell, the bands covering the whole MBZ are simply the Landau levels. Since the area, that contains a magnetic flux of h/e , is typically huge with respect to the unit cell (10 T corresponds to $\sim (20 \text{ nm})^2$ per flux quantum), this condition is approximately fulfilled. Alternatively, one can deal with infinite solids, which will provide some commensurability for each possible B . Then, the commensurable cell contains $2n$ flux quanta and m atomic unit cells with n and m being both integers larger than one. Consequently, the magnetic unit cell gets larger and the MBZ gets smaller. This leads to a backfolding of the Landau levels into the smaller MBZ, i.e., each Landau level consists of n bands. At low B , this does not matter since the bands remain nearly degenerate. But as soon as the cyclotron radius gets similar to the atomic lattice constant, the different bands are split, basically due to a different spatial distribution of the Landau level wave func-

tion with respect to the atomic lattice. This leads to a band structure development as shown in Fig. 5.15(d) and called the Hofstadter butterfly. At low field (flux) the Landau fan as in Fig. 5.10(c) is apparent, but at higher B , the Landau levels split in multiple bands with a detailed substructure, that is similar to the texture of a butterfly. The fourfold symmetry is a consequence of the tight binding Hamiltonian, that is the one of eq. (4.29) with an additional B field and, hence, not universal.

Experiments are typically performed at ~ 0.0001 flux units per atomic unit cell and are, hence, not sensitive to this structure. But artificial lattices can be tuned into the interesting region of order 0.1–1 flux units per unit cell and fingerprints of the Hofstadter butterfly have indeed been observed in the transversal conductivity $\sigma_{xy}(B)$ [117–119].

Staying with the simple Landau level picture, Fig. 5.15(a) shows the development of the Landau and spin levels at low $\vec{B}$ (same as Fig. 5.10(c)). Due to the increasing degeneracy of each level, the Fermi level discontinuously jumps down to a lower Landau level. The jump appears, if enough density of states is available in all levels below the actual one, provided that the electron density n_e is constant (Section 5.3.2). Exactly at the positions of the jumps (blue dots), all Landau levels are either completely filled or completely empty, such that Stoke's theorem can be applied and the transversal conductivity is $\sigma_{xy} = \frac{e^2}{h} \cdot n_{\text{Ch}}$. Unfortunately, these points are marginal. They appear, if n_e divided by the degeneracy eB/h is an integer. The classical Hall resistivity ρ_{xy} at these $\vec{B}$ -fields is:

$$\frac{hn_e}{eB} = n_{\text{Ch}} \implies B = \frac{hn_e}{en_{\text{Ch}}} \implies \rho_{xy} = \frac{B}{en_e} = \frac{h}{e^2 n_{\text{Ch}}} . \quad (5.49)$$

Thus, the quantized Hall resistivity predicted by topology is identical to the classical Hall resistivity $\rho_{xy} = B/en_e$ at these marginal points, where the Fermi level lies between bands (Fig. 5.15(b)). This supports the calculation, but is sincerely a disappointing result for an intricate derivation.

In other words, the quantization of transversal conductivity deduced by topology exists, but the plateaus observed experimentally can not be explained by the argument.

The riddle can be solved by including disorder. Naturally, one loses periodicity by random disorder and, thus, the MBZ disappears. In order to get a novel description of ρ_{xy} , we firstly recall that the perturbation approach leading to eqs. (5.28) and (5.32) did not rely on Bloch states. Using these two equations, we can write.

$$\hat{v}_y = \langle n | v_y | n \rangle = \langle n_0 | v_y | n_0 \rangle + ie\hbar E_x \sum_{n \neq m} \frac{\langle n_0 | v_y | m_0 \rangle \langle m_0 | v_x | n_0 \rangle}{(E_{n_0} - E_{m_0})^2} + h.c. . \quad (5.50)$$

where $|n_0\rangle$ and $|m_0\rangle$ are now states of the disordered potential. They could even be many-particle ground states and excited states of an interacting electron system, which, however, will not be discussed any further.

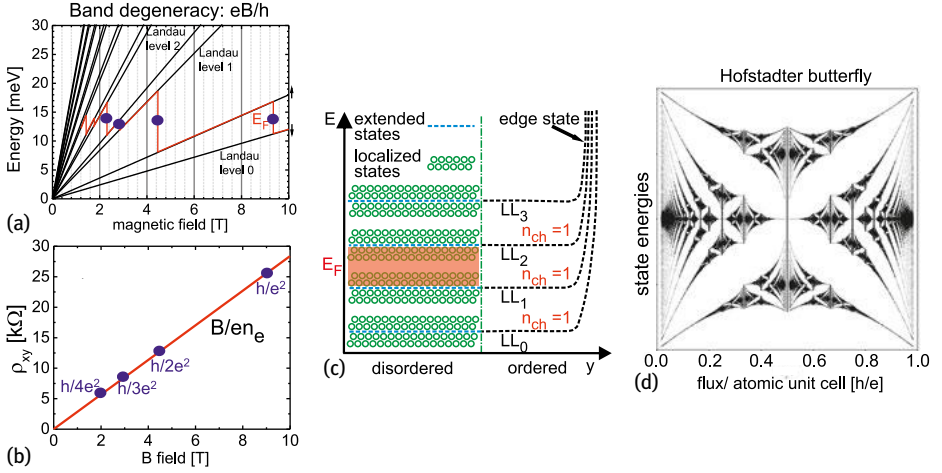


Fig. 5.15: (a) Landau and spin levels (black lines) as a function of $\vec{B}$ -field. Spin directions are marked on the right for the lowest Landau level 0. The red line is the Fermi level, if $n_e(\vec{B}) = \text{const}$. The degeneracy of each level increases with $\vec{B}$ as marked on the top and, thus, E_F exhibits jumps. The blue dots mark the positions, where all Landau levels are either completely filled or completely empty. (b) The classical transversal resistivity ρ_{xy} as a function of $\vec{B}$. The blue dots mark the values $\rho_{xy} = 1/\sigma_{xy} = \frac{h}{e^2 n_{ch}}$ where the Chern number is necessarily an integer. (c) Right hand side: Landau levels without disorder bending upwards at the rim of the sample, where they cross E_F giving rise to edge states (see also Fig. 5.11). Left hand side: disorder is added leading to extended states and localized states as explained in Section 5.3.2. The Chern number n_{ch} can be identified with the extended state, such that σ_{xy} is quantized as long as E_F is located within the localized states (brown transparent areas). (d) Hofstadter butterfly: Energy values of electron bands in a cosine periodic potential of the atoms on a square lattice as calculated by a tight binding model. A homogeneous $\vec{B}$ field is added and quantified as flux per lattice unit cell in units of h/e . (d) [120, 121].

The first term vanishes again due to eq. (5.30). The velocity operators depend on the gauge of the $\vec{B}$ field. For the Landau gauge ($\vec{A}(\vec{x}) = |\vec{B}|x\vec{e}_y$), e.g., we have $\hat{v}_x = \frac{-i\hbar}{m} \frac{d}{dx}$ and $\hat{v}_y = \frac{-i\hbar}{m} \frac{d}{dy} - \frac{eB}{m} x$.

If one takes a finite system of length L_1 and width L_2 , the presence of the magnetic field requires a gauge dependent phase shift of all wave functions across the system again being the Aharonov–Bohm phase. The general idea is now firstly to express the velocity operators in terms of these phase factors. They can be described by νL_1 for a shift of L_1 along the x direction and by βL_2 for a shift of L_2 along the y direction. Now one can perform a unitary transformation of the original wave functions $|n\rangle$ to

$$|\tilde{n}\rangle = e^{-i(\nu x + \beta y)} \cdot |n\rangle \quad (5.51)$$

This transformation uses the gauge freedom and introduces the prefactors ν and β into the velocity operators of the transformed Hamiltonian $\hat{\tilde{H}}$ such that

$$\langle n | v_x | m \rangle = \frac{1}{\hbar} \left\langle \tilde{n} \left| \frac{d\hat{\tilde{H}}}{d\nu} \right| \tilde{m} \right\rangle, \quad \langle n | v_y | m \rangle = \frac{1}{\hbar} \left\langle \tilde{n} \left| \frac{d\hat{\tilde{H}}}{d\beta} \right| \tilde{m} \right\rangle \quad (5.52)$$

This can be verified straightforwardly by transforming the Hamiltonian starting with the Landau gauge and then applying the derivation to the new Hamiltonian $\widehat{H}$. It naturally implies a description of $\widehat{v}_x$ and $\widehat{v}_y$ in terms of the phase factors at the boundaries of the sample by changing eq. (5.50) accordingly:

$$\widehat{v}_y = \frac{-ieE_x}{\hbar} \left(\sum_{\tilde{n}_0 \neq \tilde{m}_0} \frac{\langle \tilde{n}_0 | \frac{d\widehat{H}}{d\beta} | \tilde{m}_0 \rangle \langle \tilde{m}_0 | \frac{d\widehat{H}}{dv} | \tilde{n}_0 \rangle}{(E_{n0} - E_{m0})^2} - \frac{\langle \tilde{n}_0 | \frac{d\widehat{H}}{dv} | \tilde{m}_0 \rangle \langle \tilde{m}_0 | \frac{d\widehat{H}}{d\beta} | \tilde{n}_0 \rangle}{(E_{n0} - E_{m0})^2} \right) \quad (5.53)$$

Finally, we use the product rule for $\tilde{m}_0 \neq \tilde{n}_0$, i.e.:

$$\left\langle \tilde{n}_0 \left| \frac{d\widehat{H}}{d\beta} \right| \tilde{m}_0 \right\rangle = \frac{d}{d\beta} \left(\langle \tilde{n}_0 | \widehat{H} | \tilde{m}_0 \rangle \right) - \left\langle \frac{d\tilde{n}_0}{d\beta} \left| \widehat{H} \right| \tilde{m}_0 \right\rangle - \left\langle \tilde{n}_0 \left| \widehat{H} \right| \frac{d\tilde{m}_0}{d\beta} \right\rangle \quad (5.54)$$

$$= 0 - E_{m0} \left\langle \frac{d\tilde{n}_0}{d\beta} \left| \tilde{m}_0 \right\rangle - E_{n0} \left\langle \tilde{n}_0 \left| \frac{d\tilde{m}_0}{d\beta} \right\rangle \quad (5.55)$$

$$= -E_{m0} \left\langle \frac{d\tilde{n}_0}{d\beta} \left| \tilde{m}_0 \right\rangle + E_{n0} \left\langle \frac{d\tilde{n}_0}{d\beta} \left| \tilde{m}_0 \right\rangle \quad (5.56)$$

$$= \left\langle \frac{d\tilde{n}_0}{d\beta} \left| \tilde{m}_0 \right\rangle \cdot (E_{n0} - E_{m0}) \quad (5.57)$$

The same trick can be applied to the second term in the product of eq. (5.53) revealing

$$\left\langle \tilde{m}_0 \left| \frac{d\widehat{H}}{dv} \right| \tilde{n}_0 \right\rangle = \left\langle \tilde{m}_0 \left| \frac{d\tilde{n}_0}{dv} \right\rangle \cdot (E_{n0} - E_{m0}) \quad (5.58)$$

as well as to the two factors of the product behind the minus sign. With the completeness relation $\sum_{\tilde{m}_0} |\tilde{m}_0\rangle \langle \tilde{m}_0| = \mathbf{1}$, we finally arrive at

$$\widehat{v}_y = \frac{-ieE_x}{\hbar} \left[\left\langle \frac{d\tilde{n}_0}{d\beta} \left| \frac{d\tilde{n}_0}{dv} \right\rangle - \left\langle \frac{d\tilde{n}_0}{dv} \left| \frac{d\tilde{n}_0}{d\beta} \right\rangle \right] \quad (5.59)$$

or equivalently

$$\widehat{v}_y = \frac{-ieE_x L_1 L_2}{\hbar} \left[\left\langle \frac{d\tilde{n}_0}{d\phi} \left| \frac{d\tilde{n}_0}{d\varphi} \right\rangle - \left\langle \frac{d\tilde{n}_0}{d\varphi} \left| \frac{d\tilde{n}_0}{d\phi} \right\rangle \right] \quad (5.60)$$

with ϕ and φ being the boundary phases at $x = L_1$ and $y = L_2$. This gives by adding up all occupied states $|\tilde{n}_0\rangle$ ²⁷:

$$\sigma_{xy} \sim \frac{ie^2}{\hbar} \sum_{\tilde{n}_0} \left[\left\langle \frac{d\tilde{n}_0}{d\phi} \left| \frac{d\tilde{n}_0}{d\varphi} \right\rangle - \left\langle \frac{d\tilde{n}_0}{d\varphi} \left| \frac{d\tilde{n}_0}{d\phi} \right\rangle \right] . \quad (5.61)$$

The decisive argument is now that the particular choice of the boundary phases can not influence the conductivity, if the Fermi level is within localized states with vanishing amplitude, at least, at one boundary. Thus, one can write σ_{xy} also as an

²⁷ We omit the discussion of correct units in the following for the sake of simplicity.

average over the results for different boundary phases.

$$\sigma_{xy} \sim \frac{e^2}{h} \cdot \sum_{\tilde{n}_0} \int_0^{2\pi} \int_0^{2\pi} \frac{1}{2\pi i} \left[\left\langle \frac{d\tilde{n}_0}{d\phi} \middle| \frac{d\tilde{n}_0}{d\phi} \right\rangle - \left\langle \frac{d\tilde{n}_0}{d\phi} \middle| \frac{d\tilde{n}_0}{d\phi} \right\rangle \right] d\phi d\varphi. \quad (5.62)$$

The integral is formally identical to eq. (5.42). Thus, the result of the integral leads to an integer, if a continuous change of any wave function in the space of the boundary phase factors along a closed path does not change the phase factor of the wave function modulo 2π . Thus, as long as the ground state wave function is unique in boundary phase space, the transversal conductance will be

$$\sigma_{xy} = \frac{e^2}{h} \cdot n \quad (5.63)$$

with n being an integer. This is important progress, since the conductance quantization is now predicted for the typical case of a localized state at E_F . The absence of quantization is marginal and appears only when the wave function at E_F is not localized and, thus, an extended state. This is visualized in Fig. 5.15(c), where the transparent brown area marks the typical case for disordered systems with quantized Hall conductance. In the limit of an infinite sample (at $T = 0$ K), there will be only a single state in each spin-polarized Landau level, the extended state, that, if at E_F , gives rise to non-quantized σ_{xy} .

The description in eq. (5.62) makes clear that the integer n_{Ch} (Chern number) is still a geometrical property of individual wave functions, however, in the unusual space of boundary phases of the finite system. Hence, one can sum up n_{Ch} by integers of individual wave functions. It is intuitive that the localized states will not contribute to the Chern number, since they will not depend on the boundary phases. Indeed, one can show that each occupied extended state contributes a unit of one to the Chern number [122]. Thus, the Chern number is encoded in the change of the extended state to changes of the boundary conditions. One can make this even more explicit (without argument): There is exactly one node (zero point) of each extended state that crosses the whole system, if the boundary phase is changed by 2π . Thus, it is again due to the zeros of the wave functions imprinted by the B field, that the Hall conductance is quantized, but now it is the path of the zeros of the extended states in real space, which defines n_{Ch} . In Fig. 5.15(c) this is marked by the states labeled $n_{\text{Ch}} = 1$.

An important aspect of the derivation above is again the so-called bulk-boundary correspondence.²⁸ Since we deduced a finite Hall conductivity for localized, i.e., insulating states of the system at E_F , a natural question is: What are the states that carry the current? Since the bulk states are not able to carry current, the only alternatives are states at the edge of the sample. Thus, the presence of the edge states of Fig. 5.11(d) can

²⁸ Exact mathematical derivations of the bulk-boundary correspondence are lengthy, see e.g. [123]. We will discuss a more rigorous argument in Section 5.4.6.1 on 3D topological insulators.

also be derived on more fundamental, topological grounds. They are a consequence of the occupied, extended bulk states which contain a nontrivial zero due to B field. This bulk property, that is independent of the details of the disorder potential $\Phi_{el}(\vec{x})$, implies the robustness of the quantum Hall effect.

The argument can even be generalized to any topological number. An insulating material with non-trivial topological number describing the Hall conductance requires conducting states at E_F , which are located at the boundaries of the material.

5.4.5 2D topological insulators

After the topological description of the quantum Hall effect, scientists started to look for systems that exhibit a topological number describing σ_{xy} , which is different from zero, without an external magnetic field. This does not work straightforwardly, since a finite σ_{xy} would break the time-reversal symmetry (TRS) in contrast to the Hamiltonian without $\vec{B}$.²⁹ However, one can separate the Schrödinger equation into two parts for the two opposite spins of the electrons. The two spins exhibit a different spin orbit interaction, that breaks the time reversal symmetry for each spin by acting as an effective magnetic field on the electron with the corresponding spin. Thus, the two spin channels could have opposite σ_{xy} . This would still respect global time reversal symmetry, since the time-inversion inverts spin and current direction and, thus gives the same Hall voltage. The task is, thus, to find a system with a finite σ_{xy} for each spin channel.

It is natural to look for a strong spin-orbit term, e.g., proportional to $\vec{k} \times \vec{\sigma}$ with $\vec{\sigma}$ being the vector of the three Pauli matrices (page 204) representing the electron spin. Such a term naturally leads to a transversal movement of the electrons, which is opposite for the two spins. Within a band structure, the spin-orbit interaction acts much more strongly on bands, where the periodic part of the Bloch function $u_{n\vec{k}}(\vec{x})$ represents atomic states with an orbital quantum number, e.g., p-type, d-type or f-type states, than for bands without, i.e., s-type states. Thus, the different spin components of the p-type valence band of a typical semiconductor shift much more strongly than the s-type conduction band. This results, e.g., in the spin-orbit split band of the valence band of most semiconductors (Si, Ge, GaAs, CdTe, ...). The spin orbit splitting increases, moreover, with the charge number of the atoms constituting the system (Section 4.5.4), i.e., one has to look for semiconductors made of heavy atoms. If the spin-orbit interaction is strong enough, it can even lead to a crossing of the p-type valence band and the s-type conduction band as displayed in Fig. 5.16(a). Mostly, there are additional $\vec{k}$ dependent terms that mix the valence and the conduction band. Formally, one can derive such terms by perturbation theory starting from the atomic

²⁹ If one inverts the time, i.e., the direction of current, the Hall voltage would invert for constant σ_{xy} .

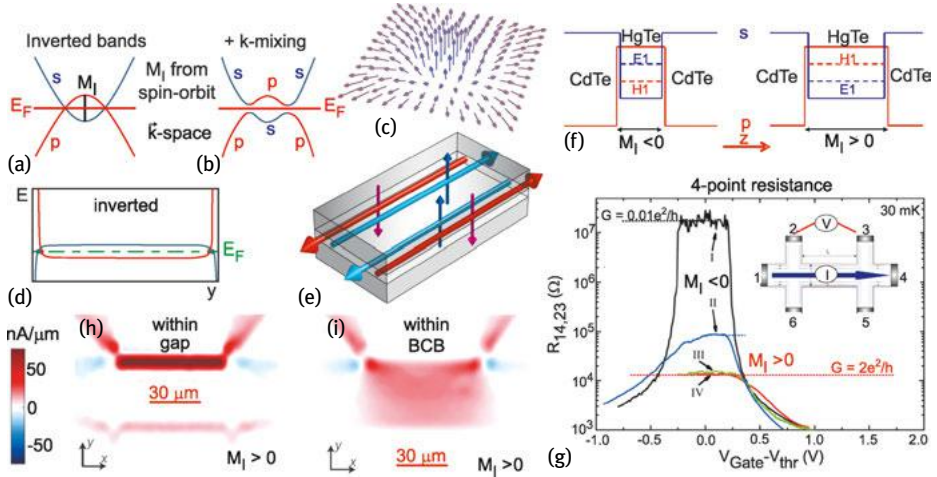


Fig. 5.16: (a), (b) Model band structure of a 2D topological insulator: (a) The spin-orbit interaction, that shifts the p-type valence band (red) upwards by M_I , leads to a band crossing with the s-type conduction band (blue). (b) An additional interaction term between the two bands (parameter A_I in eq. (5.66)) leads to a symmetric and an antisymmetric combination of the two bands, that induces an anticrossing, thus, a band gap and a changing s/p character (colors) of the resulting two bands. The Fermi level E_F is chosen to be in that band gap (red line). (c) Distribution of $\hat{d}(\vec{k}) = \vec{d}(\vec{k})/|\vec{d}(\vec{k})|$ (eq. (5.66)) in $\vec{k}$ space close to $\vec{k} = \vec{0}/m$. (d) Intuitive illustration of the edge states. While the p-type band (blue) bends downwards, the s-type band (red) bends upwards, thus, reestablishing the normal order of bands and implying a conducting state at the edge of the sample. (e) 2D illustration of the edge states (thick arrows) running in opposite directions for the two spin contributions (slim arrows). Note that two edge channels of opposite spin run in the same direction implying $\sigma_{xx} = 2e^2/h$. (f) Vertical band structure for a realization of a 2D topological insulator, i.e., a stacking of CdTe/HgTe/CdTe. The heavier HgTe exhibits a partially inverted band for large thickness ($M_I > 0$, right), i.e., the hole band H1 is above the electron band E1. For an additional confinement in the z direction, the confinement energy acts oppositely on H1 and E1, such that the normal order (E1 is above H1) can be reestablished ($M_I < 0$, left). (g) 4-point resistance $R_{14,23}$ of the stacks shown in (f) as a function of the applied gate voltage $V_{\text{Gate}} - V_{\text{thr}}$: The resistance for $M_I < 0$ (curve I) levels at $\sim 1 \times 10^7 \Omega$, i.e., at the input resistance of the measurement device. This gate voltage region marks the normal band gap. In contrast, the resistance of the $M_I > 0$ samples (curve III, IV) levels at $R_{14,23} \approx h/2e^2$ as expected for two conducting edge channels. Inset: Sketch to measure $R_{14,23} = V/I$ in (g). (h) Scanning SQUID result, that visualizes the current flow within a $M_I > 0$ sample with E_F being within the inverted band gap. (i) Same experiment, but with E_F being in the bulk conduction band (BCB). (a), (f) [124], (d), (e), (g) after [125], (h), (i) [126].

orbitals, which is known as $\vec{k} \cdot \vec{p}$ theory. Thus, the states form symmetric and antisymmetric combinations of p-type states and s-type states with different energy and, thus, prohibit a crossing of the two bands. The resulting band structure is sketched in Fig. 5.16(b). If we put the Fermi level into this anticrossing region, it is still within a band gap, albeit the character of the states in the inner part between the anticrossings is inverted with respect to the outer part of the band structure. Since p-states get a neg-

ative sign after spatial inversion and s-states are unchanged by spatial inversion, the total inversion symmetry of the product of occupied states for a certain $\vec{k}$ is different for the inner part and the outer part. We will see that this change of symmetry (sign of parity) is enough to predict quantized transversal conductance for each spin.

Next, this situation is analyzed more formally. As described for the quantum Hall effect, the exact derivations can be lengthy and will be partly skipped as marked by (without derivation) or (without argument). The details can be calculated straightforwardly, but this is not required for a general understanding.

Firstly, we separate the two spin channels according to³⁰

$$\hat{H} = \begin{pmatrix} \hat{H}_{\uparrow}(\vec{k}) & 0 \\ 0 & \hat{H}_{\downarrow}(\vec{k}) \end{pmatrix}, \quad (5.64)$$

where $\hat{H}_{\uparrow}$ and $\hat{H}_{\downarrow}$ act on the two different spins and $\hat{H}_{\uparrow}(\vec{k}) = \hat{H}_{\downarrow}^*(-\vec{k})$ in order to respect time-reversal symmetry (TRS).³¹ Now we use a two band model consisting of the s-type and the p-type band for both spin channels and write the corresponding Hamiltonian in terms of Pauli matrices σ_i and the unity matrix $\mathbf{1}$.

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (5.65)$$

as

$$\hat{H}_{\uparrow}(\vec{k}) = \epsilon(\vec{k}) \cdot \mathbf{1} + \sum_{i=x,y,z} d_i(\vec{k}) \cdot \sigma_i, \quad (5.66)$$

where the second term describes the perturbations with prefactor functions $d_i(\vec{k}) : \mathbb{R}^3 \rightarrow \mathbb{R}$. The first two Pauli matrices cover the interactions between s-band and p-band. For the lowest order, we can choose its prefactor functions to be linear in $\vec{k}$, i.e., e.g., $d_x(\vec{k}) = A_1 \cdot k_x$ and $d_y(\vec{k}) = -A_1 k_y$ with $A_1 \in \mathbb{R}$ being a constant. These terms are required to guarantee the final band gap opening (transition from Fig. 5.16(a) to (b)). They can be justified by $\vec{k} \cdot \vec{p}$ theory. The final term $d_z \cdot \sigma_z$ describes the energy difference between the two bands and has to be even with respect to inversion, if the crystal is inversion symmetric. We choose it to be $d_z(\vec{k}) = M_1 - B_1(k_x^2 + k_y^2)$. The term $M_1 \in \mathbb{R}$ describes the strength of the band inversion at $\vec{k} = \vec{0}/m$ and $B_1 \in \mathbb{R}$ describes

30 Formally, the Hamiltonian describes four bands of different total angular momentum, described by the corresponding magnetic quantum number m_j , where the $m_j = 3/2$ and $m_j = 1/2$ bands form one sector of the block-diagonal shape of the Hamiltonian and the $m_j = -3/2$ and $m_j = -1/2$ states form the other one.

31 Formally, the zeros in the off-diagonal terms require time reversal symmetry and inversion symmetry of the crystal, i.e we do not apply a Rashba term, but only an atomic spin-orbit interaction as in eq. (4.44). Then the two states $|\vec{k}, \uparrow\rangle$ and $|\vec{k}, \downarrow\rangle$ have to be identical in energy, since time reversal requires $|\vec{k}, \uparrow\rangle \rightarrow |-\vec{k}, \downarrow\rangle$ and inversion requires $|\vec{k}\rangle \rightarrow |-\vec{k}\rangle$.

the curvature of the two bands with opposite sign for each band due to σ_z . Any other details of the bands are covered by $\epsilon(\vec{k})$ and are irrelevant for σ_{xy} .³²

In order to calculate σ_{xy} by the Kubo formula (eq. (5.34)), similar to the case of the magnetic Brillouin zone (Section 5.4.2), we have to respect the periodicity of the Brillouin zone, i.e., linear terms in (k_x, k_y) are replaced by sine functions and quadratic terms in (k_x, k_y) by cosine functions (eq. (6) in [124]). Then one can show that the vector $\vec{d}(\vec{k}) := (d_x(\vec{k}), d_y(\vec{k}), d_z(\vec{k}))$, respectively, its corresponding unit vector $\hat{d}(\vec{k}) = \vec{d}(\vec{k})/|\vec{d}(\vec{k})|$, completely determines the Hall conductivity σ_{xy} , according to (without derivation):³³

$$\sigma_{xy} = -\frac{e^2}{8\pi^2\hbar} \int \int_{\text{BZ}} \hat{d}(\vec{k}) \cdot \left(\frac{d\hat{d}(\vec{k})}{dk_x} \times \frac{d\hat{d}(\vec{k})}{dk_y} \right) dk_x dk_y \quad (5.67)$$

with BZ indicating that the integral covers the whole Brillouin zone. Figure 5.16(c) shows the vector $\vec{d}(\vec{k})$ close to $\vec{k} = \vec{0}/m$. The vector $\vec{d}(\vec{k})$ is basically the vector within the Bloch sphere of the two-state space spanned by the conduction and the valence band (Section 3.4) with $\vec{k}$ being the parameter controlling the direction of $\vec{d}$. Within the center, it points upwards, but depending on the relation between M_I and B_I it can point downwards at the rim of the BZ. The periodicity requires that $\vec{d}(\vec{k})$ can only point upwards or downwards at the rim, as one immediately recognizes if one writes down the correct cosine and sine functions. Thus, the number of half windings of the $\hat{d}(\vec{k})$ vector from the center to the rim of the BZ is necessarily an integer number.

The question if $\hat{d}(\vec{k})$ points in the opposite or in the same direction at the rim and at the center of the BZ (related to the number of half windings on its way) is the decisive topological property. This integer is called the $\mathbb{Z}_2$ index, since it can have only two values. It turns out that, if $0 < M_I < 4B_I$, $\vec{d}(\vec{k})$ points in opposite directions at the rim and at the center of the BZ. This is equivalent to the existence of a crossing point of the two bands in Fig. 5.16(a), before including the anticrossing terms $d_x(\vec{k})$ and $d_y(\vec{k})$. One can show that the integral in eq. (5.67), that is also called the Skymion number of a vector field,³⁴ probes exactly that property, i.e., the number of half windings of a vector-field. Thus, if the conduction and valence band anticross and, therefore, have a different character in the inner part than in the outer part of the BZ, the Hall conductivity σ_{xy} of a single spin channel is $\sigma_{xy,\uparrow} = 1 \cdot e^2/h$, while σ_{xy} of the other spin channel

32 Due to inversion symmetry, the diagonal terms within $\hat{H}_T(\vec{k})$ can only be even in $\vec{k}$ since $\langle \psi_i | \vec{k} | \psi_i \rangle = \langle \psi_i | -\vec{k} | \psi_i \rangle$ is required. One can show that terms different in k_x and k_y do only appear in higher orders of $\vec{k}$. The off-diagonal terms have to be odd in $\vec{k}$, since they mix an odd state (p-state) and an even state (s-state). The sign of the off-diagonal term can be rationalized by writing the orbital moments of the p-states as $|p_x + ip_y\rangle$ and $|(p_x - ip_y)\rangle$.

33 The calculation uses again the Kubo formula, but without calculating the Bloch functions explicitly. Thus, the integral multiplied by the leading term $\frac{1}{8\pi^2}$ is identical to the Chern number of the Hamiltonian $\hat{H}_T(\vec{k})$ [127].

34 if multiplied by $\frac{1}{8\pi^2}$.

is $\sigma_{xy,\downarrow} = -1 \cdot e^2/h$. The effect is dubbed the spin quantum Hall effect (SQHE) and materials that exhibit the SQHE are called 2D topological insulators (2DTI) exhibiting partially inverted bands around E_F .

Next, we use again the bulk-boundary correspondence, which by the requirement that the bulk is insulating, i.e., the Fermi level is located within a bulk band gap, implies conducting edge states of a 2DTI in order to realize a finite σ_{xy} . This is sketched in Fig. 5.16(d), where the inverted bands are inverted back towards the edge in order to match the topological vacuum condition. The edge state is helical, i.e., it runs clockwise for one spin channel and counterclockwise for the other spin channel. Otherwise, it could not realize the different sign of σ_{xy} of the two spin channels (see Fig. 5.11(d) in Section 5.3.2). Thus, the opposite helicity of the two spin edge channels is protected by the different sign of $\sigma_{xy,\uparrow}$ and $\sigma_{xy,\downarrow}$, and, hence, by time-reversal symmetry (eq. (5.64)). One has one ballistic edge channel for each spin on each side running in opposite directions (Fig. 5.16(e)), as required for $\sigma_{xy,\uparrow} = e^2/h$ for both current directions for one spin and $\sigma_{xy,\downarrow} = -e^2/h$ for both current directions for the other spin.³⁵

Each spin has one ballistic channel for each direction of current flow, but these channels are on opposite edges of the sample (Fig. 5.16(e)). Each channel contributes e^2/h to the longitudinal conductivity σ_{xx} (Section 5.3.2). Hence, one can verify the presence of the edge states by the quantized longitudinal conductance of $\sigma_{xx} = \sigma_{xx\uparrow} + \sigma_{xx\downarrow} = 2e^2/h$.³⁶

This fact has been used as the first hallmark for the existence of 2DTIs. Figure 5.16(f)–(g) shows the experiment. In Fig. 5.16(f), the used samples are sketched. They consist of a heterostructure (Section 1.3.1), where two CdTe layers enclose a HgTe layer, the latter being the 2DTI with a partially inverted band gap ($M_I \approx 0.2$ eV). The band inversion is induced by the spin-orbit interaction of the heavy elements Hg and Te, i.e., $M_I > 0$. So far, there is no material known where $M_I > 4B_I$. The band gap of CdTe is much larger (1.56 eV) and not inverted. Thus, 2D layers of HgTe should exhibit $\sigma_{xx} = 2e^2/h$. Indeed, this result has been found, if the HgTe layer is thick enough and an external gate voltage V_{Gate} is applied to tune E_F into the band gap of HgTe. The inset in Fig. 5.16(g) shows a measurement setup. The measured resistance curves,

35 An explicit derivation of the edge state using an open boundary and the Hamiltonian in eq. (5.66) is given in [128], there Chapter II b. For $A_I/B_I > 0$, it results in a function $\psi_i \propto (e^{-\lambda_{1x}} - e^{-\lambda_{2x}})$ with exponential decay lengths $\lambda_{1,2} = a_x \cdot \frac{A_I}{2B_I} \pm \sqrt{A_I^2/(4B_I^2) - M_I/B_I}$ into the bulk of the 2DTI, if $E(\vec{k}) = 0$ eV and $k_y = 0/m$. The variable a_x is the lattice constant in x direction.

36 One can additionally justify that backscattering between the two channels at the same edge is prohibited, such that the channel is indeed ballistic. This is the case for impurities that respect TRS. In order to backscatter, the spin has to be rotated by 180° to fit to the spin of the counter-propagating channel. If TRS is respected, the clockwise and the counterclockwise rotation of the spin must have identical probabilities, such that these two scattering paths interfere. In spin space the rotation by 180° changes the phase by 90° . Hence, the two possible spin rotations differ by a phase of 180° and, consequently, interfere destructively, i.e., the probability for backscattering is exactly zero ([128]; Chapter II.C.1).

labeled III and IV in Fig. 5.16(g), exhibit a gate voltage region where one finds rather exactly $R_{14,23} = h/(2e^2)$.³⁷ As expected the resistance decreases monotonically, if one moves E_F into the conduction band of HgTe ($(V_{\text{Gate}} - V_{\text{thr}}) > 0.3 \text{ V}$). The counter experiment is done for a smaller thickness of the HgTe layer. This induces an additional confinement energy to the bands that is positive for the s-type band (E1) and negative for the p-type band (H1) as expected from the curvature of the bands (different sign of the effective mass). Consequently, the bands can invert back to a normal insulator ($M_I < 0$) with increasing confinement energy, i.e., reduced thickness. Reducing the thickness to 5.5 nm (curve I), the resistance within the band gap indeed dramatically increases to 20 M Ω , which is the maximum resistance that could be detected in this particular experiment [125].

The image in Fig. 5.16(e), moreover, implies that the different spin components are separated at the entrance of the 2DTI. If the current runs from the lower left to the upper right, the spin-up electrons go exclusively to the upper edge, while the spin-down electrons go exclusively to the lower edge. Filtering out, e.g., only the current from the upper edge, reveals a perfect and, thus, very efficient spin filter. This effect has indeed been observed experimentally [129] and raises hopes for a reliable spin filter not requiring any magnetic fields.

Figure 5.16(h) and (i) show the edge channels directly for the $M_I > 0$ HgTe layer. A scanning SQUID³⁸ has been used to measure the local distribution of the current within the HgTe and, indeed, the current runs only along the edges, if E_F is within the band gap (h), while it runs across the whole layer, if E_F is within the conduction band (i).

Another topological transport effect in 2D is dubbed the quantum anomalous Hall effect (QAHE). The effect is very similar to the QSHE and is found in systems that are additionally magnetic, i.e., they break TRS. One can rationalize that the exchange coupling within a ferromagnet distorts the symmetry between the two spin channels such that $\hat{H}_\uparrow(\vec{k}) \neq \hat{H}_\downarrow^*(-\vec{k})$. Hence, the Hamiltonian $\hat{H}_\uparrow(\vec{k})$ can have a partially inverted band gap implying $\sigma_{xy\uparrow} = e^2/h$, while $\hat{H}_\downarrow(\vec{k})$ has a trivial band order with $\sigma_{xy\downarrow} = 0$.³⁹ Then, the edge states of only one spin channel appear at E_F . Thus, one gets $\sigma_{xy} = e^2/h$ that is not compensated by the other spin channel, i.e., a real transversal conductivity. The first successful measurement of the QAHE, i.e., a detection of $\sigma_{xy} = e^2/h$, if E_F is within a band gap, was realized in the 2D system $\text{Cr}_{0.15}\text{Bi}_{0.18}\text{Sb}_{1.67}\text{Te}_3$ [131]. The BiSbTe al-

³⁷ The origin of the visible deviation from $R_{14,23} = h/2e^2$ is still not clear.

³⁸ A SQUID (superconducting quantum interference device) described, e.g., in detail in [130], is one of the most sensitive magnetometers detecting $\vec{B}$ -fields down to 10^{-18} T (see also Section 3.3.4ff). For this particular experiment, a SQUID is moved with piezomotors across the sample with nm precision. It detects the magnetic stray field of the current. One can reconvert the resulting $\vec{B}$ -field map into a current map, that is displayed in Fig. 5.16(h) and (i).

³⁹ Formally, the $\vec{k}$ -independent exchange interaction introduces a different M_I into $\hat{H}_\uparrow$ and $\hat{H}_\downarrow$ of eq. (5.64) such that, e.g., $M_I < 4B_I$ for one spin channel and $M_I > 4B_I$ for the other spin channel. The two spin channels, thus, exhibit a topological and a normal insulator, respectively.

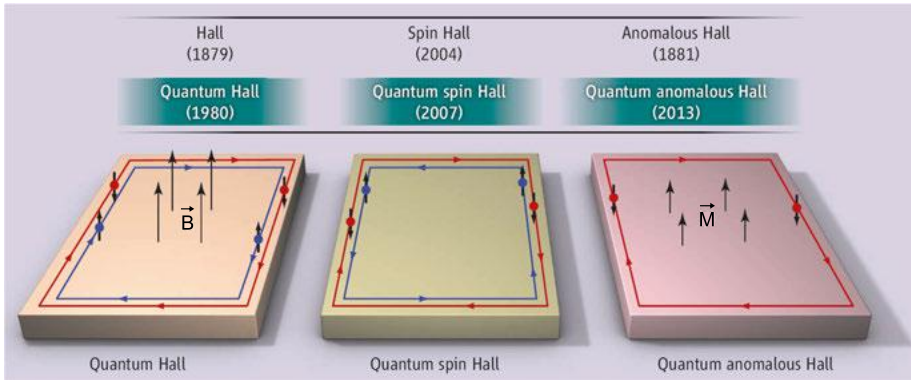


Fig. 5.17: Schematic of the three topological states corresponding to QHE, QSHE and QAHE. The edge states are marked in red and blue with corresponding propagation helicities indicated by in-plane arrows and spin directions of the moving electrons as out-of-plane arrows at the blue and red dots. $\vec{B}$ labels an external magnetic field and $\vec{M}$ an internal magnetization of the sample. The numbers in brackets above the sketches are years of experimental discovery of the corresponding effect. The upper row of terms names the classical and, thus, non-quantized effects with the year of their experimental discovery in brackets. After [132].

loy is a semiconductor again made of heavy elements. It provides a partially inverted band gap ($M_I > 0$) and the Cr is doped into the system in order to make it ferromagnetic with a magnetization $\vec{M}$ pointing out-of-plane. The additional spin dependent exchange term influences $\hat{H}_\downarrow(\vec{k})$ oppositely to $\hat{H}_\uparrow(\vec{k})$.

Figure 5.17 shows a sketch of the three different topological effects discussed so far with the corresponding edge channels, its propagation helicity and its electron spin. These samples cover only two entrances in the topological table of Fig. 5.14, namely the $d = 2$ entrances of the classes A and AII. The QHE and the QAHE have both, neither time-reversal-symmetry, nor particle-hole-symmetry, nor chiral sublattice symmetry belonging to A, i.e., the conduction electrons do not care, if their spin splitting results from an external $\vec{B}$ field or from an exchange coupling.⁴⁰ Thus, theoretically, they belong to the same symmetry class, but experimentally it is a big advantage that one can use an intrinsic sample property instead of an external $\vec{B}$ field in order to get a quantized conductance. Some researchers indeed believe that the QAHE might replace the QHE as a resistance standard in the near future.

The robustness of the QSHE and the QAHE with respect to disorder can be shown by using similar arguments as for the quantum Hall effect via the perturbative introduction of disorder. For example, one could again use a description of the transport

40 There are other effects related to electron-electron interaction, that are different for exchange interaction and external $\vec{B}$ field, but the periodic table in Fig. 5.14 ignores the electron-electron interaction for the sake of simplicity. One can show that the QSHE and the QAHE are robust even with respect to electron-electron interaction as long as the interaction is not too strong with respect to the kinetic energy of the electrons ([128]): Chapter II.C.2).

coefficients via boundary phases. More heuristically, one often argues that changing the topology requires either the inverted or the non-inverted regions of the band structure (Fig. 5.16(b)) to exchange valence and conduction band, which usually requires a strong perturbation of the Hamiltonian.

5.4.6 3D topological insulators

5.4.6.1 General analysis

Next, one can ask, if the concept of topological numbers is transferable to 3D materials. This is indeed the case as visible in the periodic table in Fig. 5.14, but the corresponding derivation is more difficult than in the former cases. At the end, however, it again boils down to the question if band orders are inverted in some parts of the Brillouin zone?

Firstly, it is intuitively clear, that we have, at least, three different $\mathbb{Z}_2$ indices for the three linearly independent 2D planes in $\vec{k}$ space, e.g., the three planes perpendicular to the k_z , k_x , and k_y direction, that are independent by their linearly independent normal vectors. We can take, e.g., the planes with $k_z = \pi/a$, $k_x = \pi/a$, or $k_y = \pi/a$ for a cubic lattice with lattice constant a , i.e., the edge planes of the Brillouin zone. Each of these planes can be described by a $\mathbb{Z}_2$ index (Section 5.4.5). The $\mathbb{Z}_2$ index of $k_z = \pi/a$ and $k_z = -\pi/a$ are identical, since these are identical planes due to the lattice periodicity. This also applies for the planes at $k_x = \pi/a$ and $k_x = -\pi/a$ as well as for the planes at $k_y = \pi/a$ and $k_y = -\pi/a$. If the bands, additionally, do not touch on the way from $k_z = \pi/a$ to $k_z = -\pi/a$, all planes perpendicular to k_z have the identical $\mathbb{Z}_2$ index, since a change of $\mathbb{Z}_2$ requires a reinversion of the bands, i.e., a closing of the band gap. In this case, the 3D crystal can be regarded as stacks of 2DTIs, where their periodic distance induces the additional Brillouin zone direction k_z . The corresponding $\mathbb{Z}_2$ index marks, as in 2D, the SQHE. Hence, $\sigma_{xy\uparrow}$ and $\sigma_{xy\downarrow}$ are still quantized in each of the layers. The spin polarized 2D $\sigma_{xy\uparrow}$ is a multiple of e^2/h , which depends on the number of layers. For the 3D Hall conductivity of each spin channel, one gets $\sigma_{xy\uparrow} = j_x/E_y = \frac{e^2}{h \cdot a_z}$ with the lattice constant a_z in the z direction by multiplying e^2/h with the density of layers along the z direction. This provides the required different unit S/m reflecting that the current density in 3D is given in A/m², while in 2D, it is given in A/m. Thus, in principle, a 3D material consisting of stacked 2DTIs can also be used as an effective spin filter. The same analysis can be provided for the planes perpendicular to the x and y directions provided the bands around E_F do not (anti-)cross anywhere within the Brillouin zone.

The topological classification of 3D materials, is, however more complex, since, firstly, band crossings between parallel $\vec{k}$ planes in the Brillouin zone are not prohibited and, secondly, since there is an additional topological index combining the linearly independent planes, that turns out to be the most robust one. Thus, 3D materials exhibit four topological indices describing transport properties: the three indices

describing the $\mathbb{Z}_2$ topology of the linearly independent planes and a general one describing the 3D crystal as a whole. To tackle this in more detail, a sketch of the general derivation of these indices will be firstly given. Afterwards, we will discuss an example Hamiltonian related to a real material that exhibits robust topological properties.

The sketch starts with a 2D material, before generalizing to 3D. Instead of calculating σ_{xy} in 2D as in the previous section, we ask directly, if edge states at E_F are necessarily present? Therefore, we regard a ribbon of an arbitrary crystalline material with finite width in x direction and infinite length in y direction. The question is, if one can transfer a spin from one edge to the other without adding energy? If this is possible by changing the Hamiltonian adiabatically, i.e., by avoiding excitations, on a loop, i.e., by ending with the same Hamiltonian as at the beginning, there is only one solution if the bulk is insulating: there must be edge states at E_F . These edge states can be differently occupied after the process without energy change. More concretely, a spin state at the Fermi level from the left edge of the ribbon has to be emptied in favor of the same spin state at the right edge again being at the Fermi level. If spin is moved without charge movement, the opposite spin has to be moved from the right edge to the left edge of the ribbon.

This means, we can prove the presence of edge states at E_F , if we find an adiabatic loop of the Hamiltonian, that pumps spins from left to right.

The following derivation shows that the possible spin transfer between the edges is a topological property of the bulk band structure of the 2D material. Thus, there is an index that tells us, if such a spin transfer is happening or not, and, thus, if edge states at E_F are necessarily present. *The derivation is quite intricate and does not immediately lead to a transparent understanding. Thus, it is not required to be read in order to understand the subsequent parts. It is, however, provided for more interested students with the clear recommendation to read it only after reading the summary of the calculation given after the bold text following eq. (5.100). Then, you can return to this point in order to follow the details of the calculation.*

The parameter of the Hamiltonian, that is tuned adiabatically in a loop, is k_y . It is tuned across the whole Brillouin zone, such that the starting point is identical to the end point.⁴¹ We concentrate on rectangular lattices for the sake of simplicity, but the arguments can be transferred to other type of lattices, too.

The 1D Brillouin zone along k_y has two time reversal invariant points called TRIMs (time reversal invariant momenta), where $-k = k$. They are at the center of the Brillouin zone, where $k = 0/m$, and at the boundary, where the periodicity requires $k = -k$. In 1D, the states at these points in k space have to be twofold degenerate according to Kramers theorem.⁴² In between, there is no requirement for such a spin degeneracy. In 2D, the degenerate states at the two lines in $\vec{k}$ space, where $k_y = -k_y$ can have differ-

⁴¹ We later show how that this can be realized within a physical system (Fig. 5.18(c) and (d)).

⁴² Time reversal inverts the spin and k . Hence, if $k = -k$, we have $E(k, \uparrow) \xrightarrow{\text{TRS}} E(-k, \downarrow) \underset{k=-k}{=} E(k, \downarrow)$.

ent k_x values. But they are still forming pairs of states with the same k_y and the same energy.

We are interested in whether a spin is transported from one edge to the other in x direction, if we move k_y from the center ($k_y = 0/m$) to the boundary ($k_y = \pi/a_y$) of the Brillouin zone.

For the calculation, we firstly return to an infinite 2D system. We start with calculating the movement of charge instead of spin, keeping in mind that spin movement is equivalent to movement of charge with different spins in opposite directions.⁴³

Therefore, we define Wannier functions for each occupied band with band index n and each lattice position R_x in the x direction, such that we can track the position of occupied states,⁴⁴ i.e.,

$$|R_x, n\rangle_{k_y} = \frac{1}{2\pi} \int_{-\pi/a_x}^{\pi/a_x} e^{ik_x(R_x-x)} |u_{\vec{k}n}(\vec{x})\rangle dk_x \quad (5.68)$$

with lattice constant a_x in the x direction and $\vec{k} = (k_x, k_y)$. By definition, the Wannier functions can look different for each k_y . One can define a charge polarization in the x direction for each k_y referring to the state at $R_x = 0$ m:

$$P_x(k_y) = \sum_n \langle 0, n | x | 0, n \rangle_{k_y} . \quad (5.69)$$

This describes how far the center of mass of the charge at this k_y value is shifted with respect to $x = 0$ m. This charge polarization can be rewritten using $\int e^{-ik_1x} \cdot e^{ik_2x} dx \propto \delta_{k_1k_2}$ with $\delta_{k_1k_2}$ being the Kronecker delta and the operator replacement $\hat{x} \rightarrow -i \cdot \frac{d}{dk_x}$.⁴⁵

$$P_x(k_y) = \frac{-1}{2\pi} \oint i \sum_n \left\langle u_{\vec{k}n} \left| \frac{d}{dk_x} \right| u_{\vec{k}n} \right\rangle dk_x = \frac{-1}{2\pi} \int_{-\pi/a_x}^{\pi/a_x} i \sum_n \left\langle u_{\vec{k}n} \left| \frac{d}{dk_x} \right| u_{\vec{k}n} \right\rangle dk_x . \quad (5.70)$$

Here $\oint$ is a closed integral due the periodic boundary conditions along k_x , i.e., $\pi/a_x = -\pi/a_x$. Next, we write the difference of the charge polarization at $k_y = 0/m$ and $k_y = \pi/a_y$ with a_y being the lattice vector in the y direction.

$$\begin{aligned} P_x(k_y = \pi/a_y) - P_x(k_y = 0) \\ = \frac{-1}{2\pi} \left(\oint_{\pi/a_y} i \sum_n \left\langle u_{\vec{k}n} \left| \frac{d}{dk_x} \right| u_{\vec{k}n} \right\rangle dk_x - \oint_0 i \sum_n \left\langle u_{\vec{k}n} \left| \frac{d}{dk_x} \right| u_{\vec{k}n} \right\rangle dk_x \right) . \end{aligned} \quad (5.71)$$

⁴³ The derivation can be found in [133].

⁴⁴ Wannier functions are used, e.g., as the starting point of tight binding calculations of periodic lattices (Section 4.5). They are centered at the lattice position R_x by the constructive overlap of waves with different wavelength at this position.

⁴⁵ Small warning: Calculation of all steps in detail can be tedious.

This describes if charge is moved for the Wannier state at $R_x = 0$ m on the way from $k_y = 0/m$ to $k_y = \pi/a_y$. Since R_x is chosen arbitrarily, the result is identical for every lattice position, which are connected to each other by translational symmetry.⁴⁶

Having the expression for the charge polarization, we can now turn to the so-called time reversal polarization. Time reversal symmetry implies that each wave function has a partner of opposite $\vec{k}$ and opposite spin with the same energy, the so-called Kramers partner. We track the development of the charge polarization of such a degenerate Kramers pair relative to each other on the way from $k_y = 0/m$ to $k_y = \pi/a_y$. Thus, we divide the Bloch states at $k_y = 0/m$ and at $k_y = \pi/a_y$ in states I and II building Kramers pairs each, i.e., having opposite k_x and opposite spin.⁴⁷ If the two sectors I and II move in opposite x directions on its way from $k_y = 0/m$ to $k_y = \pi/a_y$, the charge polarization (effective position of the charge) does not change, but the spin polarization (effective position of the spin) does change, eventually implying movement of a spin from one edge to the other.

For completeness, we write the time reversal operator $\widehat{\Theta}$, connecting the partners of a Kramers pair, explicitly:

$$\widehat{\Theta} = e^{i\pi\widehat{S}_y/\hbar}\widehat{K} \quad (5.72)$$

with $\widehat{K}$ being the complex conjugation and $\widehat{S}_y$ being the spin operator. Importantly rotating the spin from $1/2$ to $-1/2$ changes the phase factor only by $\pi/2$. Consequently, rotating the spin from $1/2$ to $-1/2$ and back to $1/2$ rotates by π , such that $\widehat{\Theta}^2 = -1$.⁴⁸ The time reversal symmetry requires as a minimal condition for the pairs at $k_y = 0/m$ or $k_y = \pi/a_y$, i.e., on the time reversal invariant lines, where $\widehat{\Theta}$ does not change k_y :

$$\begin{aligned} |u_{-k_x n}^I\rangle &= e^{i\theta_{k_x n}}\widehat{\Theta}|u_{k_x n}^{II}\rangle \\ |u_{-k_x n}^{II}\rangle &= -e^{i\theta_{-k_x n}}\widehat{\Theta}|u_{k_x n}^I\rangle. \end{aligned} \quad (5.73)$$

The consistency of these equations can be verified by the replacing $-k_x \leftrightarrow k_x$ in the second line and inserting the result into the first line. Importantly, the description includes a possible arbitrary phase $\theta_{k_x n}$, that does not influence the Kramers degeneracy, and will become decisive in the following.

With these relations, we can calculate the partial charge polarization for the sectors I and II. Starting with sector I, one defines:

$$A_{k_y}^I(k_x) := \frac{1}{i} \sum_n \left\langle u_{kn}^I \left| \frac{d}{dk_x} \right| u_{kn}^I \right\rangle \quad (5.74)$$

46 Formally, eq. (5.71) describes a boundary integral of the k space torus being closed in the periodic k_x direction (Fig. 5.13(c)). The integrations along k_y at $k_x = \pi/a_x$ and $k_x = -\pi/a_x$, that are required to form a closed path, follow the identical path in inverse directions due to periodicity and, thus, the two integrals cancel each other. The integral along a closed line could be the starting point to apply Stokes theorem, but we will not follow this path.

47 At these points $k_y = -k_y$, trivially for $k_y = 0/m$ and due to the lattice periodicity for $k_y = \pi/a_y$.

48 This observation is at the heart of the fermionic description of electrons.

and analogously for $A_{k_y}^{\text{II}}(k_x)$. Notice, that the definition is very similar to the sum of the Berry connections $A_{\text{Berry},n}(\vec{k})$ (eq. (5.44)) at $\vec{k} = (k_x, k_y)$ of all occupied bands n , but it is calculated separately for each partner sector of the Kramers pairs. With this definition, we get from eq. (5.70):

$$\begin{aligned} P_x^{\text{I}}(k_y) &= \frac{1}{2\pi} \int_{-\pi/a_x}^{\pi/a_x} A_{k_y}^{\text{I}}(k_x) dk_x = \frac{1}{2\pi} \int_0^{\pi/a_x} A_{k_y}^{\text{I}}(k_x) + A_{k_y}^{\text{I}}(-k_x) dk_x \\ P_x^{\text{II}}(k_y) &= \frac{1}{2\pi} \int_{-\pi/a_x}^{\pi/a_x} A_{k_y}^{\text{II}}(k_x) dk_x = \frac{1}{2\pi} \int_0^{\pi/a_x} A_{k_y}^{\text{II}}(k_x) + A_{k_y}^{\text{II}}(-k_x) dk_x. \end{aligned} \quad (5.75)$$

Next, we use $A_{k_y}^{\text{I}}(-k_x) = A_{k_y}^{\text{II}}(k_x) + \sum_n \frac{d\theta_{k_x n}}{dk_x}$, that can be shown by inserting eq. (5.73) into eq. (5.74).⁴⁹ Defining $A_{k_y}(k_x) := A_{k_y}^{\text{I}}(k_x) + A_{k_y}^{\text{II}}(k_x)$, we get:

$$\begin{aligned} P_x^{\text{I}}(k_y) &= \frac{1}{2\pi} \left(\int_0^{\pi/a_x} A_{k_y}(k_x) dk_x + \sum_n \int_0^{\pi/a_x} \frac{d\theta_{k_x n}}{dk_x} dk_x \right) \\ &= \frac{1}{2\pi} \left(\int_0^{\pi/a_x} A_{k_y}(k_x) dk_x + \sum_n (\theta_{k_x=\pi/a_x, n} - \theta_{k_x=0, n}) \right). \end{aligned} \quad (5.76)$$

We have chosen $\theta_{k_x n}$ such, that it changes continuously across the Brillouin zone. Importantly, the two remaining phases $\theta_{k_x n}$ can be different, since the integral covers only half the Brillouin zone in x direction. Moreover, the two phases appear with opposite sign in $P_x^{\text{II}}(k_y)$, that can be shown by repeating the calculation for $P_x^{\text{II}}(k_y)$. The difference between the two partial polarizations $P_x^{\Delta}(k_y) = P_x^{\text{I}}(k_y) - P_x^{\text{II}}(k_y)$ can be calculated more elegantly by realizing that $P_x(k_y) = P_x^{\text{I}}(k_y) + P_x^{\text{II}}(k_y)$ (eq. (5.70)) leading to:

$$\begin{aligned} P_x^{\Delta}(k_y) &= 2 \cdot P_x^{\text{I}}(k_y) - P_x(k_y) \\ &= \frac{2}{2\pi} \left(\int_0^{\pi/a_x} A_{k_y}(k_x) dk_x + \sum_n (\theta_{k_x=\pi/a_x, n} - \theta_{k_x=0, n}) \right) \\ &\quad - \frac{1}{2\pi} \int_{-\pi/a_x}^{\pi/a_x} A_{k_y}(k_x) dk_x \\ &= \frac{1}{2\pi} \left(\int_0^{\pi/a_x} A_{k_y}(k_x) dk_x - \int_{-\pi/a_x}^0 A_{k_y}(k_x) dk_x + 2 \sum_n (\theta_{k_x=\pi/a_x, n} - \theta_{k_x=0, n}) \right). \end{aligned} \quad (5.77)$$

Obviously, the phase factors survive the final subtraction $P_x^{\text{I}}(k_y) - P_x^{\text{II}}(k_y)$.

⁴⁹ We employ the sum rule for the derivative d/dk_x , applied once to the phase factor $e^{i\theta_{k_x n}}$ and once to $\widehat{\theta} u_{k_x n}^{\text{II}}$, as well as $\langle \widehat{\theta} u_{k_x n}^{\text{II}} | \frac{d}{dk_x} | \widehat{\theta} u_{k_x n}^{\text{II}} \rangle = -\langle \widehat{\theta} u_{k_x n}^{\text{II}} | \widehat{\theta} \frac{d}{dk_x} | u_{k_x n}^{\text{II}} \rangle = \langle u_{k_x n}^{\text{II}} | \frac{d}{dk_x} | u_{k_x n}^{\text{II}} \rangle$.

In the following, we clarify that $P_x^\Delta(k_y)$, i.e., the difference in charge polarization between the two time-reversal partners, is either zero or one. Thus, it can topologically classify the time reversal invariant lines at $k_y = 0/m$ and at $k_y = \pi/a_y$.

Therefore, we define the matrix $\mathbf{w}(k_x)$ with elements

$$w_{lm}(k_x) := \langle u_{-k_x l} | \widehat{\Theta} | u_{k_x m} \rangle, \quad (5.78)$$

where l and m cover the occupied states from both sectors I and II reading $l, m \in (1\text{I}, 1\text{II}, 2\text{I}, 2\text{II}, 3\text{I}, 3\text{II}, \dots)$ with $1, 2, 3, \dots$ describing the different bands. Such a matrix exists for each k_y , which we drop as an index for the sake of simplicity. The matrix consists of antisymmetric blocks of the form:

$$\begin{pmatrix} 0 & e^{i\theta_{k_x n}} \\ -e^{i\theta_{-k_x n}} & 0 \end{pmatrix} \quad (5.79)$$

for the elements belonging to the indices

$$((l, m)) = \begin{pmatrix} (2n-1, 2n-1) & (2n-1, 2n) \\ (2n, 2n-1) & (2n, 2n) \end{pmatrix}. \quad (5.80)$$

This can be verified by inserting eq. (5.73) into eq. (5.78) and using $\widehat{\Theta}^2 = -1$ as well as the orthogonality of the wave functions. The matrix does only contain the phase factors of Kramers pairs. At $k_x = 0/m$ and $k_x = \pi/a_x$, where $k_x = -k_x$, the matrix is antisymmetric. At these points, one can define the so-called Pfaffian of the Matrix $\text{Pf}(\mathbf{w})$, that exists for any antisymmetric $2N \times 2N$ matrix and is defined as

$$\text{Pf}(\mathbf{w}) := \frac{1}{2^N N!} \sum_{\tilde{\sigma} \in S_{2N}} \text{sgn}(\tilde{\sigma}) \prod_{n=1}^N w_{\tilde{\sigma}(2n-1)\tilde{\sigma}(2n)} \quad (5.81)$$

with $\tilde{\sigma}$ marking all possible permutations between the $2N$ elements and $\text{sgn}(\tilde{\sigma})$ being -1 ($+1$) for an odd (even) number of permutations. Thus, the product without permutation, i.e., with elements $w_{(2n-1)(2n)}$, is the line above the diagonal of the matrix tracing any second element (eq. (5.79)). One can show that this is the only summand of the Pfaffian, if the only non-zero elements of the matrix are in the lines above and below the diagonal.⁵⁰ Thus, we get

$$\text{Pf}(\mathbf{w}(k_x = 0)) = \prod_{n=1}^N \exp(i\theta_{0n}) \quad (5.82)$$

$$\text{Pf}(\mathbf{w}(k_x = \pi/a_x)) = \prod_{n=1}^N \exp(i\theta_{\pi/a_x n}). \quad (5.83)$$

⁵⁰ See Wikipedia: Pfaffian.

Notice that both Pfaffians are complex numbers with absolute value $|\text{Pf}(\mathbf{w})| = 1$. Inserting the Pfaffians into eq. (5.76) results in

$$P_x^I(k_y) = \frac{1}{2\pi} \left(\int_0^{\pi/a_x} A(k_x) dk_x + i \cdot \ln \left(\frac{\text{Pf}(\mathbf{w}(k_x = \pi/a_x))}{\text{Pf}(\mathbf{w}(k_x = 0))} \right) \right) \quad (5.84)$$

and, hence, we find (eq. (5.77)):

$$P_x^A(k_y) = \frac{1}{2\pi} \left(\int_0^{\pi/a_x} A(k_x) dk_x - \int_{-\pi/a_x}^0 A(k_x) dk_x + 2i \cdot \ln \left(\frac{\text{Pf}(\mathbf{w}(k_x = \pi/a_x))}{\text{Pf}(\mathbf{w}(k_x = 0))} \right) \right). \quad (5.85)$$

One can show that $\text{Pf}(\mathbf{w})^2 = \text{Det}(\mathbf{w})$ for any matrix, where the Pfaffian is defined.⁵¹ This leaves two possible signs of the Pfaffian with respect to the determinant. These different signs turn out to be the decisive distinction between the two topological classes in 2D: trivial insulator and topological insulator.

The remaining task is now to show that the first two terms are related to the determinant of the matrix $\mathbf{w}$. Therefore, we firstly rewrite the first two terms of eq. (5.85) in terms of the matrix $\mathbf{w}$ (eq. (5.78)). It is obvious from the definition of $A(k_x)$ (eq. (5.74)) that we need the derivative of w :

$$\frac{dw_{ml}}{dk_x} = \left\langle \frac{du_{-k_x m}}{dk_x} \left| \widehat{\bar{\theta}} \right| u_{k_x l} \right\rangle - \left\langle u_{-k_x m} \left| \widehat{\bar{\theta}} \right| \frac{du_{k_x l}}{dk_x} \right\rangle. \quad (5.86)$$

The minus sign comes from the fact that $\widehat{\bar{\theta}}$ and $\frac{d}{dk_x}$ anti-commute, since $\frac{d}{dk_x}$ corresponds to $i \cdot \hat{x}$ in real space and $\widehat{\bar{\theta}}$ includes complex conjugation K or, since time reversal means that the momentum k must point in the opposite direction. For the second term on the right, one can get rid of the $\langle u_{-k_x m} |$ by multiplying with $\mathbf{w}^\dagger$ according to:

$$\begin{aligned} \left(\mathbf{w}^\dagger \cdot \frac{d\mathbf{w}}{dk_x} \right)_{ol,2} &= \sum_m \langle u_{k_x o} | \widehat{\bar{\theta}}^\dagger | u_{-k_x m} \rangle \left\langle u_{-k_x m} \left| \widehat{\bar{\theta}} \right| \frac{du_{k_x l}}{dk_x} \right\rangle \\ &= \left\langle u_{k_x o} \left| \widehat{\bar{\theta}}^\dagger \widehat{\bar{\theta}} \right| \frac{du_{k_x l}}{dk_x} \right\rangle = \left\langle u_{k_x o} \left| \frac{du_{k_x l}}{dk_x} \right\rangle, \end{aligned} \quad (5.87)$$

where the 2 in the index on the left marks that only the second summand from eq. (5.86) is displayed. We used again the completeness relation $\sum_m |u_{-k_x m}\rangle \langle u_{-k_x m}| = \mathbf{1}$ as well as $\widehat{\bar{\theta}}^\dagger = -\widehat{\bar{\theta}}$ and $\widehat{\bar{\theta}}^2 = -1$.⁵² This already resembles the definition of $A(k_x)$ in eq. (5.74). In more detail, we have to take the trace of the matrix $\mathbf{w}^\dagger \cdot \frac{d\mathbf{w}}{dk_x}$ multiplied with the imaginary $1/i$.

$$A(k_x) = \frac{1}{i} \text{Tr} \left(\mathbf{w}^\dagger \cdot \frac{d\mathbf{w}}{dk_x} \right)_2. \quad (5.88)$$

⁵¹ see Wikipedia: Pfaffian. Recall that the determinant of a $N \times N$ matrix is given by $\text{Det}(\mathbf{w}) = \sum_{\bar{\sigma}} \text{sgn}(\bar{\sigma}) \prod_{n=1}^N w_{n\bar{\sigma}(n)}$.

⁵² The first equation becomes obvious considering that complex conjugation gets reversed by $\dagger$ and also the spin operator gets reversed (see eq. (5.72)).

Using the trace directly, we can get also the first term on the right hand side of eq. (5.86) into this form as shown in the following. Therefore, we use $\frac{d}{dk_x} \langle u_{k_x m} | \widehat{\Theta} | u_{k_x m} \rangle = 0$, that is an obvious consequence of orthogonality, and the fact that $\widehat{\Theta} | u_{k_x m} \rangle$ is also an eigenfunction due to time reversal symmetry. Moreover, we apply the product rule of derivatives, the clever inclusion of a $\mathbf{1}$, and the same properties of $\widehat{\Theta}$ as above. We obtain

$$\begin{aligned}
 \text{Tr} \left(\mathbf{w}^\dagger \cdot \frac{d\mathbf{w}}{dk_x} \right)_1 &= \sum_{n,m} \left\langle u_{k_x n} | \widehat{\Theta}^\dagger | u_{-k_x m} \right\rangle \left\langle \frac{du_{-k_x m}}{dk_x} | \widehat{\Theta} | u_{k_x n} \right\rangle \\
 &= \sum_{l,n,m} \left\langle u_{k_x n} | \widehat{\Theta}^\dagger | u_{-k_x m} \right\rangle \left\langle \frac{du_{-k_x m}}{dk_x} | u_{-k_x l} \right\rangle \left\langle u_{-k_x l} | \widehat{\Theta} | u_{k_x n} \right\rangle \\
 &= \sum_{l,n,m} \left\langle u_{-k_x l} | \widehat{\Theta} | u_{k_x n} \right\rangle \left\langle u_{k_x n} | \widehat{\Theta}^\dagger | u_{-k_x m} \right\rangle \left\langle \frac{du_{-k_x m}}{dk_x} | u_{-k_x l} \right\rangle \\
 &= \sum_{l,m} \left\langle u_{-k_x l} | \widehat{\Theta} \widehat{\Theta}^\dagger | u_{-k_x m} \right\rangle \left\langle \frac{du_{-k_x m}}{dk_x} | u_{-k_x l} \right\rangle \\
 &= \sum_{l,m} \langle u_{-k_x l} | u_{-k_x m} \rangle \left\langle \frac{du_{-k_x m}}{dk_x} | u_{-k_x l} \right\rangle \\
 &= - \sum_{l,m} \langle u_{-k_x l} | u_{-k_x m} \rangle \left\langle u_{-k_x m} | \frac{du_{-k_x l}}{dk_x} \right\rangle \\
 &= - \sum_l \left\langle u_{-k_x l} | \frac{du_{-k_x l}}{dk_x} \right\rangle = -i \cdot A(-k_x) .
 \end{aligned}$$

The index 1 means that we use only the first term of eq. (5.86) for the derivative. Hence, we have, using both terms of eq. (5.86):

$$\text{Tr} \left(\mathbf{w}^\dagger \cdot \frac{d\mathbf{w}}{dk_x} \right) = i(A(k_x) - A(-k_x)) . \quad (5.89)$$

Thus, we reached our goal to replace the first two terms of eq. (5.85) by an expression of $\mathbf{w}$, that now reads

$$P_x^\Delta = \frac{-i}{2\pi} \left(\int_0^{\pi/a_x} \text{Tr} \left(\mathbf{w}^\dagger \cdot \frac{d\mathbf{w}}{dk_x} \right) dk_x - 2 \cdot \ln \left(\frac{\text{Pf}(\mathbf{w}(k_x = \pi/a_x))}{\text{Pf}(\mathbf{w}(k_x = 0))} \right) \right) . \quad (5.90)$$

The next task is to rewrite the new term as an expression of determinants. Therefore, we need a few rules for determinants and for the derivative of determinants. The most simple ones are

$$\text{Det}(\mathbf{A} \cdot \mathbf{B}) = \text{Det}(\mathbf{A}) \cdot \text{Det}(\mathbf{B}) \quad (5.91)$$

$$\text{Det}(\mathbf{1} + \tilde{\varepsilon} \cdot \mathbf{B}) = 1 + \tilde{\varepsilon} \cdot \text{Tr}(\mathbf{B}) + \tilde{\varepsilon}^2 \dots$$

The latter is valid for small $\tilde{\varepsilon}$. Now we derive the derivative of a determinant explicitly:

$$\frac{d \text{Det}(\mathbf{w}(k_x))}{dk_x} = \lim_{\tilde{\varepsilon} \rightarrow 0} \frac{\text{Det}(\mathbf{w}(k_x + \tilde{\varepsilon})) - \text{Det}(\mathbf{w}(k_x))}{\tilde{\varepsilon}} . \quad (5.92)$$

With $\mathbf{w}(k_x + \tilde{\epsilon}) = \mathbf{w}(k_x) + \tilde{\epsilon} \cdot \frac{d\mathbf{w}(k_x)}{dk_x}$ and $\mathbf{B} := \frac{d\mathbf{w}(k_x)}{dk_x}$ one gets

$$\begin{aligned} \text{Det}(\mathbf{w}(k_x) + \tilde{\epsilon} \cdot \mathbf{B}) &= \text{Det}(\mathbf{w}(k_x) \mathbf{w}(k_x)^{-1} (\mathbf{w}(k_x) + \tilde{\epsilon} \cdot \mathbf{B})) \\ &= \text{Det}(\mathbf{w}(k_x)) \text{Det}(\mathbf{w}(k_x)^{-1} (\mathbf{w}(k_x) + \tilde{\epsilon} \cdot \mathbf{B})) \\ &= \text{Det}(\mathbf{w}(k_x)) \text{Det}(\mathbf{1} + \tilde{\epsilon} \cdot \mathbf{w}(k_x)^{-1} \cdot \mathbf{B}) \\ &= \text{Det}(\mathbf{w}(k_x)) \cdot (1 + \tilde{\epsilon} \cdot \text{Tr}(\mathbf{w}(k_x)^{-1} \cdot \mathbf{B})) \end{aligned}$$

with $\mathbf{1}$ being the identity matrix. This leaves us with

$$\frac{d \text{Det}(\mathbf{w}(k_x))}{dk_x} = \text{Det}(\mathbf{w}(k_x)) \cdot \text{Tr} \left(\mathbf{w}(k_x)^{-1} \cdot \frac{d\mathbf{w}(k_x)}{dk_x} \right). \quad (5.93)$$

Finally, we get rid of the first determinant by the derivative of the natural logarithm:

$$\begin{aligned} \frac{d \ln(\text{Det}(\mathbf{w}(k_x)))}{dk_x} &= \frac{1}{\text{Det}(\mathbf{w}(k_x))} \cdot \text{Det}(\mathbf{w}(k_x)) \cdot \text{Tr} \left(\mathbf{w}(k_x)^{-1} \cdot \frac{d\mathbf{w}(k_x)}{dk_x} \right) \\ &= \text{Tr} \left(\mathbf{w}(k_x)^{-1} \cdot \frac{d\mathbf{w}(k_x)}{dk_x} \right). \end{aligned} \quad (5.94)$$

Using $\mathbf{w}(k_x)^{-1} = \mathbf{w}(k_x)^\dagger$, that can be verified by inserting eq. (5.79), we see that we can replace the trace term in eq. (5.90) by a determinant and can perform the integration such that

$$P_x^\Delta(k_y) = \frac{-i}{2\pi} \left[\int_0^{\pi/a_x} \frac{d \ln(\text{Det}(\mathbf{w}(k_x)))}{dk_x} dk_x - 2 \cdot \ln \left(\frac{\text{Pf}(\mathbf{w}(k_x = \pi/a_x))}{\text{Pf}(\mathbf{w}(k_x = 0))} \right) \right] \quad (5.95)$$

$$= \frac{-i}{2\pi} \left[\ln \left(\frac{\text{Det}(\mathbf{w}(k_x = \pi/a_x))}{\text{Det}(\mathbf{w}(k_x = 0))} \right) - 2 \cdot \ln \left(\frac{\text{Pf}(\mathbf{w}(k_x = \pi/a_x))}{\text{Pf}(\mathbf{w}(k_x = 0))} \right) \right] \quad (5.96)$$

$$= \frac{-i}{\pi} \ln \left(\frac{\pm \sqrt{\text{Det}(\mathbf{w}(k_x = \pi/a_x))}}{\pm \sqrt{\text{Det}(\mathbf{w}(k_x = 0))}} \frac{\text{Pf}(\mathbf{w}(k_x = 0))}{\text{Pf}(\mathbf{w}(k_x = \pi/a_x))} \right). \quad (5.97)$$

Thus, we are at the desired stage, where the so-called time reversal polarization⁵³ P_x^Δ at either $k_y = 0/m$ or at $k_y = \pi/a_y$ depends only on Pfaffians and determinants. Moreover, these factors include only the phase factors $\Theta_{k_x n}$ of the Kramers pairs at the two time-reversal invariant momenta (TRIM) in k_x direction ($k_x = 0/m$, $k_x = \pi/a_x$). Recall that our calculation was performed for $k_y = 0/m$ or $k_y = \pi/a_y$ only, since otherwise the degeneracy leading to eq. (5.73) would not hold. Thus, only the phase vectors of the Kramers partners at the four 2D TRIMS matter.

Recall, moreover, that the Pfaffians are complex numbers with absolute value 1 (eq. (5.82)) and, by $\text{Det}(\mathbf{w}) = \text{Pf}(\mathbf{w})^2$, the determinants are complex numbers with absolute value 1, too. This implies that the natural logarithm (ln) only measures the

⁵³ It is not called spin polarization, since spin is not a good quantum number in the presence of spin-orbit interaction. The two partners are related by time reversal, hence the name.

phase of its argument (multiplied by i), that itself is only defined modulo 2π , such that the possible distinct values are $P_x^\Delta \in [0, 2)$.

Moreover, we find from the relations between Pfaffians and determinants

$$\frac{\pm \sqrt{\text{Det}(\mathbf{w}(k_x = \pi/a_x))}}{\text{Pf}(\mathbf{w}(k_x = \pi/a_x))} = \pm 1$$

and

$$\frac{\pm \sqrt{\text{Det}(\mathbf{w}(k_x = 0))}}{\text{Pf}(\mathbf{w}(k_x = 0))} = \pm 1 .$$

In complex notation, we have $+1 = e^{i0}$ and $-1 = e^{i\pi}$. Consequently, there are only two distinct results for P_x^Δ : zero or one. The question, if there is a charge polarization in opposite directions for the two partners of a Kramers pair can be distinctly answered by yes (one) or no (zero).

Of course, the branches of the square root (+ or -) have to be chosen. But if the branches are chosen to be continuous on the way from $k_x = 0/m$ to $k_x = \pi/a_x$, the question is only, if, for a distinct k_y , the two fractions between determinant and Pfaffian at the two relevant k_x points are equal or if they are different.

Thus, we can rewrite the distinction between the two possible polarizations as:

$$(-1)^{P_x^\Delta(k_y)} = \frac{\sqrt{\text{Det}(\mathbf{w}(k_x = \pi/a_x))} \cdot \text{Pf}(\mathbf{w}(k_x = 0))}{\text{Pf}(\mathbf{w}(k_x = \pi/a_x)) \cdot \sqrt{\text{Det}(\mathbf{w}(k_x = 0))}} \quad (5.98)$$

$$= \frac{\sqrt{\text{Det}(\mathbf{w}(k_x = \pi/a_x))} \cdot \sqrt{\text{Det}(\mathbf{w}(k_x = 0))}}{\text{Pf}(\mathbf{w}(k_x = \pi/a_x)) \cdot \text{Pf}(\mathbf{w}(k_x = 0))} . \quad (5.99)$$

It turns out that only the distinction between the two numbers for P_x^Δ is unique, but not its absolute value, which depends on the chosen gauge of the wave functions (without argument)⁵⁴. However, the distinction is enough to distinguish, if the time reversal polarization $P_x^\Delta(k_y)$ is changed on the way from $k_y = 0/m$ to $k_y = \pi/a_y$ or not. If it is changed, one partner of each Kramers pair is moved to the right as much as the other partner is moved to the left. This is displayed in Fig. 5.19(b).⁵⁵ Thus, a gauge-invariant topological index ν_0 reads:

$$(-1)^{\nu_0} = \prod_{i=1}^4 \frac{\sqrt{\text{Det}(\mathbf{w}(\Gamma_i))}}{\text{Pf}(\mathbf{w}(\Gamma_i))} \quad (5.100)$$

using all four non-equivalent TRIMs Γ_i as marked in Fig. 5.19(a). Notice, that the whole topological information, if polarization is changed or not on the way from $k_y = 0/m$

⁵⁴ Heuristically, one can choose the original partners either at the ($k_y = 0$) line or at the ($k_y = \pi/a$) line before asking if they change their partner towards the other TRIM line.

⁵⁵ Recall that Kramers pairs are required at $k_y = 0/m$ and $k_y = \pi/a_y$ albeit with opposite k_x values. By construction of the Wannier functions (eq. (5.68)), this implies identical lattice positions for two states with opposite spin.

to $k_y = \pi/a_y$, is encoded in the occupied wave functions at the TRIMs of the Brillouin zone Γ_i , where $-\vec{k} = \vec{k}$. Formally, one can attribute an index δ_i to each TRIM i being $\delta_i = \frac{\sqrt{\text{Det}(\mathbf{w}(\Gamma_i))}}{\text{Pf}(\mathbf{w}(\Gamma_i))}$ and multiply these indices to get the topological invariant ν_0 .

This is extremely convenient and gets even more convenient, if the system is inversion symmetric. Then, the sign of $\delta_i = \frac{\sqrt{\text{Det}(\mathbf{w}(\Gamma_i))}}{\text{Pf}(\mathbf{w}(\Gamma_i))}$ is simply the product of the parity of all occupied wave functions at the TRIM⁵⁶ (see below).

To make our argument complete in terms of an adiabatic change of the Hamiltonian along a closed loop, we have to move on from $k_y = \pi/a_y$ to $k_y = 2\pi/a_y$ such that the complete movement ends up with identical wave functions. This second movement is identical to a movement from $k_y = -\pi/a_y$ to $k_y = 0/m$. Due to time reversal symmetry, the states running from $k_y = 0/m$ to $k_y = \pi/a_y$ must run exactly the same way as states with opposite spin from $k_y = 0/m$ to $k_y = -\pi/a_y$, respectively from $k_y = 2\pi/a_y$ to $k_y = \pi/a_y$. As sketched in Fig.5.19(b), this leads to an identical movement from $k_y = \pi/a_y$ to $k_y = 2\pi/a_y$ as from $k_y = 0$ to $k_y = \pi/a_y$. Thus, the complete movement moves all spins by two lattice constants in opposite directions, i.e., it returns the system to its original time reversal polarization defined modulo 2. But, if we cut a finite ribbon out of the material, spin is effectively moved from left to right in the x direction, since the most left solid lines and the most right dashed lines in Fig. 5.19(b) cannot join with a partner from the right or the left, respectively. Simultaneously, the wave functions of the ribbon must be exactly the same by changing k_y adiabatically in one closed cycle.

Thus, we have the situation, which was needed for spin polarized edge states at the Fermi level at both sides of the insulating ribbon. We repeat that spin movement from the left to the right, without energy penalty, requires states at E_F , which are not available in the insulating bulk. Since the calculation is based on occupied wave functions of the bulk only (The matrix $\mathbf{w}(k_x)$ in eq. (5.79) is defined in terms of occupied bulk states), the edge states are a bulk property.

Since this property is not changed by moving E_F within the bulk band gap, there have to be edge states at each energy within the bulk band gap. Consequently, the edge states have to connect conduction band and valence band as shown in Fig. 5.18(b).

Here, we readopt the majority of students with a short summary of what has been derived. Firstly, we have defined pairs of states that are time reversal partners, i.e., we get the second state of the partner from the first state by applying the time reversal operator, that changes the wave vector $\vec{k}$ to $-\vec{k}$ and rotates the spin from $\uparrow$ to $\downarrow$ or vice versa. These states are called Kramers pairs. We have considered the Kramers pairs at $k_y = 0/m$ and $k_y = \pi/a_y$, since at these points one gets $k_y = -k_y$. This implies that the states are spin degenerate via

$$E(k_y, \uparrow) \underset{\text{TRS}}{=} E(-k_y, \downarrow) \underset{k_y=-k_y}{=} E(k_y, \downarrow) .$$

⁵⁶ The parity is +, if the wave function $\psi_i(\vec{x})$ does not change sign after the inversion operation and −, if the wave function changes sign.

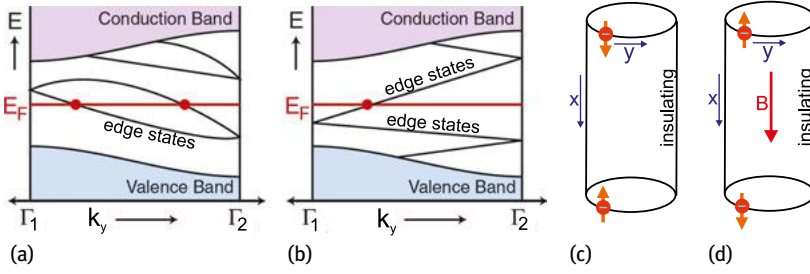


Fig. 5.18: (a), (b) One dimensional $E(k_y)$ band structures along the edge direction of a 2D ribbon of finite width in x direction and infinite length in y direction. The two topologically distinct cases of edge states (black lines) are displayed for $\nu_0 = 0$ (a) and $\nu_0 = 1$ (b). The blue and pink areas mark the projected band structure of the 2D bulk to the edge direction. Γ_1 and Γ_2 mark the time-reversal invariant points (TRIMs) where states have to be, at least, twofold degenerate. The edge states in (b) cover the whole bulk band gap as required for topologically protected edge states, while the edge states in (a) only cover part of the band gap and can not be topological. Moreover, the edge state bands in (a) exhibit two crossings at E_F and only one crossing in (b) (red points). (c), (d) Sketch of the hollow cylinder geometry without and with pierced $\vec{B}$ field in order to physically realize the spin pump. The spin exchange at the edges is realized by adding two flux quanta to the insulating cylinder with $\nu_0 = 1$. (a)–(b) [134].

Secondly, we have calculated, via construction of Wannier states in x direction (Section 4.5), if the charge of the two partners of the pair is moved in opposite directions along x , if we proceed our Hamiltonian from $k_y = 0/m$ to $k_y = \pi/a_y$. Such a movement is displayed in Fig. 5.19(b). It turns out that there are only two mathematically distinct values for this movement, which are either zero or one. These numbers are the topological $\mathbb{Z}_2$ index ν_0 . If $\nu_0 = 0$, the charges of the two partners reunite at $k_y = \pi/a_y$. If $\nu_0 = 1$, the charges of the partners move in opposite directions and they join with other partners at $k_y = \pi/a_y$ as sketched in Fig. 5.19(b). This implies, for a finite ribbon in x direction, that spin is moved effectively from the left to the right.⁵⁷

Thirdly, we have shown that, if the Kramers pair is moved in opposite directions from $k_y = 0/m$ to $k_y = \pi/a_y$, then it keeps moving in opposite directions from $k_y = \pi/a_y$ to $k_y = 2\pi/a_y$. This results in no spin movement from $k_y = 0/m$ to $k_y = 2\pi/a_y$ for an extended 2D system, since a movement by two lattice constants is mathematically identical to a movement by zero lattice constants. This sounds physically reasonable, since we move between nominally identical points in $\vec{k}$ space. Mathematically, it is due to the fact that one has a gauge freedom, that makes the movement by two lattice constants identical to the one without movement. However, if we cut out a ribbon with finite width in x direction, the moved charges at the edge of the ribbon can not return to a partner, such that the movement by two lattice constants is not identical to the movement by zero lattice constants. Within a ribbon, spin is moved from left

⁵⁷ All pairs at different atomic sites have to move identically due to lattice periodicity.

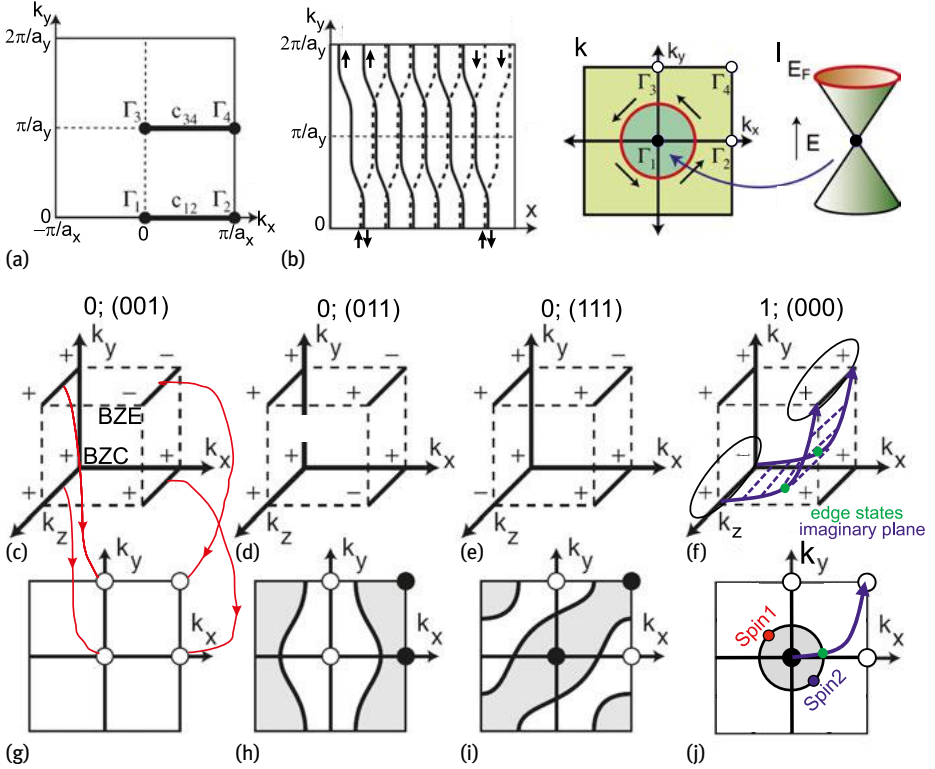


Fig. 5.19: (a) Sketch of a 2D Brillouin zone with the four time reversal invariant momenta (TRIMs) Γ_i ($i = 1, 2, 3, 4$) marked as used in eq. (5.100). The thick lines c_{12} and c_{34} are used to calculate the time reversal polarization $P_y^A(k_y)$ for the two time reversal invariant k_y values: $k_y = 0/\pi$, $k_y = \pi/a_y$. (b) Sketch of the changing polarization of Wannier states after separating them into two Kramers partners (dashed and solid lines marked by opposite spin arrows). The states switch partners on the way from $k_y = 0/\pi$ to $k_y = \pi/a_y$ and again on the way from $k_y = \pi/a_y$ to $k_y = 2\pi/a_y$. (c)–(f) Sketches of a 3D Brillouin zone (Brillouin zone center (BZC) and Brillouin zone corner (BZE) are marked in (c)) with signs of δ_i at the TRIMs marked for a particular gauge choice. Pairs of TRIMs used for the projection in (g)–(j) are connected by thick lines. The topological indices v_0 ; (v_x, v_y, v_z) are given above the images. In (f), a hypothetical 2D plane (blue), with edges in the front and back surface of the cube, is additionally drawn with the relevant time-reversal invariant lines encircled. The fact that the product of the TRIM states of these lines changes sign between the two lines requires spin polarized edge states at E_F as marked by green dots at an arbitrary (k_x, k_y) value. (g)–(j) Projected surface Brillouin zone in the k_z direction (front plane in (c)–(f)) with white (black) circles marking a $+$ ($-$) for the product of the δ_i connected by the thick lines in (c)–(f). The connection is additionally marked by the red, curved arrows between (c) and (g). Thick black lines separating grey and white areas mark the necessarily present Fermi lines of surface states. These surface states at E_F have to exist at each path from black to white dots. The projected blue plane of (f) is shown in (j) (blue line) with resulting edge state (green dot) marked. Grey (white) areas indicate the resulting areas of occupied (empty) surface states. The inverted spin directions required by time reversal symmetry are marked in (j). (k) Surface Brillouin zone identical to (j) with red Fermi line and occupied (empty) states marked in dark (bright) green. The black arrows mark a possible spin direction of the states at the Fermi line. (l) Corresponding $E(\vec{k})$ of the surface state called the Dirac cone with the marked Dirac point (black dot) and Fermi circle (red). (a)–(b) [133], (c)–(j) [135], (k)–(l) [134].

to right by going from $k_y = 0/m$ to the nominal identical $k_y = 2\pi/a_y$. This is exactly the situation discussed above, that requires states of both spins at both edges of the ribbon at E_F , if the Fermi level is within a bulk band gap. The edge states at E_F are required to enable that spin is moved from left to right without an energy penalty.⁵⁸ Hence, $\nu_0 = 1$ requires edge states for both spins at both edges.

One can show that ν_0 is identical to the spin dependent Chern index discussed with respect to 2DTIs via eq. (5.67). But since we would have to change from the description in eq. (5.67) to a more abstract one, we do not show this explicitly.⁵⁹

A more intuitive understanding can be gained via the Boltzmann relaxation model of electric current. An electric field E_y implies a change of the k_y value of all occupied states according to $eE_y = \hbar \dot{k}_y$. For a full band, this does not lead to a current in the y direction, since the global occupation of the band does not change, if all states are moved forward by Δk_y due to the periodic boundary conditions. However, if spin is moved simultaneously in the perpendicular direction to k_y as required by $\nu_0 = 1$, a transversal spin current results, hence $\sigma_{xy\uparrow} \neq 0$. Quantitatively, one can use $E_y = \hbar \dot{k}_y / e = \hbar \Delta k_y / e \Delta t$ with $\Delta k_y = 2\pi/a_y$ for the full adiabatic cycle described above. This transports one spin $\uparrow$ from left to right and one spin $\downarrow$ from right to left per unit cell in y direction,⁶⁰ such that two extra spins appear at each side as required by Fig. 5.19(b). Hence,

$$\begin{aligned} I_{\uparrow} &= -I_{\downarrow} = \frac{e}{\Delta t} \\ \Rightarrow j_{x\uparrow} &= -j_{x\downarrow} = \frac{e}{a_y \Delta t} \\ \Rightarrow \sigma_{xy\uparrow} &= \frac{j_{x\uparrow}}{E_y} = \frac{e}{a_y \Delta t} \cdot \frac{e \Delta t}{\hbar \Delta k_y} = \frac{e}{a_y \Delta t} \cdot \frac{e \Delta t a_y}{\hbar 2\pi} = \frac{e^2}{h} = -\sigma_{xy\downarrow}. \end{aligned} \quad (5.101)$$

Thus, the topological index ν_0 is equal to the spin Chern index as introduced on page 298. The difference between $\sigma_{xy\uparrow}$ and $\sigma_{xy\downarrow}$ implies that the dispersion of the states cannot be spin degenerate, since this would lead to identical Hall conductivities for both spins (Fig. 5.16(e)). Thus, the edge states are also spin polarized having a Kramers partner of opposite spin at opposite k_y .

Fourthly, we have shown that ν_0 can be calculated from the Bloch wave functions of an infinite 2D crystal. Basically, we have calculated the spin movement by the adiabatic change of the Hamiltonian within the bulk, and, finally, we have cut out a ribbon with edges. Thus, we get the properties of the edge without having an edge in the initial calculation, but just by the information, that spin is moved on the way between

58 The adiabatic charge pump, i.e., the movement of charges within an adiabatic cycle without energy penalty, has been observed experimentally within a lattice of atoms confined in the potential of standing light waves [136].

59 Some details of the complete derivation are found in [133], but they are based on a number of references quoted there.

60 The reduction to one unit cell gets obvious, while discussing Fig. 5.18 below.

$k_y = 0/m$ to $k_y = \pi/a_y$. The question, if edge states are present, is, thus, a bulk property. This implies automatically that such an edge state has to cross the whole bulk band gap of the 2D system, since the properties of the occupied bulk states do not change throughout the band gap. Such an edge state is sketched in Fig. 5.18(b) in the $E(k)$ diagram.

Fifthly, we have found that only the occupied wave functions at the time reversal invariant momenta (TRIMs), i.e., at the points in 2D $\vec{k}$ space, where $\vec{k} = -\vec{k}$, contribute to ν_0 . These TRIMs are sketched in Fig. 5.19(a) as $\Gamma_1, \Gamma_2, \Gamma_3$, and Γ_4 . The reduction of relevant $\vec{k}$ points is very convenient for a numerical determination of ν_0 .

Sixthly, we have shown that we can attribute an index to each TRIM Γ_i in the Brillouin zone being either +1 or -1. The index results from a phase factor matrix of all occupied states called $\mathbf{w}$ (eq. (5.78)–(5.80)). The resulting integer value $\delta_i = \frac{\sqrt{\text{Det}(\mathbf{w}(\Gamma_i))}}{\text{Pf}(\mathbf{w}(\Gamma_i))}$ is calculated resulting in $\delta_i = -1$ or $\delta_i = 1$. However, the value of the index δ_i is not well defined, i.e., it is gauge dependent. Only the product of the indices at the four TRIMs in the Brillouin zone is gauge independent and, thus, meaningful. If the product is -1 or +1, we have $\nu_0 = 1$ or $\nu_0 = 0$, respectively, meaning that we necessarily have spin polarized edge states with opposite chirality through the whole bulk band gap or we have not, respectively.

Finally, we describe a system with edges in the x direction, that can indeed be moved from $k_y = 0/m$ to $k_y = 2\pi/a_y$ by a physical operation. This can be done formally, e.g., by considering a hollow cylinder of the 2D material with the circumference of a single unit cell in the y direction and a finite length in the x direction (Fig. 5.18(c)–(d)). The cylinder will be pierced by an adjustable $\vec{B}$ field in the x direction. Changing the according magnetic flux by h/e introduces a phase shift of 2π per unit cell in the y direction and, thus, an increase of the wave vector by $2\pi/a_y$ as required. Thus, firstly, the electron wave functions are identical after the operation, since the wave vectors are only defined modulo $2\pi/a_y$ and, secondly, we can perform the process adiabatically without inducing energy. The latter can be seen by realizing that the field along the cylinder axis is parallel to the 2D plane of the system and, thus, cannot influence the energies of the 2D states. Consequently, we have transported spin from one edge of the cylinder to the other without changing the band structure and without energy supply exactly as required for the arguments above. The procedure is performed for one unit cell in y direction and transports one electron with spin $\uparrow$ from left to right and one electron with spin $\downarrow$ from right to left. This justifies that we have chosen $j_{x\uparrow} = \frac{e}{a_y \Delta t}$ in eq. (5.101).

Thus, if the product of phase factors of Kramers pairs (as represented by $\mathbf{w}$ in eq. (5.79)) is different at one TRIM with respect to the other TRIMs, spin is moved from left to right by an adiabatic cycle in $\vec{k}$ space. This implies the spin quantum Hall effect (SQHE) via $\sigma_{xy\uparrow} = -\sigma_{xy\downarrow} = e^2/h$, that is equivalent to two oppositely chiral, spin polarized edge state, one for each spin, at E_F (Fig. 5.16(e)).

Notice, that $\nu_0 = 0$ does not imply the absence of edge states. For example, edge states as shown in Fig. 5.18(a) can still be present, e.g., due to a particular edge chem-

istry. However, these kind of edge states do not necessarily fill the whole bulk band gap and, thus, they are distinct from the topologically protected edge states. Moreover, these edge states can be removed by modifying the edge, e.g., by H adsorption, while the topological edge states cannot be removed by a procedure modifying the edge only. The removal requires a change of the bulk properties, i.e., a change of the phase factors of occupied states at the TRIMs.

Now, we turn to the 3D case referring to the results of the intricate calculation of eqs. (5.68)–(5.100) for the 2D case. Recall that the calculation of the topological 2D index ν_0 requires knowing the signs of the multiplied phase factors of the occupied wave functions at the 4 TRIMs, i.e., at the non-equivalent points of the Brillouin zone, where $\vec{k} = -\vec{k}$. The product of the four signs is either $-$ or $+$, implying edge states via $\nu_0 = 1$ ($-$ sign) or no edge states for a topologically trivial index $\nu_0 = 0$ ($+$ sign), respectively.

Next, we transfer this approach of topological indices to 3D, again for a rectangular lattice for the sake of simplicity. We have now eight non-equivalent TRIMs as sketched in Fig. 5.19(c). They are at the center of the Brillouin zone (1) as marked by BZC, at a corner of the Brillouin zone (1) as marked by BZE, in the middle of Brillouin zone boundary lines (3), and in the middle of Brillouin zone boundary planes (3). They are marked with different signs of $\frac{\sqrt{\text{Det}(\mathbf{w}(\Gamma_i))}}{\text{Pf}(\mathbf{w}(\Gamma_i))} := \delta_i$ (for a particular gauge choice) in Fig. 5.19(c)–(f). As already mentioned, the sign is related to the inversion symmetry of all occupied wave functions at Γ_i , if the crystal is inversion symmetric. Each occupied wave function $\psi_i(\vec{x})$ gets either a factor of -1 or a factor of $+1$ after the inversion. The sign is called the parity of the wave function. The product of the parity numbers of all occupied states at Γ_i is δ_i .

In order to determine the topological index, that tells us, if a surface state is present at a particular surface, we again, firstly combine two TRIMs. This is visualized by the thick lines in Fig. 5.19(c)–(f), which are equivalent to the thick lines for the 2D case in Fig. 5.19(a). We ask the question, if the product of δ_i of the pair changes sign on the way between two pairs of TRIMs, as we did in 2D. One can imagine a 2D plane, that connects the two pairs as drawn in Fig. 5.19(f) by full and dashed blue lines. This plane must have an edge state since it connects TRIMs in the k direction along the ribbon with different pair products of δ_i across the ribbon (pairs are encircled by ellipses). The resulting edge states are marked as green points in Fig. 5.19(f). Due to TRS and $\sigma_{xy\uparrow} = -\sigma_{xy\downarrow} = e^2/h$, they have to obey $E(k_x, k_y, \uparrow) = E(-k_x, -k_y, \downarrow) = E_F$. The same applies for each possible plane between the two pairs of TRIMs. Thus, we can deform the blue plane, with edges within the two surfaces perpendicular to k_z , and always get edge states there. Since adjacent planes are similar, they will have similar edge states, implying a continuous line of edge states on the chosen $\vec{k}$ -space surface. This naturally leads to a closed contour, which is a closed line of surface states at E_F (thick black lines in Fig. 5.19(h)–(j)). Alternatively, one can argue that an interrupted line of surface states would enable the construction of a virtual plane without edge states that connects the two pairs of TRIMs with opposite product of δ_i . As mentioned,

the surface states along the closed line must have opposite spin for opposite (k_x, k_y) due to TRS, as sketched in Fig. 5.19(j).

We see that if one projects pairs of δ_i to the chosen surface, the sign of the δ_i product of the pair is decisive. This product sign is sketched by the different colors of the dots at the TRIMs in Fig. 5.19(g)–(j)⁶¹. Moreover, we learn that points with different color have to be separated by a spin polarized surface state at E_F as drawn by a thick line in Fig. 5.19(h)–(j). The fact that δ_i is a bulk property requires that a closed line of surface states exists at any energy within the bulk band gap. Hence, an $E(\vec{k})$ band of surface states connects the valence band and the conduction band. Time reversal symmetry implies that the surface states of this band have to be spin degenerate at the surface TRIMs, where $(k_x, k_y) = -(k_x, k_y)$, i.e., the spin polarized states have to merge, e.g., at $(k_x, k_y) = \vec{0}/m$. Finally, since (k_x, k_y) is a good quantum number of the surface, we expect that the band has only one surface state per (k_x, k_y) point implying an $E(k_x, k_y)$ dispersion.

A typical band of surface states obeying these requirements is sketched in Fig. 5.19(l). It exhibits a chiral spin in 2D $\vec{k}$ space. One possibility is that the spin direction is always perpendicular to $\vec{k}$, as favored by the spin-orbit interaction. Such a state, if isotropic, is called the topological Dirac cone, since it is the solution of the massless 2D Dirac equation with a reduced velocity of light.

The procedure to connect pairs of TRIMs in order to get the decisive signs of the δ_i products on a particular surface can be repeated for any surface of the crystal. It is obvious, that the case of Fig. 5.19(f), where only one of the 3D TRIMs has an inverted sign of δ_i with respect to the other TRIMs, implies surface states on any surface, since one never gets rid of the inverted sign by the pairwise products.⁶² These systems are called strong 3D topological insulators (3DTIs). However, if a pair of 3D TRIMs has an inverted sign with respect to the other TRIMs, one can find a surface, where the two inverted signs compensate, such as in Fig. 5.19(c) and (g). These surfaces, thus, do not have a topological surface state. For other surfaces, the two inverted signs do not compensate, but imply two points of different colors and, hence, at least, two connected lines of surface states as shown in Fig. 5.19(d)–(e) and (h)–(i). These systems are called weak 3D topological insulators.⁶³ Only, if all TRIMs have the same sign of δ_i , we have a trivial insulator, not implying surface states on any surface. As in 2D, however, there can still be so-called trivial surface states, that are not caused by the bulk topology, i.e., they can be removed by a change of the surface chemistry.

⁶¹ We use colors instead of + and – signs, since the product of two δ_i is not gauge invariant, while only the distinction between the products is gauge invariant.

⁶² The same applies for three inverted signs, i.e., e.g., three times $\delta_i = 1$ and five times $\delta_i = -1$.

⁶³ Since two bands of surface states are necessarily present on such a surface, one can anticipate a danger that disorder couples them, i.e., it produces symmetric and antisymmetric combinations of the wave functions. This could result in a gap and, thus, would destroy the robust surface conductivity. However, it turns out that also the conductivity of these so-called weak 3DTIs is rather robust.

In turn, the Fermi lines of a topological surface state can only be destroyed either by adding terms that break time-reversal symmetry, such that our whole analysis does not apply, or by exchanging the parity (for the case of inversion symmetric crystals) of, at least, one occupied wave function at, at least, one TRIM. Thus, one has to exchange a conduction and a valence band of different parity. In that sense, the Dirac cone is robust with respect to parameter changes of the Hamiltonian (including disorder), which do not invert the bands back into normal/trivial order. Thus, we can adapt a very similar scenario as in 2D (Fig. 5.16(a) and (b)). Inverting the p- and s-bands at one TRIM and leaving the normal order at all other TRIMs will lead to topological surface states on all surfaces. We will discuss a corresponding example in Section 5.4.6.2.

With this intuitive understanding in mind, we shortly describe the calculation of topological indices of 3DTIs. Firstly, we attribute three subscripts n_x, n_y, n_z to each δ_i . These subscripts are 0 (1), if the projection of the vector from $\vec{k} = 0/m$ to the TRIM is 0 (π/a_i) in the respective direction x, y , or z . The indices ν_j ($j = 0, x, y, z$) are then:

$$\begin{aligned} (-1)^{\nu_0} &= \prod_{n_x, n_y, n_z=0,1} \delta_{n_x n_y n_z} \\ (-1)^{\nu_{k=x,y,z}} &= \prod_{n_{j \neq k}=0,1, n_k=1} \delta_{n_x n_y n_z} \end{aligned} \quad (5.102)$$

Thus, for ν_0 , we multiply all eight δ_i . This implies a strong topological insulator, if $\nu_0 = 1 \pmod{2}$, as expected from our projection argument. If $\nu_0 = 0$, we do not necessarily have surface states on all surfaces and the other indices matter. For the three ν_k , we have to multiply four selected δ_i . If all three indices ν_k are zero, we have a trivial insulator. Otherwise, one has a weak 3DTI. The triple (ν_x, ν_y, ν_z) then marks the surface normal of the only surface without robust conductivity, i.e., without required surface states (without argument). Thus, we are back at our original imagination of 2DTIs stacked into the direction (ν_x, ν_y, ν_z) and having robust conductivity at any perpendicular surface.

5.4.6.2 Materials and experiments

To find a candidate material for a strong 3DTI, one could look for a material with band inversion at the center of the Brillouin zone, called the Γ point, which inverts back towards the edges of the Brillouin zone. This trick works straightforwardly for inversion symmetric crystals. Since we already know that spin-orbit interaction helps to invert bands, we look for heavy atoms. The combination of elements often results in band gaps due to anticrossings within the bulk, which is the second requirement. The first material being a strong 3DTI was a BiSb alloy [137], but the real breakthrough came with the materials depicted in Fig. 5.20: Bi₂Se₃, Sb₂Te₃, and Bi₂Te₃. They are inversion symmetric, consist of heavy elements, and are known to exhibit a bulk band gap. The structure is layered distinguishing the bonding in the z direction and the (x, y) -direction. Figure 5.20(c) shows a model, how the states at Γ evolve by adding interaction terms between the atoms using Bi₂Se₃ as an example (see also Fig. 4.18–4.20). The

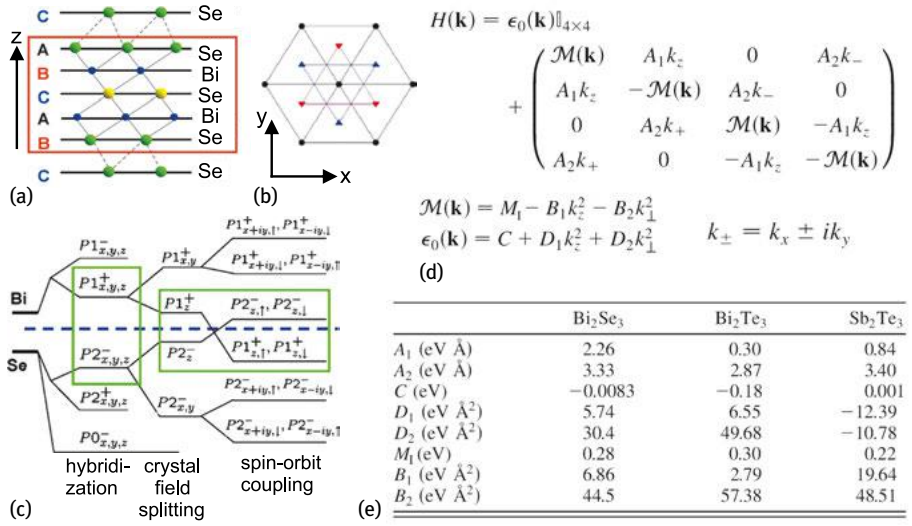


Fig. 5.20: (a) Side view of the atomic structure of the layered material Bi_2Se_3 consisting of quintuples of stacked layers (red box) in stacking sequence Se-Bi-Se-Bi-Se. The forces between the quintuples, and, thus, between adjacent Se layers, are weak van der Waals forces such that the crystal can be cleaved between the outer Se layers of the quintuple. (b) Top view of the crystal showing the ABC stacking of the hexagonal layers marked each by different symbols and colors. Sb_2Te_3 and Bi_2Te_3 as well as mixtures of these three crystals exhibit the same structure. (c) Model of the energy levels at the center of the Brillouin zone (Γ point). The p-levels of Bi and Se (first column) get first split by chemical hybridization into bonding and antibonding states marked by their parity (+, -) and their preferential orientation (x, y, z). Five p_x -, p_y -, and p_z -states result due to the five atoms of the unit cell corresponding to the five layers in the quintuple (second column). Secondly, the crystal field lifts degeneracies between p_{xy} and p_z in favor of the stronger bonds in the (x, y) plane (third column). Finally, the spin orbit coupling splits the p_{xy} states and additionally mixes all three p-states, since there are also electric fields in the z-direction, which induce the spin-orbit terms. This leads to level repulsion, e.g., between $|P1_{z\uparrow}^+\rangle$ and $|P1_{x+iy\downarrow}^+\rangle$ and, thus, to a level crossing between the states closest to E_F (fourth column). (d) Matrix description of a minimal model Hamiltonian for the structure shown in (a)–(c) describing only the central states $P2_{z\uparrow}^-$, $P2_{z\downarrow}^-$, $P1_{z\uparrow}^+$, and $P1_{z\downarrow}^+$ of (c). (e) Parameters for the model Hamiltonian deduced from first-principle calculations for three different materials. (a)–(e) [128].

main players close to E_F are p-orbitals (Bi 6p and Se 4p), since the corresponding s-orbitals and d-orbitals are completely filled. The lattice period of five hexagonal layers in the z direction gives five hybridized levels for p_x , p_y and p_z each (in sum 15), that get split due to the different strength of the bonding in the z- and in the (x, y)-direction (crystal field splitting, Section 4.6.1). The decisive step is then provided by the spin-orbit interaction (Section 4.5.4), that exchanges a state with positive parity and a state with negative parity at E_F and, thus, inverts the sign of δ_i at Γ . The dispersion of these two bands inverts these two states back on the way to the Brillouin zone boundaries, such that only the Γ point has a different sign of δ_i . Consequently, we get a strong

3DTI. The (4×4) model Hamiltonian, that includes only the bands $P2_{z\uparrow}^-$, $P2_{z\downarrow}^-$, $P2_{z\uparrow}^+$, and $P2_{z\downarrow}^+$, is given in Fig. 5.20(d). The signs + and – define the parity of the wave function. The real numbers $A_1, A_2, B_1, B_2, C, D_1, D_2$, and M_I define the terms in the matrix similarly to the ones in the 2D case (A_1, B_1, M_I in eq. (5.66)ff.).

As long as M_I is positive and smaller than B_1 and B_2 multiplied by the corresponding k^2 at the Brillouin zone boundary and, in addition, A_1 and A_2 are not both zero, we get the topological band inversion exclusively at Γ and a gap at the position of the back-inversion, thus, a strong 3DTI. Parameters for three binary materials are given in Fig. 5.20(e). They are deduced from density functional theory (DFT) calculations and all fulfill these requirements for a strong 3DTI.⁶⁴

Experimentally, the most direct proof of a strong 3DTI is ARPES,⁶⁵ that probes the band structure $E(k_x, k_y)$ of a particular surface, where the angle of the emitted photoelectrons is directly related to the $\vec{k}$ vector within the surface.⁶⁶ Modern instruments can map the $E(\vec{k})$ dispersion in one particular $\vec{k}$ direction within one shot such that mapping of a surface band structure becomes a routine task. Since the 3DTIs Bi_2Se_3 , Sb_2Te_3 and Bi_2Te_3 can be cleaved in ultra high vacuum, the preparation of the clean crystal surfaces is simple. An ARPES analyzer can, moreover, be combined with a Mott detector, which can analyze the spin direction of particular photoelectrons belonging to a selected E and $\vec{k}$. An experimental setup is sketched in Fig. 5.21(a). The photoelectrons are selected in energy by the spherical capacitor and in angle by the entrance slit position into this capacitor with respect to the light point on the sample. After the final $(E, \vec{k})$ selection by the exit slit, the electrons are accelerated and directed to a heavy metal foil (Au foil in the image), where they get scattered. Due to the strong spin-orbit interaction within the foil, the scattering direction depends on spin. For example, the spin-up (spin-down) electrons get preferentially scattered to the left (right), which after calibration allows determination of the spin polarization in up-down direction from the difference of the photoelectron current in the left and the right detector. The same can be done for the left-right spin polarization using the differences of the upper and the lower detector. The detectors are drawn in Fig. 5.21(a) as blue planes with arrows symbolizing the spin detection direction.

Figure 5.21(b) shows a calculated surface band structure by DFT with the indicated Dirac cone (surface states). The crossing point of the Dirac cone, the so-called Dirac point, is in the center of the surface Brillouin zone, which is called the $\bar{\Gamma}$ point. It is, moreover, exactly at E_F . ARPES results of the same surface are shown in the inset. One recognizes reasonable agreement between experiment and theory, except that the Fermi level in the experiment is not at the Dirac point, but within the conduction band. Hence, the material is an n-doped semiconductor. This is caused by va-

⁶⁴ Using the model Hamiltonian, one can explicitly show the presence of a single Dirac cone on each surface ([128]: Chapter III.B).

⁶⁵ Angularly resolved photoelectron spectroscopy.

⁶⁶ The method is described in detail, e.g., in [138].

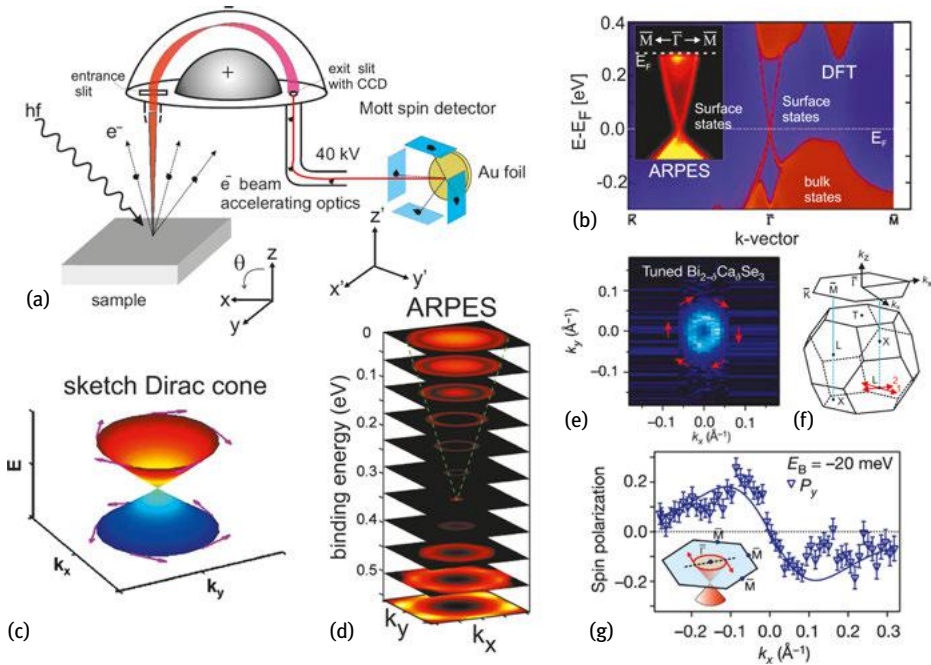


Fig. 5.21: (a) Sketch of the experimental setup for spin-polarized ARPES: hf labels the incoming photons and the red-orange areas mark the electron paths contributing to the detector signal without spin polarization. The electrons are detected by a CCD chip selecting different energies in radial direction and different angles perpendicular to the radial direction of the spherical capacitor. Polarities at the plates of the analyzing capacitor are marked. (b) DFT calculation of the surface band structure of Bi_2Se_3 at the cleavage plane. Filled, red areas are projected bulk states, while red lines are surface states as marked. Inset: ARPES data of the same surface along the ΓM direction in $\vec{k}$ space. Note the different positions of E_F in DFT and ARPES results. (c) 3D representation of a surface Dirac cone. The arrows indicate the expected spin directions of the states. (d) 3D representation of ARPES results of the cleavage surface of Bi_2Se_3 . (e) Fermi line (at $E = E_F$) of Ca doped Bi_2Se_3 as measured by ARPES. (f) Brillouin zone of Bi_2Se_3 with labeled high symmetry points and surface Brillouin zone of the cleavage plane drawn above. (g) Measured spin polarization of a cut at constant energy ($E - E_F = -20$ meV) through the $\vec{k}$ space showing that the spin polarization inverts with the sign of $\vec{k}$ as required by time-reversal symmetry. The spin polarization is strongest in the direction perpendicular to $\vec{k}$ (polarization measured in the y direction for $\vec{k}$ in the x direction) as sketched in the inset and in (e). The bad resolution in the k_x direction is due to the Mott detector which reduces the photoelectron intensity dramatically. (a), (f) [134], (b), (c) [128], inset (b) [139], (d) [140], (e), (g) [139].

cancies and antisite defects unintentionally produced during the preparation of the crystal. These defects lead to strong n-doping of the material. Thus, the predicted transport properties of 3DTIs are not expected for Bi_2Se_3 . The same is true for Bi_2Te_3 and oppositely for Sb_2Te_3 , which is strongly p-doped. However, for ARPES, this does not matter and one can map the complete Dirac cone (Fig. 5.21(d)), the Fermi line of

the surfaces (Fig. 5.21(e)) and the spin polarization including its preferential direction (Fig. 5.21(g)).⁶⁷

Using spin polarized ARPES, it has also been shown that magnetic doping, e.g., of Bi_2Se_3 with Mn atoms, that breaks time-reversal symmetry, indeed opens a gap within a Dirac cone [142], that counter-doping, e.g., with Ca, can be used to shift the Fermi level of Bi_2Se_3 to the Dirac point [139], that also thin films exhibit a gap in the Dirac cone because of the interaction of the two Dirac cones of opposite surfaces [143], such that a minimum thickness of about five quintuple layers is required to get the topological properties, and that the phase transition from a trivial insulator to a topological insulator can be tuned by material composition using, e.g., $\text{BiTl}(\text{S}_{1-x}\text{Se}_x)\text{Se}_2$ with different x [144]. Also another topologically protected state has been found by ARPES, where the pairs to be considered for the polarization difference $P_x^\Delta(k_y)$ (eq. (5.95)) are not pairs related by time reversal symmetry, but pairs related by a mirror symmetry of the crystal [145]. These materials are called topological crystalline insulators, however, with Dirac cones being only robust to disorder types or perturbations, which do not break the mirror symmetry of the crystal.

The second technique used to verify topological properties is scanning tunneling spectroscopy (STS), that measures the local density of states (LDOS), i.e., the sum of squared wave functions of the system at a particular energy E selected by the applied voltage V according to $E - E_F = eV$ [146]. One major contribution to the LDOS are standing waves, that are the overlap of incoming and scattered waves from a barrier, e.g., a step edge or a defect (Fig. 5.1(f)). Figure 5.22(a) shows a LDOS map of the cleavage surface of the 3DTI $\text{Bi}_{1-x}\text{Sb}_x$ [147]. The changing intensity across the surface (red, blue, yellow) is given by the standing electron waves originating from the scattering at potential fluctuations.

The Fourier transformation in Fig. 5.22(b) shows the $\vec{k}$ -space distribution of the standing waves. It should be compared with the $\vec{k}$ -space distribution of the states of the same surface determined by ARPES (Fig. 5.22(c)). Therefore, the possible scattering vectors $\Delta\vec{k} = \vec{k}_1 - \vec{k}_2$ between two states at $\vec{k}_1$ and $\vec{k}_2$ have to be calculated⁶⁸, and subsequently to be weighted with the density of states at the contributing $\vec{k}$ points as sketched in Fig. 5.22(d). This results in Fig. 5.22(e), that obviously deviates from the experimental data in Fig. 5.22(b). This deviation directly shows the topological character of the contributing states. As described above (footnote 36 on page 299), a direct backscattering between Kramers pairs leads to a Berry phase of π between the required possible spin rotations, being clockwise or counterclockwise, and, thus, to destructive interference. Thus, all scattering vectors combining states with $\vec{k}_1 = -\vec{k}_2$ are eliminated from the standing wave pattern by destructive interference, i.e.,

⁶⁷ The absolute values of spin polarization have to be taken with care since the polarization of the light also contributes to it via selection rules [141].

⁶⁸ To determine the standing wave, one has to calculate $|\psi_1(\vec{x}) + \psi_2(\vec{x})|^2 \propto |e^{i\vec{k}_1 \cdot \vec{x}} + e^{i\vec{k}_2 \cdot \vec{x}}|^2 \propto 1 + \cos((\vec{k}_1 - \vec{k}_2) \cdot \vec{x})$.

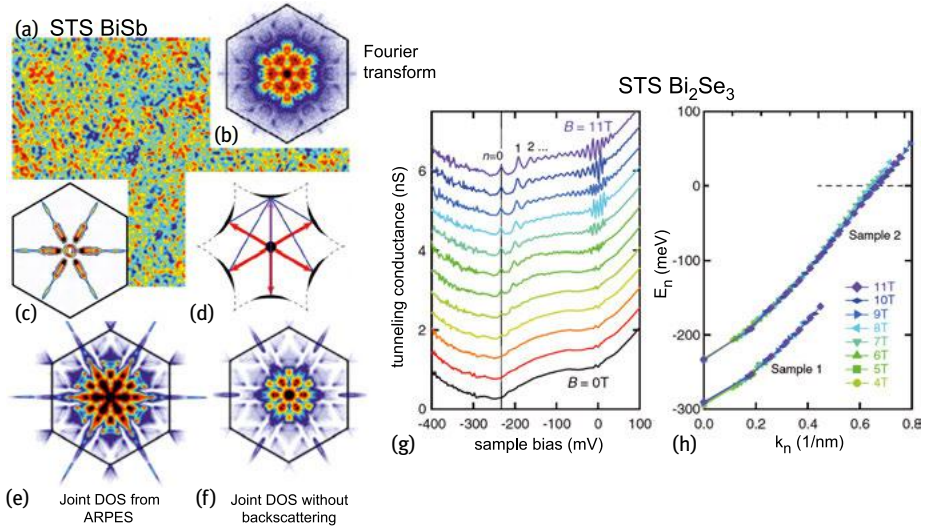


Fig. 5.22: (a) STS image of $\text{Bi}_{1-x}\text{Sb}_x$ recorded in the bulk band gap. The contrast is caused by overlapping standing electron waves scattered at multiple impurities at the surface. (b) Fourier transformation of the data in (a) representing the $\vec{k}$ space distribution of standing waves. (c) ARPES data of the same surface at the same energy representing the $\vec{k}$ space distribution of states. (d) Sketch showing how to transform ARPES data (black areas) into the Fourier transform of the STS data, namely, by defining scattering vectors in $\vec{k}$ space (red and blue arrows), which correspond to the expected wave vectors of the standing waves. Scattering vectors combining areas with large state density from ARPES contribute more than the ones combining areas with low state density. (e) Resulting expected Fourier transformation of STS images deduced from the ARPES data in (c) and the construction in (d). (f) Same as (e), but neglecting all scattering vectors combining states at $+\vec{k}$ with states at $-\vec{k}$, i.e., states that are scattered by 180° . Note that (f) fits much better to (b) than (e). (g) STS curves on the cleavage surface of Bi_2Se_3 representing the LDOS as a function of energy at different $\vec{B}$ field perpendicular to the surface as marked at the top and at the bottom curve. The numbers n mark consecutive Landau levels. (h) $E(|\vec{k}|)$ dispersion deduced from the Landau levels for two different samples. The wave number k_n corresponds to the quantized circular motion of the electron at constant velocity $|\vec{v}|$. (a)–(c), (e), (f) [147], (d) [148], (g), (h) [149].

by the helical spin character of the Dirac cone. Indeed, erasing these standing wave contributions as in Fig. 5.22(f), the experimental Fourier transformation in Fig. 5.22(b) is reproduced. Thus, STS can show directly, that the surface states of a strong 3DTI are spin helical.

Another application of STS was the indirect measurement of the $E(|\vec{k}|)$ dispersion of the surface states using Landau level spectroscopy (Fig. 5.22(g) and (h)). The experiments firstly show that the distance between Landau levels decreases with distance from the Dirac point, which is at $E_D - E_F = e \cdot V = -230$ meV. This is expected for a linear dispersion $E \propto |\vec{k}|$ that results in energies for the n th Landau level being

$E_n^{\text{LL}} \propto \sqrt{nB}$.⁶⁹ Albeit the application of $\vec{B}$ breaks time-reversal symmetry, this effect is close to negligible, since the spin splitting by the $\vec{B}$ -field is about 1 meV only, while the spin-orbit interaction is about several 100 meV. Thus, the Dirac cone might be gapped in an external field, but to a negligible amount.

Transport measurements, so far, often suffer from the intrinsic doping and the strong disorder potential of the materials. Moreover, the surface of the topological insulator is typically exposed to air and several chemicals for making Hall bar devices and contacting the surface, such that the surface chemistry is strongly changed. As pointed out above, this should not lead to a destruction of the topological surface states, but can lead to strong surface doping shifting the Fermi level at the surface in an uncontrolled manner. Thus, for reliable transport measurements, the surface of the 3DTI has to become an interface to a normal insulator. Nevertheless, seamless ambipolar transport, i.e., the change from an electron current to a hole current by gating has been observed [150] as well as the quantum Hall effect [151]. Experiments exploiting the spin helicity of the surface state by transport are under way, albeit still rather indirect [152].

The progress with weak 3DTIs is much slower, but, at least, one system has been identified and produced, where weak 3DTI properties are found. It consists of layers in a honeycomb structure made of the heavy elements Bi, I and Rh. This structure forms a 2DTI in DFT calculations with an inverted gap at the Γ point of 0.3 eV. The layers are stacked in the 3D material and are additionally separated by spacer layers of the normal insulator BiI, such that they form a weak 3DTI [153]. The material exhibits, e.g., topologically protected edge states on all step edges of the surface perpendicular to the stacking direction [154].

5.4.7 Other topological properties

There are many other fascinating effects arising from the topological considerations as described above. Most of them still await their experimental verification.

A few of them are sketched in Fig. 5.23.⁷⁰ For example, there is a so called magnetoelectric response, i.e., a static electric field can induce a robust, static magnetic field, via the Hall current circulating around the electric field, if the degeneracy of the two counter-circulating edge states is broken, e.g., by ferromagnetic layers (FM) (Fig. 5.23(a)). The same effect induces a magnetic monopole as the image of an electric monopole on top of a gapped 3DTI surface (Fig. 5.23(b)). Putting, additionally, superconductors on top of a 3DTI induces another type of gap, the superconducting gap into the Dirac cone. This gap of a so-called topological superconductor is unusual

⁶⁹ This can be verified, e.g., by using the 2D DOS of a linear $E(\vec{k})$ dispersion (eq. 1.20) and the degeneracy of Landau levels $2eB/h$ exploiting that the integrated 2D DOS must fit into the Landau levels.

⁷⁰ Arguments can be found in [128]: Chapter III.D and V.C.

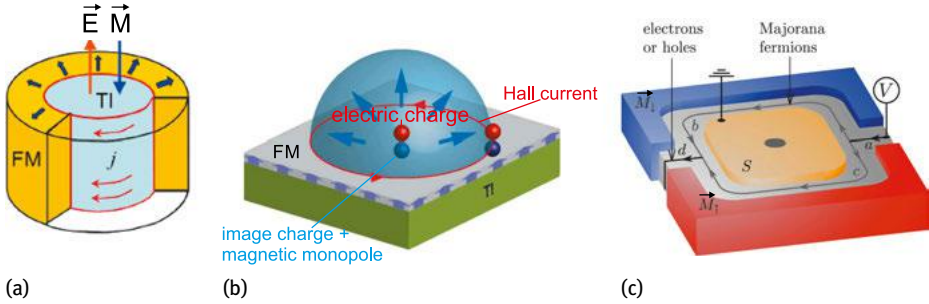


Fig. 5.23: (a) Sketch explaining the magnetoelectric effect: An electric field $\vec{E}$ in a weak 3D TI or a strong 3D TI with a gapped surface and a ferromagnetic layer (yellow, FM) on the remaining topological surfaces induces a Hall current perpendicular to the electric field. This Hall current is circular and, thus induces a magnetic field antiparallel to the electric field applied. (b) An electric monopole (red) above a gapped surface of a 3D TI, e.g., by a ferromagnetic layer (grey, FM), induces an image charge (blue) and an image magnetic monopole (blue) due to the Hall current (red line) at the interface between gapped areas of different sign. (c) Surface of a strong 3D TI (grey) covered with two magnets (red, blue) having the opposite sign of the magnetization $\vec{M}$, thus, leaving a state at the Fermi level at the interface between the two domains (black line with arrows). A superconductor (S, orange) induces Majorana channels (grey lines with arrows) at the interface between the superconductor and the ferromagnet. Applied voltage V , ground at S, and the grey dot marking a magnetic vortex in the superconductor can be used to probe unusual interference effects. (a) [155], (b) [156], (c) [157].

in the sense that the superconductor is made of only one spin species for each $\vec{k}$ rendering it a so-called spinless superconductor. One can show that states at the Fermi level appear at the edges of such superconductors. These states are equally distributed on opposite sides of the superconductor [158]. They are called Majorana modes and can be found, e.g., at the end of a 1D topological superconductor, where some experimental confirmation for the Majorana modes has been found [159]. Other hosts for Majorana modes are vortices within topological superconductors or edges between areas gapped by magnets and gapped by superconductors. The experimental proposal shown in Fig. 5.23(c) aims at the separation of an injected electron into two Majorana modes that encircle the central superconducting island, which leads to unusual interference effects.

We close this section by commenting on the robustness of the topological properties. This robustness originates from the integer values of the topological indices, which cannot be changed continuously. Hence, a qualitative change of the system is required to change its topology. For example, an occupied band has to be exchanged at one TRIM by a former empty band and the two bands must have a different character, e.g., exhibiting a different parity. However, such an exchange of bands can be realized by a minimal change of the total energy of the system. Multiple examples show that slight changes of the strain or of the stoichiometry of a specific sample can change its topological index, mostly if the system is already energetically close to a topological

phase transition. Consequently, the celebrated robustness of topological properties is, a priori, not a robustness with respect to the total energy of the system, but is related to the robustness of boundary effects, that can only be changed if the character of the material within the boundaries is changed. In this sense, all topological systems are similar to the Quantum Hall system, where the presence of the edge states is not related to any details of the edge, but can only be removed if the energetics of the bulk of the 2D system is changed.

5.5 Consequences of electron-electron interaction

5.5.1 Zero-dimensional electron systems

Next, we discuss the more complex electron-electron interaction for itinerant electrons. Corresponding electron phases can also be partly classified by topological indices, but we do not follow this path, since it gets too abstract for an intuition. The most transparent introduction into the electron-electron interaction can be given at complete quantization of the kinetic energy in quantum dots (Sections 1.3 and 3.3.1.2), i.e., within 0D systems. Such systems are also called artificial atoms, since they provide discrete energy levels similar to atoms, but are confined within much larger volumes ($10\text{ nm}^3 - 1\text{ }\mu\text{m}^3$). A similar approach has been followed for the electron-electron interaction of localized electrons in Section 4.4, where relevant effects were first described for atoms and molecules, before turning to the extended solids. We will start our approach by describing the single particle spectra of quantum dots, before we move to the consequences of the electron-electron interaction.⁷¹

5.5.1.1 Single particle spectrum

The setup of a typical quantum dot used for the study of interaction effects is shown in Fig. 5.24(a). Firstly, one grows a layered system consisting of n-doped GaAs, AlGaAs and InGaAs (Section 1.3.1). The system contains a 2D electron system confined within the InGaAs layer, that has the smallest band gap of the three materials. The thickness of the InGaAs layer is typically about 10 nm. The 2D electron system (2DES) in the InGaAs layer is filled with electrons due to adequate doping close to the surface. In order to confine the 2DES in the remaining lateral directions, lithography and etching methods can be used.⁷² One can, e.g., produce a cylinder with diameter $d_{\text{QD}} \approx 500\text{ nm}$ (Fig. 5.24(a)). A scanning electron microscopy image of such a cylinder is shown in Fig. 5.24(b). This cylinder will then be surrounded by a metal, the so called side gate. Afterwards metallic contacts to the upper and lower n-GaAs are established, which

⁷¹ We largely follow [160].

⁷² The other approach using gates is described in Sections 1.3.1.1 and 3.3.1.2.

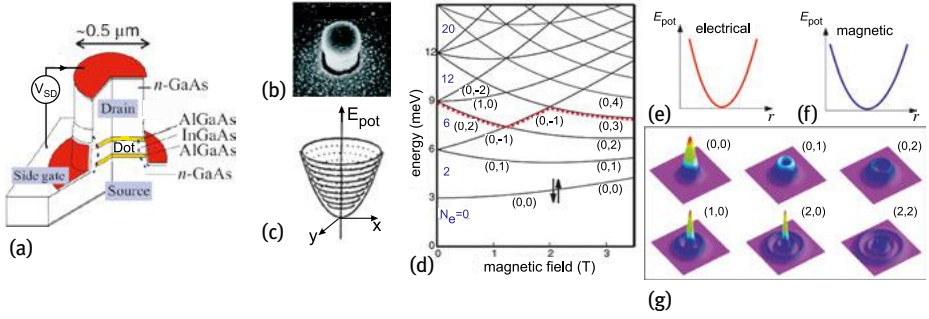


Fig. 5.24: (a) Schematic of a quantum dot device. The side gate can be used to vary the charge within the quantum dot. (b) Scanning electron micrograph of the quantum dot sketched in (a) before adding contacts. (c) Electrostatic potential within the plane of the quantum dot. (d) Single particle states of the quantum dot as a function of $\vec{B}$ field perpendicular to the plane of the quantum dot. Numbers in brackets mark radial and azimuthal quantum numbers of the states (n, m) . Blue numbers at the left mark the number of electrons N_e within the dot, if E_F is in the corresponding gap. Red dotted line marks the highest occupied state for seven electrons within the quantum dot. (e), (f) Schematic of the electrostatic and the magnetostatic potential, respectively. (g) $|\psi_i(\vec{x})|^2$ of some single particle states with labeled quantum numbers (n, m) at $B = 0$ T. (a), (b), (d), (g) [160].

act as source and drain electrodes allowing current transport across the AlGaAs tunnel barriers and, thus, through the InGaAs quantum dot. At the interface between side gate and semiconducting cylinder, negative charges appear due to the difference in work function of the two materials. These charges cause a parabolic, electrostatic potential for the electrons within the quantum dot, which is mostly rotationally symmetric within the plane (Fig. 5.24(c),(e)). The single particle states of this 2D harmonic oscillator can be calculated analytically in a $\vec{B}$ field. The central idea of the calculation is to map the Schrödinger equation of a free electron in a $\vec{B}$ field to the harmonic oscillator potential, as is typically shown explicitly within standard textbooks of quantum mechanics. The effective, harmonic potential due to $\vec{B}$, as shown in Fig. 5.24(f), leads to the equidistant Landau levels also discussed in Section 5.3.2. The squared wave functions $|\psi_i|^2$ of the states of the quantum dot at $B = 0$ T are shown in Fig. 5.24(g). Since the symmetry of the potential implies separation in radial and azimuthal coordinates, one gets states classified by radial and azimuthal quantum numbers (n, m) ($n \in \mathbb{N}^0, m \in \mathbb{Z}$). Thereby, n counts the number of antinodes in radial direction, while m belongs to the function $e^{im\varphi}$ with φ being the azimuthal angle, i.e., it is related to the orbital momentum of the quantum dot.

At $B = 0$ T, equidistant energies $E_{n,m} = (2n + |m| + 1) \cdot \hbar\omega_0$ result.⁷³ It is easy to see that the degeneracy of the energy levels increases by two, if one jumps from a

⁷³ The eigenfrequency of the quantum dot ω_0 depends on the curvature of the parabolic potential $E_{\text{pot}}(\vec{x})$ according to $\omega_0 = \sqrt{d^2 E_{\text{pot}}(\vec{x})/d|\vec{x}|^2/m^*}$.

given energy level to the next one higher in energy and considers the additional spin degree of freedom. Hence, the lowest level has a degeneracy of two, the next level a degeneracy of four, then a degeneracy of six, and so on (Fig. 5.24(d)).

Within a $\vec{B}$ field, three things are changing:

1. The additional magnetic confinement (Fig. 5.24(f)) increases the energy of all states. Formally, ω_0 is replaced by $\sqrt{\omega_0^2 + \omega_c^2}/2$ with $\omega_c = eB/m^*$. This enhancement in energy is called the diamagnetic shift.
2. The orbital momentum of the electrons is accompanied by a magnetic dipole oriented oppositely to the orbital momentum. This implies an additional energy proportional to $\vec{B}$: $E_{\text{dipole}} = -m \cdot \hbar\omega_c/2$. This is called the paramagnetic energy shift.
3. Finally, the degeneracy of the spin degree of freedom $s = \pm 1/2$ is lifted in a $\vec{B}$ field by $E_Z = s \cdot g\mu_B|\vec{B}|$ called the Zeeman energy.

Consequently, one finds:

$$E_{n,m,s} = (2n + |m| + 1)\hbar \cdot \sqrt{\omega_0^2 + \frac{\omega_c^2}{4}} - \frac{m}{2} \cdot \hbar\omega_c + s \cdot g\mu_B|\vec{B}|. \quad (5.103)$$

This is called the Fock–Darwin spectrum of a quantum dot. Assuming $g = 0$, the resulting state energies are displayed in Fig. 5.24(d). Importantly, one observes crossing points of energy levels, i.e., the seventh electron added to the quantum dot (dotted, red line in Fig. 5.24(d)) would firstly be placed into the sixfold degenerate state at $E \approx 9$ meV, before it will occupy, consecutively with increasing $\vec{B}$, the states $(0, 2)$, $(0, -1)$, and $(0, 3)$. The crossing points are decisive for the determination of the strength of particular electron-electron interaction effects (Section 5.5.1.2).

5.5.1.2 Interaction effects in quantum dots

The single particle picture as described above neglects the Coulomb interaction between the electrons. This is massively wrong, if more than one electron reside within the quantum dot. We consider a quantum dot with a typical $\hbar\omega_0 = 1$ meV for $m^* = m_e$, i.e., a potential $E_{\text{pot}}(\vec{x}) = 7.25 \mu\text{eV}/\text{nm}^2 \cdot \vec{x}^2$ and, thus, a lateral extension $2r_{00}$ of the ground state of $2r_{00} \approx 2|\vec{x}(E_{\text{pot}} = 1 \text{ meV})| \approx 11$ nm. The Coulomb interaction between the two possible electrons within this state at a dielectric constant of $\varepsilon = 1$ can be estimated classically to be $E_{\text{ee}}^{\text{cl}} \approx 1/(4\pi\varepsilon\varepsilon_0) \cdot e^2/2r_{00} = 130$ meV, i.e., two orders of magnitude larger than the confinement energy $\hbar\omega_0$. Only at much smaller confinement areas with radii of 0.1 nm (atoms), the confinement energy dominates the Coulomb repulsion. Within quantum dot experiments, the employed materials mostly exhibit $m^* < m_e$ and $\varepsilon > 1$, such that the relation $\hbar\omega_0/E_{\text{ee}}^{\text{cl}} \propto \varepsilon/m^*$ is larger (InGaAs: $m^* \approx 0.04m_e$ and $\varepsilon \approx 13$). Nevertheless, the Coulomb energy between the electrons dominates for quantum dots with diameters above ~ 100 nm.

The primary consequence of the Coulomb interaction is that it costs charging energy E_c to bring an additional electron to the quantum dot. As discussed in Sec-

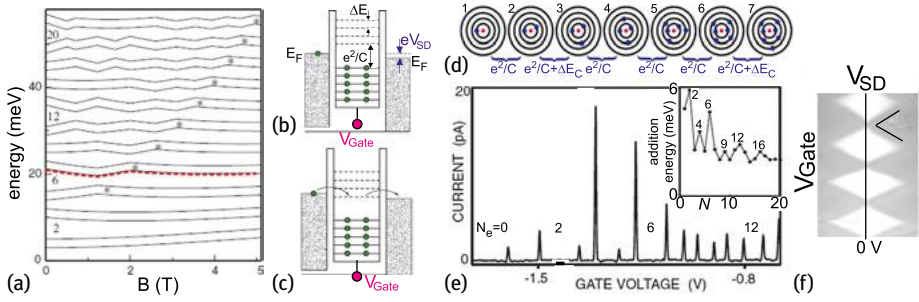


Fig. 5.25: (a) Single particle energies of a quantum dot, where the energy of the n th level is shifted by $n \cdot e^2/C$ with respect to Fig. 5.24(d). The kinks within the curves are identical to the crossing points of Fig. 5.24(d). The stars mark the last crossing point of a particular state. (b) Representation of the states within a quantum dot. The occupied single particle states are represented by lines with filled green circles. They differ by the confinement energy. The unoccupied states are additionally shifted by the charging energy e^2/C , since its occupation would cost this energy in addition to the confinement energy. Grey areas are filled electronic states up to the Fermi levels of the contacts. If no state of the dot is between the Fermi levels of source and drain electrode such that current can flow as long as $k_B T \ll e^2/2C$. The pink lever V_{Gate} indicates that the ensemble of states can be shifted by an external electrostatic potential. (c) Same as (b), but at a gate voltage V_{Gate} , where unoccupied states of the dot are found between the Fermi levels of source and drain electrode such that current can flow. (d) Schematic of quasi-atomic orbitals of a quantum dot, that correspond to the degenerate single particle states of the dot. Electrons (filled blue circles) are distributed consecutively onto the orbits (rings). The curved brackets below the orbits are marked with the energy differences between the different numbers of electron occupation of the dot at $B = 0$ T, where ΔE_C is the difference between adjacent single particle energies $E_{n,m}$. (e) Current between source and drain electrode of a quantum dot, such as in Fig. 5.24(a), as a function of gate voltage V_{Gate} at fixed voltage between source and drain $V_{\text{SD}} \leq 1$ mV. Inset: Energy distance between neighboring peaks. (f) Coulomb diamonds: the current is displayed in greyscale as a function of V_{Gate} and V_{SD} , i.e., the darker the area, the larger the current. The two black lines mark so-called excited states. (a), (d), (e) [160].

tion 3.3.1.2, one can estimate $E_c \approx e^2/C$ with C being the capacitance of the quantum dot with respect to its surrounding. Within the capacitance model, E_c is the same for all electrons. In order to get an additional electron onto the quantum dot, confinement energy and charging energy are required. They are provided by a positively charged side gate (Fig. 5.24(a)), that attracts electrons to the dot. The single particle states are, thus, energetically shifted by e^2/C with respect to each other as shown in Fig. 5.25(a). Figure 5.25(b) and (c) display the energies of different states for two different side gate voltages V_{Gate} (see also Fig. 3.3(d)–(e)). The source and drain electrodes on the left and right are filled up to its corresponding E_F (grey area). Within the quantum dot (middle), the occupied states (green circles on full lines) are separated by the single particle energies $E_{n,m}$. In principle, the lowest level would be filled by two electrons, the second lowest level by four electrons, and so on, but this is reduced within the sketch to two electrons per level for the sake of simplicity. The electron-electron interaction between the occupied states is an offset in energy being the same for all

occupied states. In order to put another electron onto the quantum dot, one has to pay the energy difference to an unoccupied single particle level and the charging energy e^2/C . The corresponding energy levels are displayed in Fig. 5.25(b) as broken lines, located above the Fermi levels of source and drain electrode, i.e., the current through the quantum dot amounts to $I \approx 0$ A at low temperature. The energy of the quantum dot states with respect to source and drain can be changed by the side gate (eq. (3.4)). In a first approximation, all the states move parallel as shown in Fig. 5.25(c). As soon as an unoccupied state is located between the Fermi levels of source and drain, an electron can tunnel onto the quantum dot, thereby shifting all other states upwards by e^2/C . Afterwards, the electron can tunnel into the drain, such that $I > 0$ A. If the originally unoccupied state is moved below E_F of the drain, the level gets permanently occupied having a distance of e^2/C to all unoccupied states again, such that the current disappears again. If an additional charging energy of e^2/C is contributed by the side gate, the next unoccupied level gets shifted between the Fermi levels of source and drain. This leads to the rather regular peaks in current as a function of side gate voltage as displayed in Fig. 5.25(e) (see also Fig. 3.3(f)). Recall that these current peaks are called Coulomb peaks. These Coulomb peaks have been exploited as charge detectors for a neighboring qubit (Section 3.3.1.3).

Increasing the voltage between source and drain V_{SD} , increases the area of side gate voltages where a state is located between the E_F 's of source and drain, such that the area of side gate voltages with $I > 0$ A continuously increases, until the source-drain voltage V_{SD} is larger than e/C and current can flow independently of V_{Gate} . Displaying the current in grey scale as a function of V_{Gate} and V_{SD} reveals the characteristic Coulomb diamonds of suppressed current as shown in Fig. 5.25(f) (see also Fig. 3.3(g)).

A closer inspection of Fig. 5.25(e) reveals an irregularity of the peak distances, which is additionally displayed in the inset. The irregularity is firstly a consequence of the confinement energies, which are often called single particle energies. If a confinement energy level, respectively an orbit of the artificial atom, is completely filled, the distances between the Coulomb peaks are larger. This is the case after two electrons for the first confinement energy, $2 + 4 = 6$ electrons for the second confinement energy, $2 + 4 + 6 = 12$ electrons for the third confinement energy, and so on. After these fillings, the next electron requires the energy penalty e^2/C for charging and the confinement energy ΔE_C , being the energy difference between the last and the next orbit. Such a successive filling of orbits is sketched in Fig. 5.25(d) including the required energies to add the next electron. The expected larger gate voltage distances of peaks after 2, 6, and 12 electrons (filled orbit) are indeed observed as most clearly shown in the inset of Fig. 5.25(e).

The inset, moreover, exhibits larger energy distances after filling 4, 9 and 16 electrons into the dot. This indicates a second consequence of the electron-electron interaction, the exchange interaction (Section 4.4). As in real atoms, it leads to Hund's rules. The first rule is that filling an electron orbit with electrons of the same spin is

more favorable than mixing the spins due to the antisymmetry of the many-particle wave function. The symmetric spin function for parallel spins implies an antisymmetric spatial wave function with larger electron-electron distances and, hence, reduced Coulomb repulsion. Thus, parallel spins are energetically favorable for the same orbit. They are also favorable for different orbits, if the gain in exchange energy is larger than ΔE_C . Because of Pauli's principle, parallel spins in the same orbit are only possible up to half filling of the orbit.

For example, within the third orbit of the quantum dot (degeneracy: 6), we can put electron 7, 8 and 9 with the same spin and, thus, with reduced electron-electron repulsion energy, respectively reduced charging energy E_C . After the 9th electron, the spin of the 10th electron must be antiparallel to 7, 8, and 9, such that the spatial wave function gets symmetric for interchanging electron 10 with electron 7, 8, or 9, leading to a maximum in E_C . This causes the peaks at 4, 9 and 16 within the inset of Fig. 5.25(e). More formally, the two-particle exchange energy of a pure two-electron system E_{Ex} is given by the first term of $\tilde{J}$ of eq (4.13) (Section 4.4) reading $E_{\text{Ex}} \propto \pm \int \int \psi_i^*(\vec{x}_1) \psi_j^*(\vec{x}_2) \cdot 1/|\vec{x}_1 - \vec{x}_2| \cdot \psi_i(\vec{x}_2) \psi_j(\vec{x}_1) d^3\vec{x}_1 d^3\vec{x}_2$. It has to be added to e^2/C with $- (+)$ sign for parallel (antiparallel) spins.

In turn, the exchange energies as well as the single particle energies of the states in the quantum dot can be estimated from the distance variations of the Coulomb peaks. Another possibility to probe the single particle states is called spectroscopy of the excited states and uses the Coulomb diamonds. Increasing the source-drain bias V_{SD} allows having two (or more) energy levels within the energy window between the two E_F 's of source and drain. This increases the tunneling current, since the probability to tunnel into or out of the dot scales with the number of available states for tunneling. Thus, if an additional state contributes by increasing V_{SD} , the current increases. This is visible as a step in the current. The step runs parallel to the onset of current within the Coulomb diamonds as displayed for one example as a black line in Fig. 5.25(f). The reason is, that the energy distance between the lowest unoccupied energy state, to be shifted into the V_{SD} window, and the next unoccupied state remains roughly the same, independent of V_{Gate} . In order to increase the visibility of these steps, one mostly displays dI/dV_{SD} . This transforms the steps into peaks, respectively into lines within a $(V_{\text{SD}}, V_{\text{Gate}})$ plot. Hence, one finds additional lines outside the Coulomb diamonds (Fig. 3.3(g)). Notice, that it does not matter, if two occupied states within the V_{SD} window increase the probability to tunnel out of the quantum dot or if two unoccupied states increase the probability to tunnel into the quantum dot, at least, as long as the two tunnel barriers are not too different. Notice further, that the tunneling probability can also be increased by other additional processes, e.g., tunneling and exciting an optical phonon at the same time within a second order process. This also leads to additional steps in the Coulomb diamond pattern, if getting energetically possible within the V_{SD} window, e.g., if eV_{SD} gets larger than the required phonon energy. Consequently, the more general term of excitation spectroscopy is mostly used.

The role of the exchange interaction can be probed more directly by applying a $\vec{B}$ field. Figure 5.26(a) shows the principle for four electrons within the quantum dot. The degenerate states at $B = 0$ T belonging to the second energy level, $(0, 1)$ and $(0, -1)$, increasingly separate with increasing $\vec{B}$. At small $\vec{B}$, the exchange interaction dominates, such that the two electrons, besides the two electrons within the state $(0, 0)$, exhibit parallel spin, thus, occupy both levels, $(0, 1)$ and $(0, -1)$. With increasing $\vec{B}$, this gets increasingly unfavorable, due to the different single particle energies of $(0, 1)$ and $(0, -1)$, until one spin flips in order to occupy the lower level $(0, 1)$, too. This happens as soon as the exchange energy E_{ex} gets smaller than the energy distance between the single particle states $(0, 1)$ and $(0, -1)$. Consequently and counter-intuitively, the magnetization (= magnetic dipolar moment/volume) of the artificial atom dramatically decreases above a certain $\vec{B}$ field.

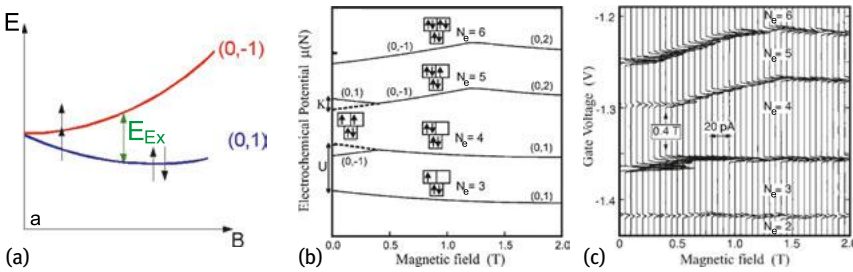


Fig. 5.26: (a) Principle of a triplet-singlet transition within a quantum dot. As soon as the energy distance between the single particle levels (blue and red line) is larger than the exchange energy E_{ex} , the two electrons occupy the same energy level with opposite spin. (b) Course of the single particle energies marked by quantum numbers (n, m) , if the exchange interaction is included in addition to the energies displayed in Fig. 5.25(a). The electron configurations for the individual states are marked using arrows to represent the spin direction within a particular single particle level (n, m) displayed as a box. (c) dI/dV_{Gate} curves recorded for different $\vec{B}$ as marked on the x axis. The state configuration can be deduced from the course of the steps. N_e marks the number of electrons within the quantum dot. The spin flip at $B = 0.4$ T is marked by a double arrow. (b), (c) [161].

The resulting course of Coulomb peaks including charging energies, single particle energies, and exchange energies is shown schematically in Fig. 5.26(b) with the number of electrons N_e present between the charging energies marked. Figure 5.26(c) shows corresponding experimental data. Both reveal a kink in slope for the charging lines (or steps) surrounding $N_e = 4$ at $B \approx 0.4$ T. This kink indicates the transition from a triplet state (two spins parallel) to a singlet state (two spins antiparallel).

The quantum dot also exhibits reverse singlet-triplet transitions, e.g., for $N_e = 2$ electrons within the dot. If the two levels $(0, 0)$ and $(0, 1)$ in Fig. 5.24(d) get closer to each other than E_{ex} , one electron will change from $(0, 0)$ to $(0, 1)$ while switching the spin in order to gain by the reduced electron-electron repulsion (exchange energy) for parallel spins.

Many other effects of electron-electron interaction can be studied within quantum dots. This includes such effects that are driven by correlation terms of the Hamiltonian, such that more than two states are involved in the corresponding interaction term (scattering process) according to $E_{xc} = u_{ijkl}\hat{c}_i^\dagger\hat{c}_j^\dagger\hat{c}_k\hat{c}_l$. The most prominent example is the Kondo effect. Generally, it describes the interaction of two electrons via two extended and two localized states. Both electrons exhibit a fermionic spin, e.g., $S = 1/2$. It turns out that a scattering process of the second order, i.e., the incoming extended state electron switches its spin twice ($\uparrow \rightarrow \downarrow$ and $\downarrow \rightarrow \uparrow$), while the localized state does the spin flip twice in opposite order ($\downarrow \rightarrow \uparrow$ and $\uparrow \rightarrow \downarrow$) reveals a diverging energy gain for the system at $T = 0$ K, similarly to the divergence found for a charge density wave in 1D (eq. (5.13), Section 5.2.1). This drives an increased density of extended states to E_F , such that the states can gain from this attractive scattering process.⁷⁴ The result is a peak in the density of states at E_F called the Kondo peak. For quantum dots, the dot itself represents the localized spin, if filled, e.g., with an odd number of electrons, and the extended spin is given by electrons from the electrodes. The process of double spin-flip (second order scattering process) can transfer an electron from the source to the drain, even if no QD state is within the V_{SD} window. Formally, a virtual excitation of a quantum dot electron into a level above E_F of the source and the drain takes place, while the itinerant electron is scattered from a source state to a drain state. Hence, the Kondo effect becomes directly visible, as a conductance, that increases with decreasing temperature in areas of side gate voltages away from the Coulomb peaks. Since the process is most favorable for identical energies for initial and final state (the localized state cannot change its energy by flipping the spin twice), the Kondo conductance is restricted, moreover, to $V_{SD} \approx 0$ mV. Finally, the Kondo effect depends exponentially on the exchange interaction between localized and extended electrons, such that it only appears if the tunnel barriers are relatively weak [162].

Since, within the quantum dots, all parameters such as V_{SD} , strength of tunnel barrier, temperature, and spin occupation of the dot are tunable, the quantum dots reveal an excellent test bed for studying the Kondo effect.

5.5.2 Electron-electron interactions in two dimensions

After this introduction into electron-electron interaction processes using artificial atoms, we will discuss electron phases of itinerant electrons employing two dimensions, that are driven by electron-electron interactions. The first two phases describe a particular arrangement driven by the Coulomb repulsion and the exchange interaction, respectively. The final phase, described in Section 5.5.2.3, exhibits completely

⁷⁴ Recall that an interaction, as described by a perturbation term, requires a scattering of the electrons between occupied and empty states. The most energy efficient way for such a scattering takes place at E_F .

novel types of quasi-particles with surprising properties as, e.g., a measurable fractional charge $Q = e/n$ ($n \in \mathbb{N}$).

Before starting, it is important to recall that a many-particle wave function $\Psi_{\text{el}}(\vec{x}_1, \vec{x}_2, \dots, \vec{x}_{N_e}, t)$ can consist of more than N_e single particle wave functions, such that the relation between electrons and single particle wave functions is not biunique. Transitions between different many-particle wave functions, e.g., after absorbing a photon (Section 2.3.1), can, thus, not be exactly mapped to single particle transitions, which eventually leads to astonishing results in measurements as, e.g., the exchange of fractional charge e/n between two electron systems.

5.5.2.1 Wigner crystal

A classically understandable example of a phase driven by electron-electron interaction is the Wigner crystal. The charge density $n_e(\vec{x})$, that evolves at low temperature, is shown in Fig. 5.27(a). A hexagonal pattern appears with one electron per maximum similar to patterns of close-packed atomic crystals. One can show that such a regular arrangement is the optimal pattern for classical electrons at $T = 0$ K.

For the sake of simplicity, we will restrict ourselves to 1D showing that the favorable position of a charged object with charge Q_1 between two identically charged objects with charge Q_2 , which are in mutual distance x_0 , is the central position at $x = x_0/2$. Therefore, we determine the minimum of the potential repulsion energy E_{pot} for Q_1 :

$$\frac{dE_{\text{pot}}(x)}{dx} = \frac{d}{dx} \left(\frac{Q_1 Q_2}{4\pi\epsilon_0} \cdot \left(\frac{1}{|x|} + \frac{1}{|x-x_0|} \right) \right) = 0 \implies x = x_0/2 \quad \frac{d^2 E_{\text{pot}}(x_0/2)}{dx^2} > 0 \quad (5.104)$$

The repulsion between the electrons, thus, keeps electrons at identical distances with respect to each other, similarly as the interaction between atoms in a solid keeps the atoms at equal distances. At $T > 0$ K and constant electron density n_e , the minimum of free energy $F = U_{\text{int}} - TS$ (U_{int} : internal energy) can favor the disordered phase because of the corresponding gain in entropy S , that eventually leads to a melting temperature T_{melt} of the Wigner crystal. Since the electrons still strongly interact above T_{melt} , a short-range order remains and one calls the phase above T_{melt} a Landau liquid or Fermi liquid (Section 4.2).

Quantitatively, the melting temperature depends on the ratio of the ordering electron-electron repulsion energy being on average $E_{\text{ee}}^{\text{cl}} \propto 1/\bar{r}_{\text{ee}} \propto n_e^{1/d}$ ($\bar{r}_{\text{ee}}$: average distance between two electrons, d : dimension of the system) and the thermal energy $E_{\text{therm}} \propto k_B T$ preferring a disordered system, such that one gets in first approximation $T_{\text{melt}} \propto n_e^{1/d}$.

Quantum mechanically, the periodic arrangement requires additional kinetic energy, since the periodic charge arrangement is realized by the superposition of plane waves with the same phase. Consequently, one needs multiples of the wave vector $k_0 = 2\pi/(2\bar{r}_{\text{ee}})$, respectively a kinetic energy of $E_{\text{kin}} \propto \hbar^2 k_0^2 / 2m^* \propto n_e^{2/d}$. Since

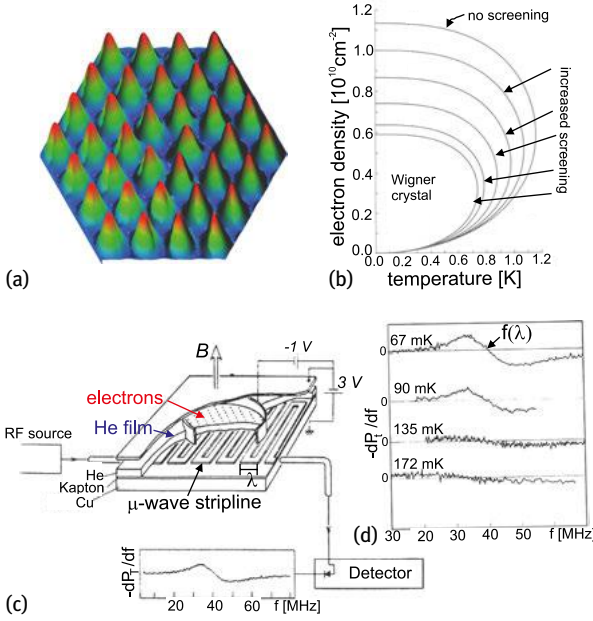


Fig. 5.27: (a) Electron density $n_e(\vec{x})$ of a 2D Wigner crystal. (b) Phase diagram of a 2D electron system. The inner curves include the screening of electron-electron interaction by the substrate. (c) Experimental setup used to probe a Wigner crystal of electrons deposited on suprafluid helium with exemplary absorption spectrum below (P_t : transmitted power, f : frequency). (d) Microwave absorption spectra as a function of frequency measured by the setup in (c) at different T as marked. The frequency $f(\lambda)$ marks the probed phonon of the Wigner crystal at $n_e = 6 \cdot 10^7 \text{ cm}^{-2}$. (b) After [163]. (c)–(d) [164].

this energy cost increases more strongly with n_e than the gain in Coulomb energy ($E_{ee}^{\text{cl}} \propto n_e^{1/d}$), a critical density n_c appears, above which the Wigner crystal is not the ground state even at $T = 0 \text{ K}$.

The resulting phase diagram for $d = 2$ is shown in Fig. 5.27(b). The different curves result from the fact that the probed 2D electron system has to be supported by a 3D crystal which screens the electron-electron repulsion ($\epsilon > 1$, $\lambda_{\text{screen}} < \infty$). Consequently, the area of the Wigner crystal within the phase diagram shrinks.

The calculated phase diagram reveals a required temperature of $T < 1 \text{ K}$ and a required electron density of $n_e < 10^{10}/\text{cm}^2$, i.e., $\bar{r}_{ee} > 100 \text{ nm}$. This differs from the above approximation $E_{ee}^{\text{cl}} > E_{\text{kin}} \implies \bar{r}_{ee} > 1 \text{ nm}$. Such a discrepancy is also known for atomic crystals. The reason is that melting does not happen by pushing individual atoms or electrons out of its equilibrium position, but starts already, if dislocations (collective shifts of electrons out of its lattice positions) within the ordered lattice become favorable in terms of free energy. This melting by dislocations results in a T_{melt} much smaller than E_{ee}^{cl}/k_B . Nevertheless, the ratio $E_{ee}^{\text{cl}}/E_{\text{kin}}$, respectively $E_{ee}^{\text{cl}}/(k_B T)$, re-

mains the decisive ratio, albeit with a prefactor of ~ 0.01 , for the determination of the phase area where the electrons crystalize.

Experimentally, Wigner crystals were primarily observed on thin films of liquid He. An experimental setup is shown in Fig. 5.27(c). Helium is firstly deposited on a flat substrate at $T < 1$ K. Helium remains a liquid down to lowest temperatures and forms a flat film (thickness ~ 100 nm) on the substrate. Onto this film, electrons are deposited that are created, e.g., by thermal emission from a hot filament. The He film is cut by a circular, negatively charged electrode which keeps the electrons within its area and can be used to change n_e . Cooling the whole film below $T < T_{\text{melt}}$ can now create a 2D Wigner crystal of electrons on the He film. To probe the existence of this Wigner crystal, an elegant method measures the lattice vibrations of the crystal, which disappear above T_{melt} . Therefore, a meandering metallic stripe (μ -wave stripline) with distance between the stripes of $L \approx 20$ μm is positioned below the He film. On the stripe, one applies an ac voltage with tunable frequency f . This ac voltage periodically attracts and repels the electrons being about 3 μm above the stripe, such that the electron lattice gets periodically compressed and expanded. If this excitation is in resonance with a vibration of the lattice, i.e., if f matches the frequency of the phonon of the Wigner crystal with wave length $\lambda = L$, the applied ac voltage loses energy and, thus, the ac resistance of the metallic stripe increases. One measures the transmitted power $P_t(f)$ along the stripe as a function of f . More precisely, one determines the ratio of $P_t(f)$ with and without the electrons on top of the He film and differentiates this ratio with respect to f by lock-in technique in order to improve the signal-to-noise ratio. The result for different T but fixed n_e is shown in Fig. 5.27(d). A resonance appears below $T = 0.1$ K at $f \approx 35$ MHz, that indicates the phonon frequency of the Wigner crystal at $\lambda = 20$ μm . The resulting $f(\lambda)$ nicely agrees with calculations of the phonon dispersion of a 2D Wigner crystal.

A schematic of a Wigner crystal on top of a He film deposited on a metal is shown in Fig. 5.28(a). Because of the attraction of the electrons by the image charge within the metal, the He is repelled around each electron such that a little valley appears. If the electron lattice gets moved by an electric field, the valley structure will be moved with it, such that the viscosity of the He film contributes to the electrical resistance of the Wigner crystal. The corresponding resistance is shown in Fig. 5.28(b) as measured with the experimental setup sketched in the inset. One can apply a voltage pulse to electrode B1 and measure the time delay until it appears at electrode B3. In practice, the phase difference $\Delta\varphi$ between an ac voltage coupled to B1 and the resulting ac voltage at B3 is determined. The additional grounded electrode in between reduces the direct capacitive coupling between B1 and B3. The measured time delay $\Delta t = \Delta\varphi/(2\pi f)$ can be used to determine the average drift velocity $v_d = L_{13}/\Delta t$ of the Wigner crystal with L_{13} being the distance between B1 and B3. Using the applied voltage amplitude to numerically calculate the electric field $|\vec{E}|$ within the Wigner crystal, and knowing n_e , the specific conductivity reads $\sigma = n_e e v_d / |\vec{E}|$. This determines the resistance of the Wigner crystal after considering additional geometrical factors. Figure 5.28(b) shows

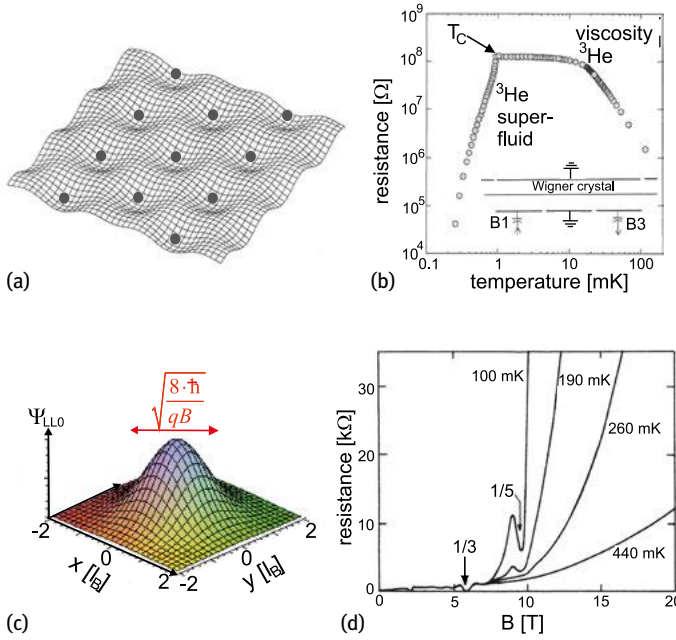


Fig. 5.28: (a) Schematic of a Wigner crystal of electrons (grey balls) on a liquid He film, deposited on a metallic substrate. (b) Resistance of a Wigner crystal on top of liquid He as a function of T , $n_e = 2 \cdot 10^8/\text{cm}^{-2} \Rightarrow T_{\text{melt}} = 0.27$ K. Inset: Schematic of the experimental setup. (c) $\psi_i(\vec{x})$ of a single particle state within the Landau level with lowest energy. (d) Resistance of a 2D electron system within GaAs as a function of $\vec{B}$ field perpendicular to the plane of the electron system at different T , $n_e = 4 \cdot 10^{10} \text{ cm}^{-2}$, electron mobility $\mu = 3 \cdot 10^5 \text{ cm}^2/(\text{Vs})$. Fractional numbers with arrows indicate the filling of the lowest spin-polarized Landau level (filling factor ν). (b) [165], (d) [166].

that the resistance at high temperature decreases with increasing T , since the viscosity η of the ^3He decreases with T ($\eta \propto T^{-2}$). Moreover, it drops dramatically below $T \approx 1$ mK that marks the transition of ^3He into the suprafluid phase that exhibits $\eta = 0 \text{ Ns/m}^2$. Thus, obviously, the Wigner crystal can also be used to probe properties of the underlying He film.

Within solids housing 2D electron systems, the Wigner crystal has not been unambiguously observed at $B = 0$ T.⁷⁵ This is primarily due to the electrostatic disorder potential induced by defects, that competes with $E_{\text{ee}}^{\text{cl}}$. In addition, the 2D electron systems with lowest disorder are housed in GaAs/AlGaAs heterostructures (Section 1.3.1), that exhibit a large dielectric constant $\epsilon = 13$ decreasing $E_{\text{ee}}^{\text{cl}}$ and a low effective mass $m^* = 0.07$ increasing E_{kin} . Consequently, the Wigner crystal is expected at

⁷⁵ Some evidence for Wigner crystal formation has been found in 1D electron systems in solids housed, e.g., in carbon nanotubes [167].

$n_e < 10^8/\text{cm}^2$, respectively $\bar{r}_{ee} > 1\text{ }\mu\text{m}$, $E_{ee}^{\text{cl}} < 0.1\text{ meV}$, and $T_{\text{melt}} < 10\text{ mK}$. In turn, one can roughly estimate that a potential disorder with amplitude 0.1 meV and length scales of $1\text{ }\mu\text{m}$ would destroy the Wigner crystal in favor of an arrangement profiting by placing the electrons into the minima of the electrostatic disorder potential (Section 5.3.1).

In order to create a Wigner crystal in solids at larger density, one has to reduce E_{kin} . The most simple way is applying a $\vec{B}$ field perpendicular to the 2D plane. If $\vec{B}$ is large enough to put all electrons in the lowest spin polarized Landau level ($\vec{B}[T] > n_e/(2.4 \cdot 10^{10}/\text{cm}^2)$)⁷⁶, the kinetic energy of all electrons is identical. It is simply the zero point energy $\hbar\omega_c/2$. The corresponding probability density of a single particle state in the lowest Landau level is shown in Fig. 5.28(c). It is a 2D Gaussian function with a full width at half maximum of about the cyclotron diameter $d_c = 50\text{ nm}/\sqrt{B[T]}$. As soon as $d_c \ll \bar{r}_{ee}$, the Wigner crystal does not require any kinetic energy anymore. Additionally, the antisymmetrization of the spatial many-particle wave function for parallel spins favors a disjunct distribution of the corresponding Gaussian single particle states. Thus, using $B = 10\text{ T}$, a Wigner crystal should be stable at about $n_e < 2 \cdot 10^{11}\text{ cm}^{-2}$. In reality, it turns out that the Wigner crystal in a $\vec{B}$ field competes at the smallest potential disorders with particular liquid phases that reduce the total electron repulsion energy E_{ee} even better than the Wigner crystal (Section 5.5.2.3). Consequently, the Wigner crystal appears only at fillings of the lowest spin polarized Landau level of less than $1/5$. Since this Wigner crystal is typically pinned at the smallest potential disorders, very similar to the charge density waves of Section 5.2.1, it exhibits an insulating conductivity $\sigma(T) \propto \exp(-E_{\text{pin}}/k_B T)$ (E_{pin} : pinning energy at a defect). This is indeed found, e.g., in the data of Fig. 5.28(d) for $n_e = 4 \cdot 10^{10}\text{ cm}^{-2}$. At $B > 10\text{ T}$, corresponding to a filling of the lowest spin polarized Landau level by less than $1/5$, the resistance strongly increases with decreasing temperature. The additional observation that the resistance gets strongly reduced at sufficiently large voltage (not shown), corresponding to a depinning of the Wigner crystal from the defects as well as the observed frequency dependence of the resistance corresponding to a phonon excitation of the Wigner crystal (as in Fig. 5.27(d)) confirms the existence of the Wigner crystal [165].

5.5.2.2 Stripe phase

A second electron phase in 2D electron systems in a $\vec{B}$ field driven by electron-electron interaction is the so-called stripe phase. It is strongly influenced by the exchange interaction and appears, if an energetically higher, spin-polarized Landau level is half filled. Surprisingly, electrons within the stripe phase prefer to be closer to each other than given by their average distance in order to reduce their overall repulsion. To understand this counterintuitive behavior, we firstly recall the behavior of electrons in

⁷⁶ The degeneracy of a spin polarized Landau level is $n_{\text{LL}}/2 = eB/h$ (Section 5.3.2).

a $\vec{B}$ field without electron-electron interactions as described in Section 5.3.1 and 5.3.2. The $\vec{B}$ field implies Landau and spin levels with energies $E_{n,s} = (n + 1/2) \cdot \hbar e B / m^* + s \cdot g \mu B$ with $n = 0, 1, 2, \dots$ and $s = \pm 1/2$. The levels are broadened by the disorder potential (Fig. 5.10(g)). The corresponding density of states is shown again in the lower part of Fig. 5.29(a). The ratio between electron density n_e and the degeneracy density of a spin-polarized Landau level $n_{LL}/2 = e \cdot B/h$ is called the filling factor

$$\nu := \frac{n_e}{n_{LL}/2} = \frac{n_e}{eB/h}. \quad (5.105)$$

Recall that $n_{LL}/2$ is identical to the areal density of pairs of magnetic flux quanta penetrating the 2D system. For $\nu = 9/2$, the lowest two Landau levels are completely filled and, additionally, half of the lower spin level of the third Landau level ($n = 3$, $s = -1/2$) is filled with electrons. One could expect that the electrons within the ($n = 3$, $s = -1/2$) level distribute themselves in regular lateral distances as in a Wigner crystal (Section 5.5.2.1) in order to reduce their electron-electron repulsion energy. However, the charge densities of individual electronic states in Landau levels with $n > 0$ are ring structures (Fig. 5.29(b)) and not Gaussian functions making their ordering less straightforward.

The radius of the ring corresponds to the quantized cyclotron radius $r_c = \sqrt{2n+1} \cdot l_B$ and its width is about the magnetic length $l_B = \sqrt{\hbar/eB}$. The resulting Wigner crystal at half filling (e.g., $\nu = 9/2$) is sketched in Fig. 5.29(c). The boxes mark areas containing a flux of h/e with density $n_{LL}/2$ and the red rings mark the occupied states of the half-filled Landau level. Obviously, the states overlap strongly, i.e., the resulting direct Coulomb repulsion, called Hartree energy E_{Hartree} (eq. (4.12)),

$$E_{\text{Hartree}} = \iint |\psi_i(\vec{x}_i)|^2 \frac{e^2}{4\pi\epsilon\epsilon_0 \cdot |\vec{x}_i - \vec{x}_j|} |\psi_j(\vec{x}_j)|^2 d^2\vec{x}_i d^2\vec{x}_j \quad (5.106)$$

deviates strongly from the Coulomb repulsion of point charges in the center of the boxes. A more detailed calculation reveals that the Coulomb repulsion energy barely changes if two overlapping rings are pushed closer together. The screening of the electron-electron repulsion by electrons from lower Landau levels, moreover, implies that the Coulomb repulsion energy drops dramatically, if the distance between two electron ring centers gets larger than $2r_c$. Consequently, the arrangement of the rings in stripes, as shown in Fig. 5.29(d), does increase the repulsion with neighbors in the stripe only slightly, but, at the same time, reduces the interactions with electrons in neighboring stripes more strongly. It, thus, appears favorable to separate the stripes as much that rings of neighboring stripes do not overlap anymore.

Notice that the total charge density $n_e(\vec{x})$ is only slightly changed by this rearrangement. The amplitude of the resulting charge density wave perpendicular to the stripes (Fig. 5.29(a), top), is only 10–20% of the average charge density within the half-filled Landau level. One can calculate more precisely, that the optimal wave length of the charge density wave is $\lambda \approx 2.7 \cdot r_c$ as marked in Fig. 5.29(e), where only the centers of the electron rings are marked as dots. For electrons in the third Landau level at

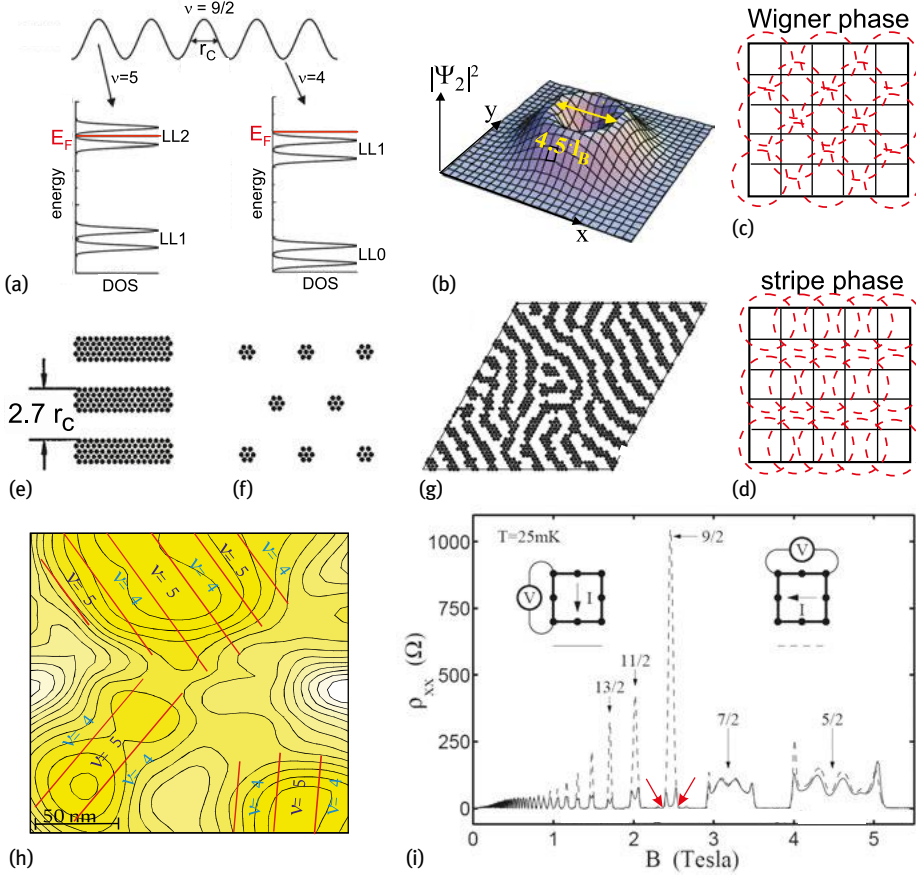


Fig. 5.29: (a) Schematic of the stripe phase. Top: Charge density $n_e(x)$ perpendicular to the stripes at $\nu = 9/2$ with marked cyclotron radius r_c . Bottom: Local density of states corresponding to the areas marked by arrows with indicated local filling factor. (b) $\psi_i(\vec{x})$ of a state in Landau level LL2. (c), (d) Schematic of the ordered electron rings in a Wigner crystal (c) and in the stripe phase (d) for half filling of a spin polarized Landau level LL2, i.e., at $\nu = 9/2$. (e) Sketch of the stripe phase obtained at half filling of higher Landau levels, where the dots symbolize the center of the rings in (c), (d). (f) Bubble phase observed at a quarter filling of a spin polarized Landau level. (g) Numerically simulated stripe phase on a triangular lattice. The stripes meander which indicates that the system is not in the energetic ground state but only close to it. (h) Sketch of a disordered stripe phase (red lines with indicated local filling factor ν in between the lines) in the presence of an electrostatic disorder potential as displayed in color scale with black equipotential lines. (i) Longitudinal resistance of a 2D electron system in GaAs as a function of $\vec{B}$. The resistance is measured for current along two perpendicular directions (full and dashed lines) as marked in the insets, $n_e = 2.3 \cdot 10^{11} \text{ cm}^{-2}$, $\mu = 9 \cdot 10^7 \text{ cm}^2/(\text{Vs})$. Fractional numbers with arrows mark the filling factor ν . Red arrows indicate likely bubble phases. (e)–(g) [168], (i) [169].

$B = 10$ T one finds $\lambda \approx 60$ nm. Quantitatively, the repulsion term E_{Hartree} of the stripe phase at this λ does not differ from E_{Hartree} of a Wigner crystal, that would imply that a mixture of both should appear in experiment. The stripe phase is, however, favored by the exchange interaction⁷⁷

$$E_{\text{Ex}} = \int \int \psi_i^*(\vec{x}_m) \psi_j^*(\vec{x}_n) \frac{e^2}{4\pi\epsilon\epsilon_0 \cdot |\vec{x}_n - \vec{x}_m|} \psi_j(\vec{x}_m) \psi_i(\vec{x}_n) d^2\vec{x}_m d^2\vec{x}_n. \quad (5.107)$$

For parallel spins of electrons, as present in a spin polarized Landau level, E_{Ex} contributes with a negative sign decreasing the electron-electron repulsion. Consequently, the optimal arrangement favors small E_{Hartree} and large E_{Ex} . A large E_{Ex} , however, requires a direct overlap of the contributing wave functions (eq. (5.107)), i.e., $\int \int \psi_i^*(\vec{x}_n) \cdot \psi_j(\vec{x}_n) d^2\vec{x}_n \gg 0$, contrary to E_{Hartree} , that barely depends on the overlap at small distance. Thus, E_{Ex} favors the strong overlap of the wave functions within the stripe phase. The stripe phase is, thus, a consequence of both, the strong reduction of the direct Coulomb repulsion E_{Hartree} for large distances between the electrons ($> 2r_c$) and the increasing exchange interaction for increasing overlap of the occupied wave functions, that is favorable for electrons with parallel spin.

The quantitative comparison between the different phases requires complex, partly numerical calculations [168, 170] and will not be described in detail. In short, firstly one transforms the Schrödinger equation of all electrons in a way, such that the completely filled Landau levels are approximated by a wave number ($q = |\vec{q}|$) dependent dielectric constant $\epsilon(q)$. For the electron-electron interaction within the half-filled Landau level, one obtains $\epsilon(q) = \epsilon\epsilon_0 \cdot (1 + 2/(qa_B^*)) \cdot (1 - J_0^2(qr_c))$ with effective Bohr radius $a_B^* = \epsilon m_e / m^* \cdot 0.5 \text{ \AA}$ and zeroth Bessel function J_0 . The completely filled levels are, thus, described as a polarizable background. Then, one calculates wave vector dependent interaction energies E_{Hartree} and E_{Ex} for the half filled Landau level using the Fourier transform of the interaction potential $\Phi_{\text{el}}(q) = 2\pi e^2 / \epsilon(q) q$. This results in an oscillating Hartree term being approximately (besides a q -independent term): $E_{\text{Hartree}}(q) \propto 1/(2 + q \cdot a_B^*) \cdot J_0^2(qr_c)$. The exchange term is positive for all $q > 0$, i.e., it always prefers the arrangement in stripes, while E_{Hartree} determines the distance between the stripes.

The approximate ground state derived from a numerical calculation for the arrangement of the electron rings on a triangular lattice for a half filled spin-polarized tenth Landau level is shown in Fig. 5.29(g). One recognizes that the stripes slightly meander. This might be due to the finite size effects in the simulation or due to the fact that the simulation did not find the absolute ground state. It also shows that a bending of the stripes is easily possible, implying that the stripes can follow a moderate

⁷⁷ Recall that the electron-electron interaction energy contains terms of the form $E_{ee,ijkl} = \int \int \psi_i^*(\vec{x}_m) \psi_j^*(\vec{x}_n) \frac{e^2}{4\pi\epsilon\epsilon_0 \cdot |\vec{x}_n - \vec{x}_m|} \psi_k(\vec{x}_m) \psi_l(\vec{x}_n) d^2\vec{x}_m d^2\vec{x}_n$ including four different single particle wave functions. But mostly, the two terms with only two single particle wave functions E_{Hartree} ($i = k, j = l$) and E_{Ex} ($i = l, j = k$) are relevant. All other terms are called correlation terms (Section 4.4).

disorder potential as sketched in Fig. 5.29(h), where areas of large filling are placed in potential minima. In addition, the stripe order costs entropy leading to a melting transition at higher T . For typical electron densities $n_e = 10^{11} - 10^{12}/\text{cm}^2$, the melting temperature can be estimated to be $T_{\text{melt}} \approx 1 - 10$ K.

Experimental evidence for the stripe phase is shown in Fig. 5.29(i). The resistance of a 2D electron system with high mobility μ appears to be largely anisotropic at filling factors $9/2, 11/2$, etc. Temperature dependent measurements at these ν show metallic conductivity in one direction and insulating conductivity in the perpendicular direction. This can be easily rationalized by the stripe phase. Within the transition regions between different integer filling factors of the stripes (Fig. 5.29(a)), edge type states appear at E_F following the stripe directions (Section 5.3.2). They can transport current along the stripes, but not perpendicular to it. Thus, one gets metallic behavior along the stripes and insulating behavior perpendicular to the stripes.

In order to measure this, one needs very low T (25 mK) and high electron mobility μ such that the stripes are oriented in the same direction over macroscopic distances. For example, the electrodes for the experiments shown in Fig. 5.29(i) are 5 mm apart and the mean free path of the electrons within the 2D system is 0.5 mm at 25 mK. Most likely, some effects related to the crystal structure prefer one of the two directions for the orientation of the stripes by small differences in potential energy, but this is not finally clarified.

The combination of large overlap between neighboring wave functions (large E_{Ex}) and large distances between other neighboring wave functions (small E_{Hartree}) can, of course, also be realized differently as shown, e.g., in Fig. 5.29(f). In fact, calculations predict that such phases, called bubble phases, can appear for Landau levels at $\nu \in [n + 1/n, n + 0.4]$ with $n > 2$. These bubble phases are similar to the Wigner crystal, but with many electrons within a density maximum (bubble). The reason, that they are more stable than the stripe phase, is that the electrons at the edges of the electron islands can avoid the Hartree repulsion from neighboring islands more effectively than electrons can avoid each other within the stripe phase. With increasing filling factor, the islands start to overlap. This then favors the stripe phase requiring less free space in order to avoid overlap of electrons from different entities. So far, there are some indications of the existence of these bubble phases close to integer filling factors $\nu \approx n$, that, moreover, seem to appear and to disappear consecutively as a function of ν , but an unambiguous proof is missing.⁷⁸ At very small differences to integer fillings $\nu - n < 1/n$, the bubble phase with one electron per bubble appears in theory and is identical to the Wigner crystal discussed in Section 5.5.2.1.

Notice that the stripe phase appears for a half filled band due to the ring like wave functions within the band. This should be contrasted with the Mott–Hubbard transi-

⁷⁸ Micro-wave experiments and non-linear $I(V)$ characteristics indicate, moreover, that the peaks marked by red arrows in Fig. 5.29(i) are related to insulating bubble phases.

tion, which also appears for a half filled band, but for a band, that consists of nearly point-like electron wave functions (Fig. 4.14 in Section 4.5.3). Hence, details of the wave functions can be decisive for the electronic phase and can lead to very different transport behaviors (anisotropic metal versus insulator).

5.5.2.3 Fractional quantum Hall effect

Within the lowest Landau level ($\nu < 1$), we already found the Wigner crystal at $\nu < 1/5$ as an insulating state (Section 5.5.2.1). Experimentally, a number of minima of the longitudinal resistance accompanied by plateaus of the Hall resistance are found at larger ν as shown in Fig. 5.30(a). The plateaus exhibit $R_{\text{Hall}} = h/e^2 \cdot 1/\nu$ with ν being the filling factor at the corresponding minimum of the longitudinal resistance. The minima appear at rational filling factors $\nu \in \mathbb{Q}$ with odd denominator.⁷⁹ They are also

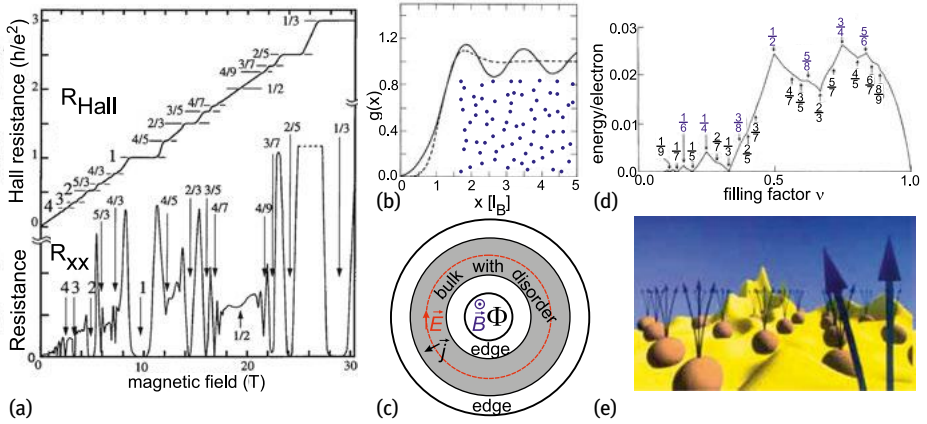


Fig. 5.30: (a) Longitudinal resistance R_{xx} and Hall resistance R_{Hall} of a 2D electron system at high mobility $\mu \approx 10^8 \text{ cm}^2/\text{Vs}$, $n_e = 2 \cdot 10^{11}/\text{cm}^2$. Fractional numbers at the plateaus of R_{Hall} and at the arrows marking the minima of R_{xx} are filling factors ν . (b) Spatial correlation function $g(x)$ of a Wigner crystal (full line) and of the Laughlin liquid at $\nu = 1/3$ (dashed line) with x being the distance between two electrons. Inset: Snap shot of the electrons (blue points) in a Laughlin liquid according to numerical simulations. (c) Corbino geometry (ring area between thick lines) with edge (white areas) and insulating bulk (grey area) due to disorder. The ring is pierced by a magnetic flux Φ , respectively a $\vec{B}$ field perpendicular to the plane. The red line marked with electric field $\vec{E}$ and crossed by current density $\vec{j}$ is used for the integration in eq. (5.113). It explains the appearance of fractional charge (see text). (d) Numerically calculated energy per electron of the ground state as a function of ν . The energy is given in units of $e^2/(4\pi\epsilon\epsilon_0 l_B)$. Filling factors with odd (even) denominator are given in black (blue). (e) Sketch of the stable quasiparticles (electron + three fluxes of h/e) at $\nu = 1/3$. The yellow background is a sketch of the electrostatic disorder potential. (a) [171], (b)–(d) [172], (e) [173].

⁷⁹ More exotic electron phases with minima at even denominators exist, e.g., at $\nu = 5/2$, but are beyond the scope of this book.

somehow symmetric around $\nu = 1/2$. The effect is called the fractional quantum Hall effect (FQHE). It can be explained by a new type of quasiparticles arising from the many-particle wave function of the 2D electron system.

The explanation of the FQHE is based on the preference of certain electron-electron distances in $\vec{B}$ field, that can be understood semi-classically. The Coulomb repulsion is rotationally symmetric and implies that two electrons circle around each other within the $\vec{B}$ field. Since the phase of their single-particle wave functions ψ_i is varied by encircling a magnetic flux and the phase of ψ_i has to be unambiguous, i.e., it must be a multiple of 2π per circle, one obtains preferential distances $\bar{r}_{ee} = |\vec{x}_n - \vec{x}_m|$ between two electrons n and m encircling each other: $\bar{r}_{ee} \approx l_B \cdot \sqrt{2n}$ ($n \in \mathbb{N}$). This argument can be transferred to a system of many electrons encircling each other in pairs at the preferential distance. Consequently, one finds electron densities n_e that are favorable, if the number of fluxes h/e is identical to a multiple of the number of electrons. The corresponding electron phases are called incompressible phases, since they exhibit a band gap for excitations implying that a change of electron density by a minimal amount costs a finite energy penalty. Such a system with band gap is an insulator that explains the minima in R_{xx} straightforwardly (Section 5.3.2). This semi-classical idea would predict minima at fractional numbers $\nu = 1/2, 1/3, 1/4, \dots$, but it does not explain all minima observed in experiment and wrongly predicts minima at fractional numbers with even denominators. The latter is due to the fact that it neglects the fermion character of the electrons implying an antisymmetric exchange statistics.

A more detailed argument has been given by Bob Laughlin. He used three requirements from general physical principles in order to guess a many-particle wave function, that allowed a reasonable quantitative description of the most prominent minimum at $\nu = 1/3$.

The requirements are:

1. Since electron-electron interaction as well as $\vec{B}$ field are rotationally symmetric in the 2D plane, the many-particle wave function $\Psi_{el}(\vec{x}_1, \dots, \vec{x}_{N_e})$ must be an eigenfunction of the total orbital momentum.
2. Since electrons are fermions, the many-particle wave function must be odd with respect to an exchange of two coordinates. Since the parallel spins in the lowest spin polarized Landau level ($\nu < 1$) imply an even spin function, the spatial wave function has to be odd.
3. If the electron-electron interaction at large enough $\vec{B}$ is significantly smaller than $\hbar\omega_c$, the many-particle wave function consists of single-particle wave functions $\psi_i(\vec{x})$ of the lowest Landau level⁸⁰

$$\psi_i(\vec{x}) = \exp\left(\frac{x^2 + y^2}{4}\right) \cdot \left(\frac{d}{dx} + i \cdot \frac{d}{dy}\right) \cdot \exp\left(-\frac{x^2 + y^2}{2}\right). \quad (5.108)$$

⁸⁰ For the sake of simplicity, we neglect normalization factors of the wave functions, that are irrelevant for the following arguments.

In order to make the representation simple, we use the coordinates $z := (x + iy)/l_B = |z| \cdot \exp(i\varphi)$. Then, the arguably most simple function obeying all three requirements for N_e electrons is:

$$\Psi_{m,\text{el}}(z_1, z_2, z_3, \dots, z_{N_e}) = \prod_{k=1}^{N_e} \prod_{j < k} (z_j - z_k)^m \cdot \exp\left(-1/4 \cdot \sum_{l=1}^{N_e} |z_l|^2\right). \quad (5.109)$$

This function exhibits a quantized orbital momentum of $L_{\text{tot}} = \hbar \cdot \frac{N_e(N_e-1)}{2} \cdot m$ (without derivation) and is antisymmetric with respect to exchange of two coordinates, if m is odd. Implicitly, we use the idea that all electrons within the many-particle wave function are identical (m is the same for each pair (j, k)) and that the electron-electron repulsion, which has to be minimized, interacts between pairs of electrons only, i.e., the polynomial factors consist of differences of only two coordinates. An analysis of this function reveals that it exhibits a lower total energy than the Wigner crystal for most ν . This can be rationalized by regarding the correlation function⁸¹ $g(x)$ shown for $m = 3$ in Fig. 5.30(b) (dashed line). While the exchange-correlation hole around the selected electron can be approximated by a x^2 function for the Wigner crystal (solid line), it is a x^{2m} function for the Laughlin wave function (dashed line). Consequently, electrons in the Laughlin function are less close to each other. This results in a smaller Coulomb repulsion energy. Additionally, the Laughlin function shows barely a minimum at the position of a neighboring electron, while the Wigner crystal shows long range oscillations of $g(x)$ indicating that the Laughlin wave function represents an electron liquid, while the Wigner crystal is an electron solid. The inset of Fig. 5.30(b) is a typical snapshot of the electrons within the liquid as obtained by numerical calculations [172]. It reveals the long range disorder in the electron distribution and the preferentially large distance between neighbors by naked eye.

Interpreting the square of $\Psi_{m,\text{el}}$ of the Laughlin liquid as a classical probability distribution of particles at finite temperature T within a potential energy E_{pot} , one can understand the preference of filling factor $\nu = 1/3$. Therefore, one writes:

$$|\Psi_{m,\text{el}}|^2 = e^{-E_{\text{pot}}(z_1, \dots, z_{N_e})/k_B T} \quad (5.110)$$

with

$$E_{\text{pot}}(z_1, \dots, z_{N_e}) := -2m^2 \sum_{k=1}^{N_e} \sum_{j < k} \ln |z_j - z_k| + m/2 \cdot \sum_{l=1}^{N_e} |z_l|^2 \quad \text{and} \quad k_B T := 2m. \quad (5.111)$$

Here, $k_B T = 2m$ is chosen such that E_{pot} becomes identical to the potential of a classical 2D plasma made out of N_e charges with charge m . The charges pairwise repel

⁸¹ A correlation function $g(x)$ describes the probability to find one of the other electrons in a distance x from a selected electron.

each other via a logarithmic function (first term on the right of $E_{\text{pot}} \dots = \dots$).⁸² Moreover, they are confined in a parabolic background potential (second term on the right of $E_{\text{pot}} \dots = \dots$). The plasma is, hence, accompanied by a homogeneous circle of charges with opposite sign. The charge density of this background is given by $1/(2\pi l_B^2)$, if one includes the constants correctly within eq. (5.109) and calculates the potential of a homogeneously charged disk.⁸³ These background charges keep the charges m within the area. It is energetically favorable to keep the plasma electrically neutral on small length scales, i.e., each area of a flux of h/e gets $1/m$ from the charge carriers with charge m . Consequently, the ideal charge density is $N_e = N_\phi/m$ with N_ϕ being the number of magnetic flux portions of strength h/e . Thus, we found the preferential filling factor to be $\nu = 1/m$ (Fig. 5.31(b)). Each additional m -type charge or each additional flux h/e (background charge) would locally charge the equivalent plasma. This is energetically unfavorable.

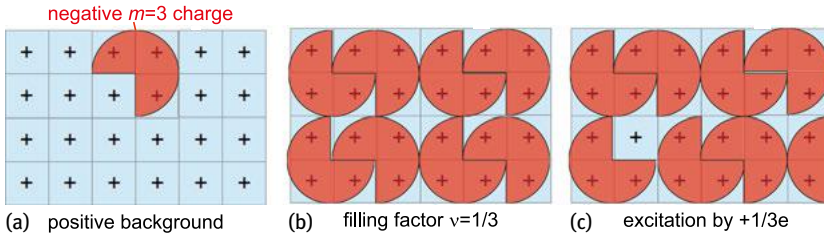


Fig. 5.31: Visualization of the plasma analogue of the fractional quantum Hall effect close to $\nu = 1/3$. (a) Positive background (blue) with individual $+1$ charges and a single $-m$ charge (red object) representing an electron for the case $m = 3$. (b) Favorable filling of the -3 charges at $\nu = 1/3$. (c) Charge excitation by a third of a -3 charge, that represents the excitation by charge of $e/3$, slightly away from $\nu = 1/3$.

Consequently, the Laughlin liquid is most stable at filling factors $1/m$ with an odd integer m . This phase is separated by a band gap from the phase with an additional charge. This has also been found numerically, i.e., the 2D electron system at $\nu = 1/3, 1/5 \dots$ is an insulator with a band gap. Hence, it requires a minimum energy to add an additional quasiparticle. Typical excitation energies at $m = 3$ are $\Delta E \approx 0.027e^2/(4\pi\epsilon\epsilon_0 l_B)$.⁸⁴ In fact, the strongest minimum of $R_{xx}(B)$ (Fig. 5.30(a)), that implies the largest excitation gap, is observed at $\nu = 1/3$. At lower $\nu \leq 1/5$, the

⁸² The logarithmic term becomes plausible by using Maxwell's equation in 2D reading $\oint \vec{E} d\vec{s} = Q/\epsilon\epsilon_0$ leading to $|\vec{E}| \propto (r - r_0)^{-1}$ and $E_{\text{pot}} \propto \ln(r - r_0)$, where $r - r_0$ is the distance to a particular charge.

⁸³ One can show that a homogeneous circle of charge leads to a parabolic confinement potential (see also Section 5.5.1.1, Fig. 5.24(c)).

⁸⁴ Using $\epsilon = 13$ as for GaAs and $B = 10$ T implying $l_B = 9$ nm, we obtain $\Delta E = 0.3$ meV.

Wigner crystal often gets energetically favorable, such that our theory only applies for $\nu = 1/3$ and partly, depending on the details of the system, for $\nu = 1/5$.

Interestingly, the charged excitations of these ground states carry a charge of $Q = e/m$ ($e = 1.6 \cdot 10^{-19}$ C). Intuitively, this is the charge, that can be moved between the flux boxes (Fig. 5.31(c)). More formally, this gets evident, if one analyzes the ring geometry in Fig. 5.30(c), that is pierced by a current carrying coil in the center marked by the magnetic flux Φ and the corresponding $\vec{B}$ field pointing out of the plane. If the interior of the 2D sample (inner ring) is insulating as for $\nu = 1/m$ or due to disorder forming localized states (Section 5.3.2), the tensor of resistivity reads:

$$\boldsymbol{\rho} = \begin{pmatrix} 0 & B/(n_e e) \\ B/(n_e e) & 0 \end{pmatrix}. \quad (5.112)$$

Changing Φ adiabatically, i.e., slowly, with the help of the coil implies by Maxwell's equations (eqs. (2.41)–(2.44), eq. (1.1) (Ohm's law)):

$$\begin{aligned} \frac{d\Phi}{dt} &= \iint \frac{d\vec{B}}{dt} d\vec{A} = -\oint \vec{E} d\vec{s} = -\oint \boldsymbol{\rho} \cdot \vec{j} d\vec{s} \\ \Rightarrow \frac{d\Phi}{dt} &= -\rho_{xy} \oint \frac{d\rho_{c\perp}(\vec{x})}{dt} d|\vec{s}| \Rightarrow \Phi(t) - \Phi(0) = -\rho_{xy} \cdot \Delta Q_{\perp}. \end{aligned} \quad (5.113)$$

The corresponding contour of the closed integration path (red line) and the vectors $\vec{E}$ and $\vec{j}$ are sketched in Fig. 5.30(c). Naturally, $\vec{E}$ is perpendicular to $\vec{B}$ by Maxwell's equation and $\vec{j}$ is perpendicular to $\vec{E}$ by the exclusively non-diagonal elements of $\boldsymbol{\rho}$. Finally, we used the definition of the current density $|\vec{j}| = d\rho_c/dt$ with charge density $\rho_c(\vec{x})$ (Section 1.2.1) and considered that the charge flow is perpendicular to the integration contour ($\vec{E}$) via the non-diagonal matrix $\boldsymbol{\rho}$. This is symbolized by $\rho_{c\perp}$ and the total charge $\Delta Q_{\perp}$ traversing this contour (red line). Thus, the addition of a magnetic flux causes a charge transport $\Delta Q_{\perp}$ through a closed ring with arbitrary radius within the insulator. Adding a flux of $\Phi(t) - \Phi(0) = h/e$ implies a change of the single particle wave functions by the irrelevant factor $e^{i2\pi}$. Thus, the transport of charge across the ring, i.e., from the inner edge to the outer edge or vice versa, is equivalent to an excitation of the system without changing the contributing $\psi_i(\vec{x})$ and without changing the energy, if the process is performed adiabatically, i.e., slowly. The Laughlin wave function describing the interior of the ring, consequently remains unchanged. With $n_e = 1/m \cdot eB/h$ at $\nu = 1/m$ using the degeneracy of a spin polarized Landau level eB/h (Section 5.3.2, page 278), we find

$$\Delta Q_{\perp} \stackrel{\text{eq. (5.113)}}{=} \frac{h}{e\rho_{xy}} \stackrel{\text{eq. (5.112)}}{=} \frac{hen_e}{eB} = \frac{he^2B}{heBm} = \frac{e}{m}. \quad (5.114)$$

The edge channels, thus, exchange fractional charges e/m by redistribution of the electron wave functions during the adiabatic cycle and, consequently, can also transport fractional charges e/m . This fits with the intuition that, e.g., the positive charge in

Fig. 5.31(c) can be moved through the lattice by adequate rearrangement of the $m = 3$ negative charges.

A more formal quantum mechanical way to explain the fractional charges is to add the flux h/e to Laughlin's wave function (eq. (5.109)). Such an additional flux must be represented by an additional zero for each single particle wave function, since surrounding the flux gives a phase factor of 2π . This would lead to a diverging kinetic energy after being contracted to a point, except if this point has a weight of zero (Section 5.4.2, Fig. 5.13(a)). The additional flux can, thus, be represented by a factor $\prod_{j=1}^{N_e} (z_0 - z_j)$ in front of Laughlin's wave function (eq. (5.109)). However, this factor differs from the usual factor of an additional electron with index k , $\prod_{j=1}^{N_e} (z_j - z_k)^m$, in a way that adding an additional electron to the system would be equivalent to adding three such fluxes. Thus, adding a flux of h/e to the 2D electron system, which is an allowed operation, one formally adds a fractional charge of $Q = e/m$.

The fractional charge of the quasi-particles can be directly measured by the square of the amplitude of the current noise S_I with units A^2/Hz . Therefore, one uses a setup as sketched in Fig. 5.32(a). One drives a current from source to drain, i.e., along the upper edge channel. Within the constriction, that can be changed in width by the two marked gate voltages, charge carriers can be scattered from the upper edge channel to the lower one and, thus, can be detected at the electrode labeled "Noise." The strength of the noise of this current is given by the arbitrary back scattering of individ-

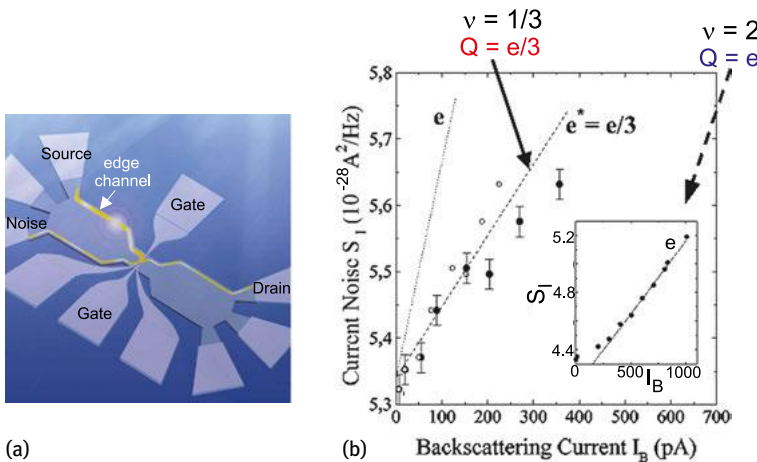


Fig. 5.32: (a) Sketch of the sample used to measure the fractional charge within the fractional quantum Hall regime. The voltage is applied between source and drain and drives a current through the edge channels of the 2DES, while the backscattered current from the constriction at the gates is analyzed via the contact "Noise." (b) Square of the amplitude of the current noise per frequency interval S_I as a function of the current I_B detected at the electrode "Noise" at filling factor $\nu = 1/3$ (main image) and filling factor $\nu = 2$ (inset). Lines mark expected noise levels for different charges of the quasiparticles as marked. (b) [174].

ual charges at the constriction. The noise at given backscattered current gets reduced, if the individual charges get smaller, since then the current flows in smaller pieces and, hence more continuously. Figure 5.32(b) shows indeed that the current noise at $\nu = 1/3$ is significantly smaller than expected for an individual charge of e and can be fitted assuming a charge of $Q = e/3$. This result is not contradicting the quantization of the elementary charge in units of e , but appears due to the correlated motion of all electrons within the system. This correlated motion makes $Q = e/m$ a measurable quantity.

Introducing disorder, as sketched in Fig. 5.30(e), leads to insulating behavior, even if ν slightly deviates from $\nu = 1/m$. The argument is similar as in the integer quantum Hall regime (Section 5.3.2). The excited quasiparticles with charge e/m (electrons or holes) are localized in the disorder potential (Fig. 5.30(e)), such that they cannot move through the system and, hence, they cannot transport charge. Only the edge channels contribute to the conductivity. As for the integer QHE (Section 5.3.2), they form a ballistic, one-dimensional conductor, that, however, transports particles with fractional charge $Q = e/m$. Since each ballistic channel contributes a conductivity of $Q \cdot e/h$ (derivation at the end of Section 5.3.2 or in Section 1.4.1), we get a Hall resistance: $R_{\text{Hall}} = h/eQ = h/e \cdot m/e = hm/e^2$. This explains the plateau value in R_{xy} around $\nu = 1/3$ (Fig. 5.30(a)). With the help of the symmetry of electrons and holes within the lowest Landau level, it also explains the plateau around $\nu = 2/3$, that represents the lowest Landau level occupied by $1/3$ with holes.

But what about the other minima in R_{xx} and plateaus in R_{Hall} ? Figure 5.30(d) shows the energy of the two-dimensional electron system in a $\vec{B}$ field as resulting from a numerical calculation for a few electrons. It displays the ground state energy per electron as a function of ν . Besides the minima at $\nu = 1/3$ and $\nu = 2/3$, minima appear at other ν , that, at least, partly correspond to the ν of the $R_{xx}(B)$ -minima in Fig. 5.30(a).

A possible explanation for the additional minima is a favorable arrangement of the quasi particle excitations (charge: $e/3 = \text{flux } h/e$) that are added to the favorable state at $\nu = 1/3$. Formally, one repeats the argument of the plasma analogue for the remaining excited quasiparticles not belonging to the $\nu = 1/3$ state. These quasiparticles move in an equivalent background of electrons. Since the excited quasiparticles are formally similar to bosons⁸⁵, one obtains a Laughlin wave function (eq. (5.109)) of them with even exponent, that is called k to distinguish it from the odd exponent m . Hence, even integers k of the inverse filling factor N_e/N_{qp} (N_{qp} : quasi particle number) are preferred within the plasma analogue. A favorable filling factor is, thus, defined

85 Using the wave functions of the excited quasiparticles of the form $\prod_{j=1}^N (z_0 - z_j)$ does not lead to a negative sign after exchanging two quasiparticle terms z_i , i.e., they are bosons. As a side remark, we mention that the real exchange statistics of the $\nu = 1/3$ quasiparticles with charge $e/3$ is fractional, i.e., an exchange of two quasiparticles reveals a phase factor $e^{in/q} \neq \pm 1$. This is called anyon statistics [175].

by (N_Φ : number of fluxes h/e):

$$N_e = k \cdot N_{qp}, \quad N_\Phi = mN_e + \alpha_\pm \cdot N_{qp}, \quad (5.115)$$

with $\alpha_\pm = \pm 1$ in order to describe electron and hole excitations of the $\nu = 1/3$ state on the same footing. Solving for the filling factor $\nu = N_e/N_\Phi$ reveals

$$\nu = \frac{kN_{qp}}{mN_e + \alpha_\pm N_{qp}} = \frac{k}{mk + \alpha_\pm} = \frac{1}{m + \alpha_\pm/k}. \quad (5.116)$$

This novel stable state can again be excited by fractionally charged quasiparticles with symmetric exchange statistics, such that one gets a hierarchy of favorable filling factors:

$$\nu = \frac{1}{m + \frac{\alpha_{1,\pm}}{k_1 - \frac{\alpha_{2,\pm}}{k_2 - \frac{\alpha_{3,\pm}}{k_3 - \dots}}}} \quad (5.117)$$

with k_i (m) being even (odd) integers and $\alpha_{i,\pm} = \pm 1$.

This indeed explains many of the states (plateaus) observed in the FQHE regime, but fails to explain its hierarchy, i.e., states of higher hierarchy in this model appear at larger temperature than states of lower hierarchy.

The correct hierarchy can instead be obtained within the so-called composite fermion description. Therefore, one realizes that Laughlin's wave function for $\nu = 1/3$ can be rewritten as $\prod_j \prod_{i < j} (z_i - z_j)^{m-1} \cdot \Psi_1^{LL}(z_1, z_2, \dots, z_{N_e})$ with Ψ_1^{LL} being the favorable wave function for $\nu = 1$ (Laughlin's wave function for $m = 1$ or Slater determinant of the single particle wave functions of the lowest Landau level). Formally, one takes the many-particle wave function of the lowest Landau level, that is insulating (Section 5.3.2), and adds two fluxes of h/e to each electron for $m = 3$. The system at $\nu = 1/3$ is ergo a system of electrons with two attached fluxes h/e , that fill exactly one Landau level. This corresponds to an effective $\vec{B}$ field B_{eff} for the new quasiparticles, consisting of one electron and two fluxes of h/e , that is exactly $1/3$ of the real $\vec{B}$ field or $B_{\text{eff}} = B - B_{1/2}$ with $B_{1/2}$ being the B field at $\nu = 1/2$. Thus, the two fluxes of h/e , caused by the concerted motion of the other electrons around a particular electron, compensate an external magnetic field of $B_{1/2}$. One can easily show that the attachment of a flux of $2 \cdot n \cdot h/e$ ($n \in \mathbb{N}$) to an electron does not change the fermionic statistics [176]. Intuitively, it is still a fermion since the exchange of two coordinates in the additional factor $\prod_j \prod_{i < j} (z_i - z_j)^{2n}$ reveals $+1$, such that the exchange statistics is not changed. A system with electron density of one electron per flux $2h/e$ is, thus, a system of novel fermions, consisting of two fluxes h/e and one electron, with $B_{\text{eff}} = 0$ T. These new fermions are called composite fermions. At a real external $\vec{B}$ field of x fluxes of h/e ($x \in \mathbb{R}$) per electron the composite fermions are prone to $x - 2n$ fluxes of h/e per composite fermion. For $\nu = 1/3$ and $n = 1$, one gets one such flux per composite fermion and, thus, a completely filled lowest Landau level.

Formally, the addition of flux to the electrons is performed via the so called Chern–Simons gauge transformation. More intuitively, electrons at $\nu = 1/2$ encircle each

other in a way that the external field is perfectly compensated for each electron. Applying an electric field then moves the composite fermions similar to free electrons, i.e., they behave like Bloch waves. However, the whole encircling environment has to be moved with the electron in order to keep the effective $\vec{B}$ field at zero. This leads to a larger inertial mass for the composite fermions than for the bare electrons. Indeed, the mass m_{CF}^* depends on the density of the electrons at $\nu = 1/2$ according to $m_{\text{CF}}^* \propto 1/l_B \propto \sqrt{n_e}$, respectively, the Fermi circle of the composite fermions exhibits a Fermi wave vector of $k_{\text{F}}^{\text{CF}} \propto 1/l_B$. These composite fermions have been found by experiments at $\nu = 1/2$. For example, applying a very small magnetic field to the composite fermions leads to a movement of the composite fermions in relatively large circles (diameter: $\sim 1 \mu\text{m}$) as expected from the Lorentz force [177].

The number of composite fermions N_{CF} is fixed by the electron number at the field $B_{1/2}$ for filling factor $\nu = 1/2$ according to $N_{\text{CF}} = 1/2 \cdot \frac{eB_{1/2}}{h} A$. Here, A is the sample area and eB/h is the degeneracy of a spin polarized Landau level. By deviating the B field from $B_{1/2}$, the composite fermions effectively feel the field $B_{\text{eff}} = |B - B_{1/2}|$ implying an effective number $N_{\tilde{\Phi}}(B_{\text{eff}})$ of flux quanta $\tilde{\Phi}$. Consequently, the composite fermions form Landau levels and show quantum Hall plateaus around effective integer filling factors $\tilde{\nu} = N_{\text{CF}}/N_{\tilde{\Phi}}(B_{\text{eff}}) \in \mathbb{N}$ (eq. (5.105)).

Next, we want to deduce the corresponding real filling factors $\nu = \frac{n_e}{eB/h}$ (eq. (5.105)). We employ $N_{\tilde{\Phi}} = \frac{B_{\text{eff}} A}{h/e}$ and $N_{\text{CF}} = 1/2 \cdot \frac{eB_{1/2}}{h} A$. This leads to

$$\tilde{\nu} = \frac{N_{\text{CF}}}{N_{\tilde{\Phi}}(B_{\text{eff}})} = \frac{eB_{1/2}}{2h} \cdot \frac{h}{eB_{\text{eff}}} = \frac{B_{1/2}}{2 \cdot |B - B_{1/2}|} \quad (5.118)$$

$$\Rightarrow |B - B_{1/2}| = \frac{B_{1/2}}{2\tilde{\nu}} \quad (5.119)$$

$$\Rightarrow B = B_{1/2} \cdot \frac{2\tilde{\nu} \pm 1}{2\tilde{\nu}}. \quad (5.120)$$

Since $n_e = \frac{1}{2} \frac{B_{1/2}}{h/e} = \nu \frac{B}{h/e}$ (Section 5.3.2), one gets $\frac{B}{B_{1/2}} = \frac{1}{2\tilde{\nu}}$ leading to

$$\nu = \frac{\tilde{\nu}}{2\tilde{\nu} \pm 1}. \quad (5.121)$$

Obviously, the state at $\nu = 1/3$ corresponds to the filling factor $\tilde{\nu} = 1$ with + sign for the composite fermions, such that the initial idea of the derivation naturally reappears. Moreover, the apparent symmetry of the resistance minima around $\nu = 1/2$ (Fig. 5.30(a)) and the fact that the minima get deeper at increasing B distance from $\nu = 1/2$ can be naturally explained. The former is due to the symmetry between negative and positive B_{eff} for the composite fermions and the latter is due to the increased absolute value of B_{eff} increasing the energy distance between adjacent Landau levels of the composite fermions.

Such fractional states, explained by Landau levels of composite fermions, can also be observed in higher Landau levels (see, e.g., the minima at $\nu = 4/3$ and $\nu = 5/3$ in Fig. 5.30(a)), where the completely filled spin polarized Landau level is just a polarizable background (Section 5.5.2.2). The argument also applies for other even numbers

of fluxes h/e attached to the electrons as, e.g., four such fluxes per electron, revealing quasi-particles without effective $\vec{B}$ field at $\nu = 1/4$.

The system of composite fermions shows, in addition, collective behaviors known from normal electrons, such as ferromagnetism with domain patterns or superconductivity driven by Cooper pairs of the composite fermions. This provides the advantage that the collective effects can be switched on and off by either changing n_e via a gate voltage or by changing $\vec{B}$.⁸⁶

The predictive power of the composite fermion model, thus, appears to be enormous, albeit it is based only on a well-guessed wave function (eq. (5.109)). The FQHE is another excellent example showing that the electron-electron interaction can provide completely unexpected behavior as, e.g., the fractional charge. Since there is no way to solve the corresponding Schrödinger equation by ab-initio methods, the results have not been predicted prior to their experimental observation. Moreover, they needed to be rationalized by a mixture of electrodynamic and quantum mechanical arguments prior to the development of adequate models, that support the arguments quantitatively.

Another prominent example of this kind is the description of interacting electrons in 1D without $\vec{B}$, that is called the Luttinger liquid description. Here, the low-energy excitations are either collective spin density waves or collective charge density waves. These two types of excitations exhibit different velocities, such that inducing an electron into the system at one side leads to a different velocity of its charge degree of freedom and its spin degree of freedom. Hence, the charge degree of freedom arrives earlier on the other side of the system than its spin counterpart. However, describing this behavior more formally by so-called bosonization of a model Hamiltonian is beyond the scope of this book and can be found, e.g., in [178–180].

5.6 Summary

The last two chapters of the book have described several electronic properties, that would not exist within the well established description of the electrons by Bloch waves. They arise because of additional terms in the many-particle Schrödinger equation describing electron-phonon, electron-disorder, spin-orbit, or electron-electron interaction. The resulting electron phases such as charge density waves, integer quantum Hall phases, Mott–Hubbard insulators, or fractional quantum Hall phases do not drop immediately from the Schrödinger equation, which is too complex to be solved rigidly. Often, such phases have been discovered by chance and are described theoretically much later. The most prominent example is superconductivity, discovered in 1911 and explained in 1957, albeit the required Schrödinger equation has been known

⁸⁶ Conventional superconductivity can also be switched off by a $\vec{B}$ field.

since the late 1920s. The high temperature superconductivity, discovered in 1986, remains unexplained up to now, i.e., even the question if electron-phonon interaction, electron-electron interaction or both dominate the pairing of the Cooper pairs is not clarified unambiguously. The Kondo effect, briefly described in Section 5.5.1.2, was discovered in 1934 and required 30 years before being explained by a second order spin dependent electron-electron interaction. Of course, also the opposite way of a theoretical prediction, which was difficult to be found in experiments, appears. For example, the concept of a Luttinger liquid was introduced in 1963, while first experimental hints of its existence were found only in the 1990s, that are still not unambiguous.

Importantly, the rigorous understanding of individual effects can have far reaching consequences. Most strikingly, the mathematical understanding of the exact quantization of the quantum Hall effect in terms of topological indices opened the whole new field of topology in solid state physics. This field continues to predict novel physical properties of well studied materials by theoretical analysis. Interestingly, these properties have been mostly overlooked in previous studies or partly have been found without any reasonable explanation. This tells us that adequate analysis of the Schrödinger equation might still bear novel possibilities for the exploitation of electrons, albeit it has been tackled for about a century.

In other words, albeit the describing equation for electron systems in solids is known exactly since nearly 100 years, the investigation of such systems still provides groundbreaking results due to the impossibility of a rigid solution of the Schrödinger equation, neither analytically nor numerically. It is easy to imagine that combinations of the different interaction terms might lead to additional insights in the near future.